TENDER AND CONTRACT DOCUMENTS

FOR

PROJECT: G+4P+20TYP+R RESIDENTIAL BUILDING @ 1341148 , AL MAMZAR.

OWNER: MS. HESHAM ABDULLA QASSIM AL QASSIM, ABDULLA AL QASSIM

VOLUME -I – MECHANICAL
(M.E.P. SPECIFICATIONS)

OWNER: MS. HESHAM ABDULLA QASSIM AL QASSIM, ABDULLA AL QASSIM	**CONSULTANT:** URBANHABITAT URBAN HABITAT ARCHITECTURAL & CONSULTANT BUSINESS BAY , DUBAI, UNITED ARAB EMIRATES

SECTION -1

PART A – MECHANICAL WORKS

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SECTION 1:
GENERAL REQUIREMENTS.
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GENERAL:

- This specification is applicable to MEP works. However, it is to be read in conjunction with all other contract documents including all document issued for Main Contract.
- The term 'Consultant' used in this specification means 'Engineer'.
- This section defines the minimum mechanical services installation requirements the Montrose Development, Dubai. The section shall be read in conjunction with the complete specification requirements: however, reference shall be given to Division 20 – General MEP Requirements, and all other MEP divisions of the specification documentation.
- The specification, including the technical schedules is to be read in conjunction with the drawings,
- which jointly describe the contract works. Where a discrepancy is identified between the requirements of the specification and the drawings, the specification is to take precedence and the discrepancy notified to the Engineer for the necessary action to be implemented.

TECHNICAL REQUIREMENTS

- The following Clause identifies the major areas for which Contractors design/design development is required. This specification section is to be read in its entirety for other areas
- Major Areas for Contractor Design/Design Development
- Provide design for primary and secondary support system for all MEP and associated equipment. Also access ladders, walkways, gantries and guardrails to suit the MEP installation.
- Re-calculate the pump and fan resistances based upon the Shop drawings and the actual equipment selected to ensure that they meet the performance requirements for the plant and equipment selected.
- Design the provisions of anchors, bellows, compensators, loops and bends to accommodate thermal expansion and contraction for the pipe work systems.
- Produce shop drawings and co-ordination drawings as required by the MEP specification.
- Develop acoustic treatment to meet the noise and vibration criteria specified.
- Produce Builders work drawings of all openings, bases, etc.

LIAISON WITH AUTHORITIES

- The Contractor shall ensure approvals are received from all Local Authorities associated with the HVAC systems. Authorities in question shall include, but not be limited to, the following: -
- **Dubai Civil Defence (DCD)**
- **Dubai Municipality (DM)**
- **Dubai Electricity and Water Authority (DEWA)**
- **Telcom Regulations**

CIVIL DEFENCE AUTHORITY

- It is the responsibility of the Contractor to comply fully with the requirements and regulations of the local Civil Defence Authority, liaise with them and obtain their final approval of drawings and installation. All relevant drawings and equipment shall be approved by Civil Defence prior to commencement of work and all time and expenses incurred in gaining approvals shall be included in the tender (including any equipment, material or system required by the Civil Defence Authority but not mentioned explicitly in these documents).

DUBAI MUNICIPALITY

- It is the responsibility of the Contractor to comply fully with the requirements and regulations of the Local Authority, liaise with them and obtain their approval of drawings and installation.

DUBAI ELECTRICITY AND WATER AUTHORITY

- It is the responsibility of the Contractor to comply fully with the requirements and regulations of the Local Authority, liaise with them and obtain their approval of drawings and installation.

TELCOM REGULATIONS

- It is the responsibility of the Contractor to comply fully with a) the requirements and regulations of the Local Authority, liaise with them and obtain their approval of drawings and installation.

QUALIFICATION OF CONTRACTOR:

- The scope of MEP work on this job includes variety of high quality Equipment and Systems, as detailed in the specification and drawings. It is important to achieve the intended quality of performance for long term functioning of the services installation.
- Each Services Sub-contractor must have the technical expertise to undertake this quality of work with similar demonstrably completed projects in the United Arab Emirates. The Consultant reserves the right to inspect any or all of the Services Sub-contractor's designated similar projects and to approve, or otherwise, the Contractor on the basis of the work so demonstrated. The Consultant may request certificates of satisfaction from Consultants or Owners of other projects done by the Sub-contractor.
- The Consultant is not obliged to give reasons for any acceptance or rejection of any Sub-contractor, or whether any rejection is due to past performance or to present staff limitations or otherwise.

DESIGN AND INTENT:

- Contract Drawings and Specification provide indicative design and intent for the system requirements.
- Based on the contract documents, Contractors shall prepare the shop drawings, and carry out installation and commissioning of all required systems to meet design intent for the job.
- Provide all items, articles, materials, operations, sundries, labour, supervision, guarantees, allowances for overhead and profit, etc., shown or implied in any of the contract documents. Include equipment, tools, scaffolds, and other incidentals as may be required.
- Consider the Specifications as an integral part of the work together with the drawings. Consider any item or subject omitted from one, but mentioned or reasonably implied on the other as properly and sufficiently indicated and provide the same under the work of this division.

EXAMINATION OF SITE AND DRAWINGS:

- Visit the site of the proposed works and obtain all information as to existing conditions and limitations and all proposed works on adjacent sites and in adjacent areas, which might affect the works on this site, whether by Private Individuals or by Government Authorities or others.
- Examine the specifications and all drawings including the specifications and drawings of all other divisions before bidding and again before commencing any portion of the works.
- Neither the Client nor the Consultant will be responsible for any claim for extra work or expense resulting from the failure of the Contractor to be fully aware of site conditions, drawings implications, etc..

CONTRACT DRAWINGS:

- The drawings for services works are performance drawings, diagrammatic, and intended to convey the scope of work and indicate general arrangement and approximate locations of apparatus, fixtures, pipe and duct runs, etc. The drawings do not intend to indicate Architectural or Structural details.
- Do not scale drawings. Obtain accurate dimensions to structure and architectural items from drawings of those trades. Confirm by site measurement. Verify location and elevation of all

services (Water, Electrical, Telephone, Sanitary, Storm Drainage, Gas, etc.) before proceeding with the work.

- Install all ceiling mounted components (Diffusers, grilles, detectors, light fixtures, emergency lights, fire detectors, loudspeakers, camera points, etc.) in accordance with the reflected ceiling drawings, which are to be prepared by the Contractor and coordinated with all trades. These must be submitted for approval and be approved before any work commences for installation of MEP items in false ceiling.
- Leave space clear and install all work to accommodate future materials and/or equipment as indicated and/or supplied by another Division of work of the Contract. Install all pipe runs, conduit runs, cable trays, etc., to maintain maximum headroom and clearances, and to conserve space in shafts and ceiling spaces and under floors, and to provide adequate space for service and maintenance.
- Confirm on the site the exact location of outlets and fixtures. Confirm also location of outlets and fixtures provided by any other Division of work under the Contract.

BUILDER'S WORK DRAWINGS:

- Prepare 1:50 scale drawings in conjunction with all trades concerned, showing sleeves and openings for all passages through structure and all insert sizes and locations, equipment bases, pump pits, anchors, inertia slabs, floor and roof curbs, wall openings, trenches, pertaining to MEP works.

SHOP DRAWINGS:

- Submit shop drawings and samples for materials and equipment as listed in this and in each subsequent section.
- Present a schedule of shop drawings after award of the Contract and not later than what is specified in the Contract documents, indicating the anticipated date when the drawings will be submitted for review. Assume full responsibility for timely submission of all drawings. Allow a minimum of three weeks for the Consultant's review.
- Show all MEP services items at 1:50 scale on the layout drawings and at 1:20 scale details for shafts, plant areas, pump room, etc.. Base all dimensions of equipment on the approved make and models. Co-ordinate with all other trades for location, routes and levels. Show all requirements of maintenance spaces, access doors and cleanouts.
- Assume responsibility for accuracy of equipment dimensions related to space available, accessibility for maintenance and service, compliance with inspection authorities codes.
- The submission of samples will be subject to the same procedure as that of shop drawings.
- The Consultant's review shall not relieve the Contractor from responsibility for deviations from the Consultant's drawings and specifications, unless he has, in writing, called the Consultant's attention to such deviations at the time of submission of drawings. The Consultant's review shall be construed to apply to, and only to, general arrangements and shall not relieve the Contractor from the entire responsibility. Any approval by the Consultant shall be on the understanding that any item submitted shall be ordered with options and modifications to fully meet the specification. Any fabrication, erection, setting out or other work done in advance or receipt of stamped drawings shall be done entirely at the Contractor's risk and cost.

AS BUILT DRAWINGS

- Keep one complete set of approved MEP shop drawings at site office.
- Show on those prints the installed inverts of all services entering and leaving the building. Dimension underground services at key points of every run in relation to structure and building. Record all elevations for underground services in relation to floor level of the building and give reference datums to Municipal bench marks.
- Indicate exact location of all services left for future work. Show and dimension all work

embedded in the structure.

- At the end of job execution, revise the negatives of the original shop drawings to incorporate any and al changes, done at site during execution of job so that the "As Built" drawings shall indicate the exact "As Built" condition of all services.
- Before preliminary handing over, two (2) sets of prints of these "As Built" drawings shall be submitted to Engineer for approval. Upon approval four (4) sets of prints shall properly bound and handed over to Client as part of Preliminary Handing Over of the job.

OPERATION AND MAINTENANCE MANUALS

- At the time of Preliminary Handing Over, Contractor shall handover to Client four (4) bound sets of the operation and maintenance manuals. These shall consist of, but not limited to following:
- List of all equipment installed.
    - General layout.
    - Wiring diagram of control panels.
    - Operation instructions, include start-up and shutdown procedure.
    - Maintenance instructions, including preventative maintenance instructions for components of the equipment.
    - Lubricating instruction and recommended cycle of lubrication.
    - Complete parts lists, showing manufacturer's name catalogue number, as recommended by manufacturer for all system equipment, necessary for two years operation.
- The Contractor is to prepare and submit one draft copy for Consultant's approval prior to submitting 4 Nos. final sets.
- The section dealing with complete systems shall be subdivided into each service with a ready means of reference and detailed index. The function and manner of operation of each system shall be clearly described together with illustrations and line diagrams in schematic form showing the location and function of control valves, items of equipment and spaces or areas, which are served by these items. The colour coding and identification systems employed shall be explained.

PROTECTION AND STORAGE

- Protect the buildings and structures from damage due to carrying out of this work.
- Protect all mechanical work from damage. Keep all equipment always dry and clean.
- Cover all openings in equipment and materials. Cover all temporary openings in ducts and pipes with polyethylene sheets or caps until final connection is made.
- The quality of such cover must be determined with due regard to how long it may be until final connection.
- Be responsible for and make good any damages caused directly or indirectly to any walls, floors, ceilings, woodwork, brickwork, finishes, services, roads, gardens, etc.

CONTRACTOR'S STAFF AT SITE

- The Sub-contractor will maintain at site, as necessary for the performance of the Contract, qualified personnel and supporting staff, with proven experience in erecting, testing, and adjusting projects of comparable nature and complexity.
- Before commencing work the contractor will submit details of the proposed Engineers and Supervisors, including copies of their certificates. If in the Consultant's opinion the proposed Engineers or Supervisors are not adequately qualified or are otherwise unacceptable, the onus is on the Contractor to submit alternates until such approval is given.
- Where the Contractor's or Sub-contractor's staff is, or becomes during the Contract deficient in the choice of adequate or acceptable personnel, the Contractor is to remedy the situation by

appropriate measures.

- Approval of the Contractor's or Sub-contractor's Engineers or staff shall in no way prevent the withdrawal of that approval at any time during the Contract should the Consultant so desire. In the event of such disapproval then the Contractor will be required to rectify the position as stated above with 14 days.
- In the event of any negligent or severely detrimental behaviour the Consultant has the right to order the removal from site of any Engineer, Supervisor, or worker on a "forthwith" basis.
- The Contractor will remedy the situation on a similar forthwith basis and in any event will not be allowed to proceed with any area of work, which as a result is left without proper supervision or technical backup until such time as the Consultant is again satisfied by the measures taken.

STANDARDS AND AUTHORITIES

- All equipment's and installations shall be in accordance with local codes and regulations. In general, all equipment and installation shall also be in accordance with either British Standard Specifications or a European or North American specification as detailed in the Contract documents.

WORKMANSHIP

- Install equipment, ductwork, conduit and piping in a workmanlike manner to present a neat appearance and to function properly to the satisfaction of the Consultant. Install ducts and pipes parallel and perpendicular to the building planes. Install all piping and ductwork concealed in chases, behind furring, or above ceiling, except in unfinished areas. Install all exposed systems neatly and group to present a neat appearance.
- Install all gauges, thermometers, etc., to permit easy observance.
- Install all equipment and apparatus, which requires maintenance, adjustment, or eventual replacement with due allowance for this.
- Install control valves to guarantee proper sensing. Shield elements from direct radiation and avoid placing them behind obstructions.
- Install all panels and boards, etc., to permit easy operation.
- Include in the work all requirements of Manufacturers as shown on their drawings.
- Replace all work unsatisfactory to the Consultant without extra cost and to the standard required by the Consultant. This applies to any item which is found to be defective in service during the maintenance period.

MATERIALS AND EQUIPMENT

- For all works, use only new material and equipment. For materials not specified, use high quality commercial items. If a Manufacturer is specified on a similar item, then the same Manufacturer may be proposed for the non-specified items. Obtain approval of all other Manufacturers from the Consultant.
- "New" is defined as newly manufactured, "state of the arts", tested and proven item of equipment. Items which have been held in stock for any extended period of time by either the Manufacturer or the Supplier will be rejected.

CLEANING

- Each day as the work proceeds and on completion, clean up and remove from the premises all rubbish, surplus material, equipment, machinery, tools, scaffolds, and other items used in the performance of the work. Clean out dirt and debris and leave the buildings broom clean with no stains and in a condition acceptable to the Consultant.
- Where electrical items form part of the visible finish in the rooms, protect from over-painting, etc. and give all items a final cleaning before handing over of the project.

ACCESSIBILITY

- Each item of equipment shall be located to be accessible for maintenance or repair without removing adjacent structures, equipment, piping, ducts, or other materials. For the large Air Handling units, the Contractor shall ensure that these can be built up on site from components which can be taken into the area.
- Any item of equipment needing maintenance shall be located to be accessible for maintenance or repair without removing adjacent structures, equipment, piping, ducts, or other materials.
- Cleanouts shall be located to permit rodding of all drain lines. These shall be located wherever possible external to occupied areas, and to minimize spillage problems during rodding.

INSPECTION, TESTING AND ADJUSTING

- All the works provided as part of this contract shall be inspected and commissioned in accordance with all relevant specifications, and codes of practice and to the entire satisfaction of the consultant.
- The contractor shall employ the services of a specialist company approved by the engineer regularly engaged in providing a testing balancing and commissioning service and who has been in continuous business for not less than 7 years employing fully trained staff having not less than 3 years dedicated experience.
- A senior experienced commissioning engineer with minimum dedicated experience of 7 years shall be responsible for supervising and directing the activities for the testing and commissioning team
- Carry out all tests specified. Carry out all tests required by authorities having jurisdiction. Test equipment to the requirements of, and where necessary, in the presence of the manufacturer.
- Provide all equipment’s, labour, instruments, loading devices, incidentals and pay for all fuel, power and sundries required to carry out the tests.
- All installations shall be inspected and tested in sections as the work proceeds and on completion as composite systems and it shall be noted that the consultant or any of the other relevant authorities may require to inspect or test any equipment during manufacture at the manufacturer’s works. All necessary arrangements shall be made as part of this contract.
- This will generally not apply to specified items unless specifically noted in these documents, but may be necessary for alternate equipment, should this be considered at all.
- All tests shall be arranged in cooperation with the consultant and his engineer as all other concerned parties and shall be subject to at least five (5) days’ notice in writing of the time, location and nature of the test to be performed. No test shall be considered valid unless the consultant is present.
- All necessary skilled and unskilled labour shall be provided for attendance during the tests (including pre- and post- test activities) and the test media shall be provided and subsequently disposed of except where specifically stated otherwise.
- The testing and adjusting is the contractual responsibility of the contractor but actual performance of the tests is expected to be the sole responsibility of the sub contractor.
- Have all testing and balancing performed by only persons who are thoroughly versed in this type of testing and balancing and with proven ability. Submit names, complete with experience records and references for approval of the consultant.
- Any defects occurring at any time during the test duration shall be made good and a complete re-test shall be carried out, all at no cost to the contract.
- Where failure occurs during a test, inspection or commissioning procedure which results in damage to the building fabric and/or any services not provided as part of this contract, or

requires subsequent builder's work to be carried out, carry out all such repair work to the entire satisfaction of the consultant at no cost to the owner or the consultant.

- All test points shall be provided which are necessary to carry out the specified tests and commissioning procedures including facilities for measuring or monitoring temperature,
- pressure, pressure drop, volume flow, in-duct sound, power sound pressure, humidity or other relevant conditions in both air side and water side systems. Such points shall be fitted with removable plugs, flanges or other appropriate and approved devices.
- Prepare test report forms for each test to be performed and submit these to the consultant at least two weeks prior to the commencement of any test.
- Only test after the system installation has been completed and the system has been put into continuous operation. Perform the testing, adjusting and balancing when outside conditions are commensurate with the design conditions for the given system. Add dummy loads to the system if outside and inside conditions are less severs than the specified points.
- Prepare a complete list of instruments for each test containing for each instrument:
- Name of instrument manufacturer.
- Scale and full-scale accuracy.
- Date of last calibration test.
- Name of last calibrating company.
- All instruments and consumables, such as recording paper, necessary for conducting the tests shall be provided, including but not limited to the following:
- Tube type velometer.
- Slopping tube manometer.
- Pitot tubes of various lengths.
- Mercury in glass thermometers.
- Weekly recording thermometers.
- Weekly recording relative humidity meter.
- Anemometer for diffusers with collector.
- Ammeter, voltmeter, wattmeter.
- Power factor meter.
- Insulation tester.
- Earth loop impedance tester.
- Tachometer.
- Duplicate signed test certificates shall be provided after each test, which will be countersigned by the consultant who witnesses the test. The test certificate shall give the following particulars:
- Apparatus or section under test.
- Maker's number (if any).
- Nature, duration, and conditions of test.
- Result of test.
- No test shall be valid until the certificate is provided.
- Duplicate copies of test certificate carried out at manufacturer's works shall be forwarded to the

consultant for approval prior to dispatch of the article to site.

- No section of the works shall be insulated or in any other way concealed prior to testing and inspection and subsequent concealment where applicable shall only take place following written authority from the consultant.
- All necessary facilities, measuring and recording instruments including test pumps and gauges for inspection, testing and commissioning requirements shall be provided and shall be checked or calibrated as necessary before use.
- The Consultant reserves the right to call for a demonstration of the accuracy of any instruments provided.
- All representatives present during inspection, testing and commissioning shall be fully conversant with the system concerned and the method of system and instrument operation.
- Manufacturer’s or Specialist Sub-contractor’s representatives shall attend where specifically indicated elsewhere in the specification or where necessary to ensure full service and cooperation is available to the Consultant to enable the works to be tested and commissioned in accordance with the requirements of the specification.
- All necessary precautions shall be taken to safeguard against damage during inspection, testing or commissioning. Any damage so caused shall be made good at no cost to the contract.
- All tests shall last for the minimum time period stated or longer if necessary to ensure all sections have been fully examined as required by the test.
- All performance tests shall be carried out initially prior to the Consultant being requested to witness that tests and thereby avoid unnecessary re-tests being required.
- To enable testing and commissioning to be carried out the Consultant will, on request, provide the necessary information concerning design intent and all certificates shall indicate any variation in terms of the actual and percentage deviation from the design value.
- All items with electrical supplies shall be certified ready for use by the Electrical Sub- contractor and shall fulfill DEWA Requirements.
- Submit all test reports in triplicate to the Consultant for his review and records. Perform all tests called for in other sections of this Division. Unless specified otherwise test, adjust and balance the following systems:
- Electrical switchgear and control gear. All low voltage systems.
- All extra low voltage systems. Pumps.
- Air Handling Units.
- Fan Coils.
- Piping Systems.
- Tanks and Receivers.
- Air Supply Systems.
- Motors and Starters.

PUMPS

- Electrical inputs.
- Motor speed.
- Inlet and outlet pressures. Inlet and outlet temperatures. Flow rate.
- Noise levels.

- Automatic changeover facilities function test.

FANS AND MOTORS

- Fan rpm.
- Motor rpm.
- Motor voltage to all phases.
- Motor current to all phases. Static pressure across fan. Air flow through fan.
- Air temperature at fan suction and discharge. Verify fan rotation.
- Noise measurements for 6 central frequencies for the full range of speed, flow, pressure generated.
- Power input at maximum condition.

FAN TEST CERTIFICATES

- Duplicate type test certificates shall be provided for fans, indicating performance curves, sound levels, etc., when tested in accordance with B.S. 848.

FOR CENTRAL SYSTEM COMPONENTS

- Static pressure across filters. Static pressure across coils. Air flow at each main duct.
- Air temperature before and after each foil. (both DB and WB).
- Fresh air and return air quantities.
- Noise measurements at 6 central frequencies.

AIR DISTRIBUTION SYSTEMS

- For HVAC systems adjust and balance all system components ensuring that the test data are within three percent of design requirements for the overall system and within five percent for each component. Repeat balancing and adjusting until the requirements are met. Submit a test report listing the measured data versus design data for the following sections:
- Air Leakage.
- Air Distribution. Grilles and Diffusers.

AIR LEAKAGE TEST

- Air leakage tests shall be conducted on all ventilating systems before ductwork is enclosed. The tests shall be sustained for 30 minutes at the design pressure plus 50% during which time the leakage, if any, must not increase. The air shall be released to zero and the section of ductwork immediately retested at the designed pressure for that section without any increase in leakage. Blanks shall be inserted at each flange as the sections are tested, and these shall be left in position until the installation is complete when they shall be removed.
- The blanks at the flange joints shall be removed and the flanges resealed, the main fans shall be operated at the designed volume and pressure. Under this condition each flange joint shall be inspected with a soap and water test.
- The total leakage of air for the complete system when measured at the fan discharge shall not exceed 3% of the total air volume, and the aggregate of the progressive section tests shall not exceed the permitted total leakage.

AIR DISTRIBUTION TESTS

- Smoke tests shall be conducted on all ventilating systems to ensure correct air diffusion from outlet terminals.

GRILLES AND DIFFUSERS

- Air flow rate.
- Manufacturer’s k factors.
- Noise measurements.

ELECTRIC MOTOR TEST CERTIFICATES

- Duplicate type test and abbreviated tests shall be provided for electric motors in accordance with BS.4999 as appropriate. For motors in excess or 40 KW routine individual test certificates shall be provided in accordance with BS.4999.

STARTER AND CONTROL GEAR CERTIFICATES

- Duplicate type test certificates shall be provided for all starters and control gear in accordance with BS.587, and for control panels as a whole, routine individual high voltage tests shall be carried out as allowed in BS 587.

TEMPORARY USAGE

- The Consultant reserves the right to use any piece of mechanical equipment, device, or material installed under this section for such reasonable lengths of time as may by required to make a complete and thorough test of same before the final completion and acceptance of the work.
- Such tests shall not be construed as acceptance of any part of the work, and it is agreed and understood that no claim for damage will be made for any injury or breakage to any part or parts of the above due to these tests, where caused by weakness, inaccuracy of parts, or by defective materials or workmanship.

INSPECTION OF THE WORK

- The representatives of the owner and the consultant will make periodic visits to the site during construction to ascertain that the work is being executed in reasonable conformity with all plans and specifications but will not always execute quality control. Each subcontractor must maintain the quality control as intended in the contract documents.
- Correct all deficiencies immediately as noted during field inspections.
- Request in writing that a final inspection of all services. Do not issue this written request until:
  - All deficiencies noted during job inspections have been corrected.
  - All systems have been balanced and tested and are ready for operation.
  - All balancing reports have been submitted and reviewed.
  - All instruction manuals have been submitted and reviewed.
  - The cleaning up is finished in all respects.
  - All spare parts and replacement parts specified have been provided and receipt of thesame acknowledge.

COMMISSIONING OF SERVICES

- All piping systems shall be flushed, chemically cleaned, and then filled with treated water, or appropriate fluid, vented as necessary, and brought to operating conditions and the flows then regulated to the design values.
- All water circuits shall be balanced by means of the regulating valves provided and flow rates shall be determined on a temperature and pressure drop basis. Flow through pumps shall be measured by relating the pressure drop across the pump to manufacturers test curves. A copy of the test curve indicating the final operating point shall be forwarded to the consultant.
- All refrigerant systems shall be vacuumed, purged and charged according to A.S.H.R.A.E. standard methods. All safety controls and interlocks will be tested as part of the commissioning procedures.
- Flow through coils shall be adjusted to give the design temperature difference. In general water systems shall be balanced in accordance with the guidelines specified in NEBB manual. Upon

completion of balancing and testing operations for heating and cooling systems, temperature measurements shall be taken in all rooms and the readings tabulated in schedule form together with hourly ambient temperature readings taken over the measuring period. Two copies of the schedule shall be forwarded to the consultant.

- All ventilating systems shall be commissioned in accordance with the procedures recommended in the AABC manual. Duplicate schedules of commissioning results shall be forwarded to the consultant and shall detail the recorded air volume and percentage deviation from design air volume, for each air input and extract terminal. Wet and dry bulb temperature measurements shall be taken in all rooms served by air supply systems and the results indicated on the schedule of commissioning results.
- This all shall be recorded together with external ambient wet and dry bulb temperatures measured at hourly intervals over the measuring period.
- Sound pressure levels (dBA) shall be measured in all rooms containing supply or extract terminals, all plant rooms, all rooms immediately adjacent to plant rooms and all room located above or below plant rooms. The consultant may at his discretion request a spectrum band analysis of sound pressure in any locations.
- All automatic controls shall be commissioned by the control suppliers.
- Fault conditions shall be simulated and all alarms and safety devices shall function correctly. Such proving tests shall be carried out in the presence of the consultant and certificates shall be provided specifically detailing all check procedures, which have been carried out.
- Following proving test all installations shall be left in working order ready for handover.
- All electrical systems shall be tested and commissioned in accordance with IEE regulations and local authority codes.

CORRECTION AFTER COMPLETION

- Remedy all work in accordance with the General Conditions of Contract for a period of one year from the date of acceptance of the completed work.
- Attend immediately to any and all the defects occurring during the period defined above and repair in a manner to prevent recurrence. This Contractor is responsible for all works required by other trades necessary to repair the works of this section, or necessary to repair damage caused by the failure of any part of this section.
- Instruct all suppliers and manufacturers that guarantees on equipment will commence when the completed work is accepted and not from the date the equipment is put into operation. In the event that this condition is omitted by the supplier, or if subsequent cost to the Owner is involved, this Contractor shall be liable to the full extent of the guarantee as indicated in the specification.

WORK COVERED IN OTHER DIVISIONS

- The Services Contractor is to be aware of the overall scope of the job and must coordinate with other trades and other Contractors on site as necessary to ensure a complete mechanical installation. The contractual obligation on the General Contractor to coordinate does not relieve the Sub-contractors of the obligation to also coordinate and be aware.

PAINTING AND IDENTIFICATION

- Painting shall be provided for un-insulated ducts, pipes, supports, metal bases, etc.. Also any equipment with damaged paint shall be repainted to original paint standard and colour.
- All surfaces to be painted shall be prepared by thoroughly cleaning and removing all rust, grease, oil, dirt and surface corrosion, using wire brush, emery paper and/or degreasing medium as required. The paint shall be applied in accordance with the maker’s instructions and the type of paint to be used shall be in accordance with the following:

- Ferrous surface, one coat of Calcium Plumbate metal primer plus wash primers as necessary, followed by 2 undercoats and one finishing coat.
- Non-Ferrous surface, one coat of zinc chromate primer plus wash primers as necessary, followed by 2 undercoats, and one finishing coat.

- After finished painting is completed, identify each piped and ducted service. Locate identification and flow arrows:
- Behind each access door.
- At each change of direction on all joining pipes and ducts.
- At not more than 10 meters apart in straight runs of exposed pipes and ducts, but on both sides of sleeves.
- At not more than 10 meters apart in straight runs of pipes and ducts behind removable enclosures such as lay-in ceiling but on both sides of sleeves.
- Above each floor or platform for vertical exposed pipes, preferably 1.5 meters above floor or platform level.
- p.v.c. tape identifying bands will be accepted.
- Alternately use stencils and stencil paint onto all piping and ductwork. Use letters a minimum of 30mm high. After completion of the works provide to Owner usable stencils for each service. Use wording shown in the description section of the legend on the Mechanical Drawings, or as instructed in writing by the Consultant.
- Wherever insulation is to be painted, the paint used shall comply with all the fire resistance requirements for insulation finish, and shall be carried out by the insulation Sub-contractor.
- In all cases the actual grade of paint to be used shall be suitable for the operating surface temperature and shall be approved by the maker for the application concerned. In certain cases the grade of finishing coat may not required the application undercoats, in which case these may be omitted, provided that the Consultant's approval in writing is obtained beforehand.
- All insulated or un-insulated pipework in concealed positions shall be identified by means of 75mm wide identification bands, painted neatly on, at right angles to the pipe axis at intervals not greater than 3 meters.
- In addition the name of the service and pipe diameter shall be stenciled on in a visible position with an arrow indicating the direction of flow. Flow and returns shall have the letter "F" or "R" added to the identifying name.
- The identifying band colours and the finishing colour of the services to be painted shall be agreed with the Consultant prior to application, but for tendering purposes shall be in accordance with the colours and procedures given in B.S. 1710.
Conditioned Air Supply	Blue
Conditioned Air Return	Purple
Fresh Air	Green
Toilet Exhaust	Brown
Kitchen Exhaust, concealed	Red
Chilled Water Supply	Blue
Chilled Water Return	Purple
Condensate	White
- Ductwork shall not be painted but shall be prepared for painting by others where installed in

exposed positions as defined in the General Technical Clauses.

- Ductwork shall be identified in accordance with the procedures laid down in H.V.A.C. Code of Practice No. DW 141.
- All equipment located in concealed positions shall have a nameplate secured to the item giving the following information:
- Equipment reference number (as indicated on the record drawings). System.
- Room / Area served. Duty / Output information.
- The name plate shall be 100 x 100 mm approximately, of white plastic 3mm thick with the above information engraved in black lettering and the plate shall be secured by screws, bolts, clips, etc., as appropriate to the item concerned.
- This plate is in addition to any name plate supplied by the manufacturer of the item giving detailed specification information for the equipment.

INSERTS, SLEEVES, ESCUTCHEONS AND CURBS

- Use only factory made, threaded or toggle type inserts as required for supports and anchors, properly sized for the load to be carried. Place inserts only in portions of the main structure and not in any finishing material.
- Use factory made expansion shields where inserts cannot be placed, but only where approved by the Consultant and for light weights.
- Do not use power activated tools except with the written permission of the Consultant.
- Supply and locate all inserts, holes, anchor bolts, and sleeves in good time when walls, floors, and roof are erected.
- Pass insulation unbroken where pipe or duct is insulated. Size sleeves to provide adequate clearance all around.
- Provide UPVC pipe sleeves for all pipes crossing floors or walls.
- Pack all sleeves between the insulated pipe and the sleeve or where un-insulated between the pipe and the sleeve with loose fiberglass insulation. Seal the annular space as follows:
- For all horizontal sleeves in exposed areas, use a seal equal or better fire rated than the wall to be sealed.
- For horizontal concealed sleeves through fire walls and through walls separating areas of different air pressure, use a permanently resilient silicone based sealing compound.
- For all vertical sleeves through roofs, janitor’s closets, equipment rooms, use permanently resilient silicone based sealing compound, non-inflammable and waterproof.
- Ensure that the seal is compatible with floor and ceiling finishes. Check the room finishing schedules for further details and clarify if necessary with the Consultant.
- Use the following sleeving for ducts:
- The minimum thickness of duct material passing through a sleeve shall be 1.3mm (18 Gauge).
- For rectangular duct openings through walls and floors a removable hardwood box-out shall be provided of the required size. Softwood or plywood will not be acceptable.
- Through fire walls, build fire dampers into wall, or make detailed fixing in accordance with Consultant’s instruction.
- Through floors where ducts are not furred in or enclosed in a duct shaft, provide 100 mm high and 100 mm wide watertight concrete curbs, with 25 mm chamfered edges all around. Extend

sleeves where used flush to top of curb. Read Concrete Specification before proceeding. Through floors where duct is enclosed in a duct shaft or furred in, provide the watertight curbs at the extreme top and bottom only.

- Cover exposed floor and wall pipe sleeves in finished areas with satin finish chrome or nickel plated solid brass or with satin finished stainless steel escutcheons with non- ferrous set screws. Split cast plates of the screw locking type may not be used. Do not use stamped steel friction type split plates.
- Through roofs, provide curbs and sleeves as shown on drawings and to suit flashing requirements.
- After ducts are installed, pack the opening and seal the packing as follows:
- Use fiberglass insulation for packing except through curbed concrete floors where asbestos braid must be used.
- Seal the asbestos braid packing in openings through floors with permanently resilient silicone base non inflammable waterproof compound. Press duct supports firmly down into caulking before bolting it down to curb.
- Through all vertical walls seal the fiberglass packing using permanently resilient silicone based sealant.
- Brace duct sleeves and box-outs to retain their position and shape during the pouring of concrete and other work.
- Provide bracing for each duct at every passage through structure to prevent sagging.
- Cover exposed duct sleeves and openings in exposed areas. Use 100 mm galvanized steel escutcheons in the form of a duct collar. Over curbs extend the collar 30 mm down the side of the curb, similar to counter flashing fix collar in place with Cadmium plated screws.

ACCESS PANELS AND DOORS

- Install all concealed mechanical equipment requiring adjustment or maintenance in locations easily accessible through access panels of doors. Install systems and components to result in a minimum number of access panels. Indicate access panels on as-built drawings.
- Provide the respective Division of work with panels, doors or frames, complete with all pertinent information for installation.
- Ensure that access doors are installed in a manner to match the building grids where applicable.
- Prepare detail drawings showing location and type of all access doors in coordination with other trades before proceeding with installation and hand these to the Contractor to obtain approval.
- Size all access doors to provide adequate access and commensurate with the type of structure and architectural finish. Should it be necessary for persons to enter, provide a minimum opening of 600 x 600 mm.
- Ensure proper fire rating of access doors in fire separations.
- Lay-in type ceiling tiles, if properly marked may serve as access panels.
- Provide panels in glazed tile walls of 2.6 mm thick 304 alloy stainless steel, no 4 finish, with recessed frame secured with stainless steel, countersunk, flush-headed screws.
- Provide panels in plaster surfaces with dish shaped door and welded metal lath, ready to take plaster. Provide a plastic grommet for door key access.
- Provide other access doors of welded 2.6 mm thick steel, flush type with concealed hinges, lock and anchor straps, complete with factory prime coat. Submit details to Consultant for approval.

GUARANTEES

- The Contractor will guarantee all material and workmanship for at least one year after preliminary take over by the Owner.
- All warrantees from equipment suppliers will be vested in the Owner, regardless of whether the Contractor who supplied the equipment is still associated with the project or not.
- Guarantees will be full guarantees and will include all overhead, profit, incidental charges and sundries.
- Where damage is caused to any other item by any failure of the item guaranteed, then the guarantee shall also include the costs incurred in rectifying that damage.

MAINTENANCE

- Maintenance is defined as the Contractual Liability to maintain the equipment in working condition, the regular checks and servicing of equipment during maintenance period to keep the equipment in best working order. This maintenance period shall be as defined in conditions of Contract.
- Regular maintenance shall be as necessary, but in any event not less frequently than monthly.
- In the event that the owner has his own staff, the Contractor is still to check monthly and advise on any problems and is still to assume responsibility.
- At the time of Preliminary Handing Over, the Contractor shall run training sessions for Client's operating staff to explain the functions of all systems and equipment and the operating procedures.
- The preventive regular maintenance and faults maintenance shall be entirely Contractor's responsibility during the maintenance period.
- Contractor's maintenance team shall be deployed to carryout weekly checks of all equipment and controls, monthly cleaning of all filters and necessary greasing and lubrication. A visit schedule shall be submitted and agreed with Engineer before Preliminary Handing Over.
- All fault and failure complaints shall be attended by Contractor as soon as possible, but, repairs must be carried out within maximum of 12 hours after the reported failure.
- Contractor shall repair/replace all equipment, controls and systems, provide new coat of paint for equipment and piping at the end of maintenance period and offer entire MEP installation for checking and approval of Engineer and Client before issue of Final Handing Over Certificate.

GREEN BUILDING REQUIREMENTS:

- Contractor is responsible to ensure and implement all the green building specification related to MEP works as per the Green Building specification Volume which supersedes the mechanical specifications if any contradiction.
- Contractor should review the complete design documents and give his comments to engineer prior quoting, if not comments, Contractor will be held responsible for any rectification required and to run the system successfully without any extra charge.
- All Materials should be supplied from the main factories in the country of origin of the Manufacture, any deviation from this, like supplying materials assembled in another different country under a license or another name is not accepted, unless Client approved the same and Contractor submitted the related cost saving and the sole opinion of the Engineer.
- Contractor shall perform the testing and commissioning of HVAC system by a third party approved by the Engineer.
- Any additional items required by local authorities up to the tender date / Final Quote, or necessary to operate all systems, but not shown in the tender drawings should be added by Contractor without extra charge.

- Contractor should stick to manufacture list of the engineer strictly, even if the same is contradicting with the Green Building proposed supplier, subject to case by case and to the sole opinion of the Engineer.

END OF SECTION

SECTION 2

VENTILATION AND AIR CONDITIONING EQUIPMENT

SECTION 2 - VENTILATION AND AIR CONDITIONING EQUIPMENT

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SCOPE OF WORKS

- The works shall include supply, installation, testing, commissioning and guarantee of the air conditioning and ventilating systems as schematically shown in the tender and contract drawings and as referenced in the specifications. The works will also include calculations as modifications to suit actual site conditions.
- The contractor will also include all overheads at office and site, labor, sundries, plant, appliances and consumables both for the works and for testing, commissioning and during the guarantee period.
- The tender drawings are intended to indicate the scope of works but the contractor is to confirm exactly all equipment including additional safety and performance factors as noted.
- By virtue of carrying out a complete check of all design information the contractor is to assume full responsibility for the correct functioning of the system and to carry any liability or other insurance or guarantee as may be necessary to protect all parties in this regard.
- The Contractor shall supply, install, test and commission the complete mechanical services
- installation as described within the complete specification documentation and as indicated on the tender design drawings and equipment schedules.
- B. The engineering services included in the works shall include but not be limited to the supply, delivery, positioning, installation, setting to work, testing and commissioning of the following systems:

- Air conditioning
- Ventilation and exhaust
- Car park smoke ventilation and extract
- Basement smoke extract system
- Corridor smoke management systems
- Stair & lift pressurization
- Thermal insulation
- Water treatment
- Acoustic and vibration control
- Building management system
- Testing and commissioning

DESIGN AMBIENT CONDITIONS

Maximum outside dry bulb temperature	46 Deg. C
C Minimum outside dry bulb temperature	6 Deg.
C Coincident outside wet bulb	6 Deg.
C Wind speed, for A.C. load calculations	15 KPH
Haze factor	0%
Coincident outside wet bulb	29.5 Deg.

- Solar loads to be based on maximum instantaneous values in appropriate direction for consideration of the worst exposure calculations, and to be based on worst cumulative block load when checked on an hourly basis from 10 am to 4 pm over the period July to September, coincident with the maximum outside ambient as listed above.

CODES, STANDARDS & REGULATIONS

- A. The design, specification and installation of the Mechanical services will be in accordance with
- good engineering practice and will conform to, but not limited to, the following Standards, Codes
- of Practice and Regulations applicable to Dubai:
- British Standards.
- National Fire Protection Association (NFPA) Standard Regulations.
- CIBSE Guidelines.
- ASHRAE Guidelines.
- BSRIA Guidelines.
- Heating and Ventilating Contractors Association DW 144 & DW 172.
- Dubai Civil Defence (DCD) Guidelines & Regulations.
- UAE Fire & Life Safety Code of Practice.
- Dubai Municipality Green Building Regulations & Specifications.
- Dubai Municipality Regulations.
- Telcom Regulations

DESIGN INTERNAL CONDITIONS

Summer dry bulb	23.0 deg + 1ºC
Relative humidity	50 % + 5 %
Noise Level	
Bedroom	NC 25-30
Lobbies, corridors	NC 35-40
Living, dining room	NC 30-35
Prayer hall	NC 35 -40

DESIGN CRITERIA

Duct Velocity	Units
High velocity	12 M/Sec.
Low velocity, supply	6 M/Sec.
Low velocity, return	5 M/Sec.
Terminal branches	3.5 M/Sec.
Grille face (average)	2.5 M/Sec.
Kitchen exhaust	6M/Sec.

DESIGN CRITERIA AND STANDARDS

GENERAL

- The design of the Engineering services is based on the parameters and design data stated in the following clauses and these shall not be changed or amended without prior written notice from the Engineer.

AMBIENT DESIGN CONDITIONS

- For the design of all thermal loads and selection of equipment the following ambient conditions shall apply:

PERIOD	DBT / WBT
Summer	46°C / 29°C
Winter	12 °C / n/a

INTERNAL DESIGN CONDITIONS

- The following schedule states the average internal summer and winter design conditions for the spaces indicated based upon the design ambient temperature stated elsewhere.
- The average space temperature in mid-season will vary between the two temperature limits.

ROOM NAME	ENVIRONMENTAL DESIGN CONDITION
	SUMMER
	TEMPERATURE
Apartment Living Areas	24 °C ± 2°C
Kitchen	24 °C ± 2°C
Bedrooms	24 °C ± 2°C
Bathrooms	26 °C ± 2°C
Corridors	24 °C ± 2°C
Lift Lobbies	24 °C ± 2°C
Main Entry Lobbies	24 °C ± 2°C
Public Toilets	26 °C ± 2°C
Plant rooms	28 °C
Electrical rooms with risers	23 °C ± 2°C
Main Telephone room	22 °C ± 2°C
Telephone rooms with risers	22 °C ± 2°C
GSM rooms	22 °C ± 2°C
LV rooms	23 °C ± 2°C
Retail	24 °C ± 2°C
F&B Dining area	24°C ± 2°C
F&B Kitchen	28°C
Car parking	u/c
Transformer Room	u/c
Generator Room	u/c
RMU Room	u/c

u/c - uncontrolled
* No direct cooling will be provided. Area will be cooled by the infiltrated air from
adjoining cooled space as part of exhaust ventilation.

- No winter heating is provided / considered.
- There shall be no direct humidity control within the building; however, the design is based upon maintaining the humidity within the above limits. All outdoor air will be de-humidified prior to being supplied to each space and subsequently cooled to the desired space conditions using individual fan coil units.

AIR INFILTRATION

- Generally the building shall be pressurized with respect to outside by approximately 10% to assist with limiting infiltration of unconditioned ambient air, such that the maximum air infiltration to the building shall not exceed 10m3/h·m2 of façade at an applied differential pressure of 50Pa in accordance with Dubai Municipality Green Building Regulations & Specifications.

- An infiltration rate of 0.35 ACH was used to calculate the building cooling loads for all spaces with external facades.

VENTILATION RATES

- Ventilation rates to all spaces shall be in accordance with ASHRAE 62.1 standard requirements.

BUILDING FABRICS

- Calculations of thermal loads have been based upon the following thermal transmittance values for wall construction, glazing and roof construction, as well as shading coefficient for glazing.

BUILDING ELEMENT U-VALUE	W/(M2·K)
External Wall	0.35
Roof	0.25
Glazing (including frame)	2.1
Shading coefficient	0.25

NOISE CRITERIA

- The building services shall be selected and installed such that the noise ratings identified in the acoustic division of specification are not exceeded in the occupied space unless stated differently elsewhere. The contractor is to take responsibility for ensuring the noise criteria
- are achieved in all areas by calculation prior to the production of shop drawings and ensuring all installations comply with standard details and industry best practice. Claims resulting from poor acoustic performance due to low quality installation will not be entertained.
- Noise levels generated by the building services to the external environment are to be in accordance with the requirements of the Local Building Municipality or as otherwise specified.
- Internal and external acoustic surveys and reports shall be provided to confirm compliance with the criteria set out in this specification.
- Attenuators shall be provided on all HVAC systems in order to achieve the space NC ratings set out in the acoustic specification. Based on approved equipment selections and
- duct layouts, the Contractor shall with the aid of a specialist select, supply and install attenuators (and / or other necessary acoustic measures) in all HVAC systems as necessary to meet specified noise ratings. Duct lining shall not be used in place of attenuators unless specifically approved by the Engineer.
- Anti-vibration devices, inertia bases etc. shall also be provided as applicable to all plant in order to achieve specified noise ratings or limit break out of noise or vibration to surrounding areas. If specified noise ratings are exceeded in plant rooms the Contractor shall at his own cost provide necessary noise break out rectifications to ensure adjacent room noise levels are achieved as specified, subject to approval of the Engineer. All plant and equipment shall be submitted with relevant noise data and resultant room level noise calculations shall be submitted to the Engineer for approval. The approval of all plant and equipment shall be subject to the above specified room noise levels being achieved. The Contractor shall take remedial action at his own time and cost where space noise levels exceed those specified.

CONNECTIONS

- Only one liquid line, one suction line and one power supply connection shall be required for each unit.

DUCT SIZING

- Calculation and selection of duct sizes was based on current CIBSE calculation methods and the following maximum velocity criteria (subject to noise criteria stated elsewhere).

Duct Type	Maximum air Velocity m/sec		
Area	Critical	Normal	Uncritical
	(Bedrooms etc.)		(Plant rooms, parking areas,)
Riser	5	7.5	10
Above Ceiling 3 5 6	3	5	6
Opening Supply 1.5 2.5 3	1.5	2.5	3
Opening Extract 2 3 4	2	3	4

- Ductwork classification shall be as defined in HVCA DW/144.
- The length of flexible ductwork shall be limited to the minimum possible and shall not exceed 1.5m. Flexible ducting shall be sized to a pressure of not more than twice the equivalent sheet metal ducting.

PIPE SIZING

- Calculation and selection of pipe sizes shall be based on current CIBSE calculation methods.
- Chilled water pipe sizes shall be based on criteria set out below:

Pipe diameter mm	Maximum velocity-m/s	Maximum pressure drop-Pa/m
$ø \leq 50$	1	300
$65 \geq ø \leq 150$	2	300
$ø \geq 200$	3	300

- The pipe sizing shall also be subject to noise criteria stated elsewhere.

PRESSURE RATING OF EQUIPMENT AND MATERIALS

- The Contractor shall ensure that all equipment and materials provided have suitable pressure rating to operate within the installed systems and the systems specification.
- The table below outlines the required pressure rating of HVAC equipment. These pressure ratings overwrite the ratings included within standard sections of the specification. The pressure ratings quoted are subject to final system configuration on site and Engineer's approval.

EQUIPMENT	PRESSURE RATING
Fan coil units	PN 10
Air handling units	PN 10
Plate heat exchangers	PN 10
Chilled water pipework, valves and accessories	PN 10
Chilled water pumps	PN 10

REFRIGERANTS

- It is a requirement of DM Green Building Regulations & Specifications that the refrigerants used on the project should have zero ozone depletion potential (ODP) or global warming potential (GWP) less than 100.
- All equipment running on refrigerants, such as DX split units, shall comply with the requirement.

SPLIT UNITS (DUCTED TYPE)

- Furnish and install air-cooled condensing units in the locations and manner shown on the plans. The units shall be properly assembled and tested at the factory.
- Nominal unit electrical characteristics shall be 380 volts, 3 phases 50 Hertz as per DEWA approval.
- Condenser coil shall be of nonferrous construction. Coil shall have aluminum plate fins, mechanically bonded to seamless copper tubes. Coil shall be circuited for sub cooling. Coils shall have heresite or equal approved anti-corrosion coating.
- The outdoor unit casing shall be weatherproof with zinc coated galvanized steel.
- The cabinet shall be exterior enamel paint finish and weather resistant. Outdoor unit cabinet finish shall have been tested in accordance with ASTM B-117 salt spray test procedure for a minimum of 1000 hours .
- For smaller capacities 4 ~ 10 HP, Outdoor unit shall be equipped with heat sink to cool the PCB. For 8~66 HP, the cooling of the inverter PCB shall be conducted by heat sink (high pressure liquid refrigerant via heat exchanger attached to rear side of inverter PCB).
- For 8 ~ 66 HP (Top discharge), the system shall be equipped with multi sensors (temperature & humidity) at the outdoor units, sensing the outdoor air temperature and relative humidity. It shall automatically adjust the refrigerant's target evaporating pressure / temperature by compressor logic control in order to precisely match the building sensible and latent load as per the variation in ambient temperature and humidity
- The Indoor Outdoor unit combination on the system shall be between 70~130%. Also the actual capacity of the Outdoor unit shall be equal to or greater than the sum of specified room loads.
- The sound pressure level shall not be more than 69.3 dB(A) at normal operation measured horizontally 1m away & 1m above the ground. Further the outdoor unit shall have night silent operation to reduce the noise level by 4 db(A).
- For capacities between 8~66 HP(Top discharge units), The Outdoor unit shall be able to perform the Auto Dust removal function, during which, the condenser fan will rotate in reverse direction to blow away dust and sand accumulated over the condenser coil

CONDENSER FANS AND MOTORS.

- Unit shall be furnished with direct driven, propeller type fans arranged for vertical discharge for units at roof, and horizontal discharge for units at ceiling of ground floor. Condenser fan motors shall have inherent protection, and shall be of the permanently lubricated type, resiliently mounted. Each fan shall have a safety guard. Controls shall be included for cycling fan(s) for intermediates operation.
- The condenser coil fins shall be corrosion resistant for coastal application(Black fin coating with minimum salt spray test of 2000 hours. The test shall be Carried out by reputed third party test laboratory like UL as per testing standards ISO 21207 Method B (Salt contaminated condition + severe Industrial or traffic Environment)

COMPRESSOR(S)

- For capacities between 4~6 HP, the compressor shall be single DC Inverter Rotary compressor.
- For capacities between 8~66 HP, The outdoor unit shall have one or more DC Inverter control scroll compressor capable of modulating frequency as per load requirements.
- For capacities between 8~66HP, The compressors shall be high pressure side shell (HSS) to

improve the COP by reducing power input with the help of better oil circulation. The old technology LSS compressor shall not be acceptable.

- The Inverter frequency range shall be between 10~165 Hz. The compressor shall be equipped with 6 internal pressure relief valves to safeguard the compressor during low loads. All the control options shall be factory installed and tested for reliability.
- Compressor shall be covered in an acoustic lining jacket to maintain low noise level as per manufacturer standards. Each module of outdoor unit shall have reverse phase and phase failure protection.
- The compressor shall be equipped with special bearing to operate longer time with less lubrication. This helps in enhancing the durability and reliability of compressor.
- The compressor motor shall be class B insulation.

INDOOR UNITS

- The capacity of air handling units shall be not less than that scheduled on drawings. The units shall be draw-through single zone units with return air plenum.
- The space available for units shall be confirmed with the Civil and Architectural proposals of the main contract and sizes of units will necessarily fit into the spaces available. Where necessary the units may be built up on site, subject to acceptance of the finished units for warranty purposes by the original supplier and his local agent.
- The unit casing and panels shall be constructed from heavy gauge galvanized steel with backed enamel paint. The casing and panel shall be insulated with 1/2" (13mm) thick fiberglass having density of 2.0/cubic ft. (32.0 kg/m3) and thermal conductivity of 0.23 BTU.in/ft² ºfh) 0.033 w/m²k).
- The insulation shall meet latest NFPA fire requirements. The insulation shall be extended all around duct connection collar.
- Drain pans shall be of heavy gauge welded and then galvanized steel construction and shall have drain connections on both sides. Drain pans shall be internally lined with 25 mm of closed cell foamed in place insulation, which shall extend up to the top edges, and 25 mm of similar insulation on the outside. In the event that any condensation problems appear after installation, the contractor shall apply greater thickness of insulation to the underside and edges. This is an addition to any other remedial measures required by the consultant. Any stacked or double height coils shall have separate drain pans to reduce carry over, such pans will be piped to the main drain, not into the lower pan.
- Fans and shafts shall operate well below their critical speeds. Each shaft assembly shall be statically and dynamically balanced before shipment from the manufacturer. If any out-of-balance is apparent on site the consultant reserves the right to require new factory-supplied units to be provided until satisfactory balance is demonstrated.
- Fan wheel shall be centrifugal double inlet forward curved belt driven type. Statically and dynamically balanced, complete with shaft and self-aligning.
- Fan bearings shall be designed for a life of at least 200,000 hours; factory pre- lubricated and equipped with standard grease fittings with extended tube lines if necessary to terminate outside the unit. Fan drives shall be rated at 150% of the maximum motor horse power of the units and shall be equipped with adjustment for belt tension.
- Cooling coils shall be fabricated from nominal 3/8” (10mm) size copper tubing expanded into fins to give a mechanical bond. Return bends shall be die formed. Headers shall be heavy section seamless copper tubing. All joints shall be silver brazed. Fitting shall include plugged vent and drain taps for each section. The fins shall be aluminium.
- The coil shall be designed to operate at 200 psi and 300 degrees Fahrenheit (149 degrees centigrade). The performance shall be not less than that listed in the schedules.
- Coils shall have a minimum of 4 rows for re-circulation service and 8 rows for fresh air service.

- System will be constant speed, but with adjustable drive pulleys to give plus or minus 10% from the rated design speed.
- Unit operation shall be controlled by wall mounted digital indoor thermostats having ON/OFF control, temperature set point and space temperature indication.
- Colours and finishes of the thermostat shall be to approval of the architect's.
- All AHU's shall have flat panel type filter of 1" thickness in standard modular sections with aluminium filter media. (Through away filter to be used only for commissioning of system).
- Initial pressure drop to be 60 Pascal at 2.5 M/sec. Final pressure drop 180 Pascal with dust holding capacity of 800 gm/M2. ASHRAE gravimetric efficiency 84%..
- The outdoor unit shall be equipped with BLDC inverter motor, capable of modulating frequency / speed as per the ambient temperature and discharge pressure / temperature.

CONTROL SWITCH

- Each unit shall have a wall mounted electronic control switch with a system on-off switch, room temperature control thermostat and 3-speed fan control.

REFRIGERANT PIPE WORK

- Refrigerant pipe wok shall be hard copper tubing to ASTM B88 type L.
- In case of multiple compressors each compressor shall have non return valve to Isolate the compressor and to avoid transfer of refrigerant and carbon entering in another compressor in case of motor burn out.
- The main EEV (Electronic Expansion Valve) in the outdoor shall be controlled by main Control PCB located inside the outdoor unit. Every outdoor shall have sub-cooler to increase the sub-cooling temperature and evaporator cooling capacity.
- Each outdoor unit shall have Accumulator to control the volume of refrigerant entering in the compressor to secure the efficiency.
- Automatic refrigerant charging function shall be applied to the system; it will operate when the system is encountered with problems such as refrigerant leakage or parts failure etc. Automatic refrigerant charging function will operate with DIP switch setting and will stop system operation when optimal refrigerant amount is charged to the system to prevent additional refrigerant flowing into the system.
- **Recommended copper pipes Material:**
- Seamless phosphorous deoxidized copper pipe Wall Thickness: Comply with the relevant local and national regulations for the designed pressure 3.8MPa (551psi). We recommend the following table as the minimum wall thickness.
- **<u>Recommended minimum Wall thickness</u>**

Outer diameter [mm]	6.35	9.52	12.7	15.88	19.05	22.2	25.4	28.58	31.8	34.9	38.1	41.3
Minimum thickness [mm]	0.8	0.8	0.8	0.99	0.99	0.99	0.99	0.99	1.1	1.21	1.35	1.43

- Interconnecting refrigeration pipe work between the indoor and outdoor +Slave 1& 2 recommended to layout on galvanized GI tray. Pipe work sizing, layout, fittings, etc shall be in strict accordance with the manufacturers design and installation requirementsThe pipe work shall be of refrigerant quality copper to half hard temper. Soft tempered pipe work may be used where the pipe diameter is ¼"or 3/8". Long radius bends shall be formed using pipe bender. The use of short radius pre formed bends and elbows should be avoided to minimize pressure drop and possibility of leaks. Pipe work shall be carried out manufacturer's design and installation instructions and in accordance with longest possible of copper wire shall be utilized to minimize joints on site. Appropriate refrigeration installation tools must be utilized to avoid the use of elbows, dry Nitrogen must be in the system brazing (no cold brazing will be allowed).

SPLIT UNITS (SMALL CAPACITIES DECORATIVE) .

- Unit shall be factory assembled, wired, piped and run tested
- Unit shall be designed to be installed for indoor application.
- Unit shall be attached to an installation plate/bracket that secures unit to the wall.
- Unit shall be capable to be installed with heat pump VRF system.
- The depth of the unit shall not exceed 265mm.
- Unit case shall be manufactured using ABS material and has a white finish.

Cabinet Assembly:

- Unit shall have one supply air outlet and one return air inlet.
- Unit shall be equipped with factory installed temperature thermistors for:
  - a. Return air
  - b. Refrigerant entering coil
  - c. Refrigerant leaving coil
- Unit shall have a factory assembled, piped and wired electronic expansion valve (EEV) for refrigerant control.
- Unit shall have a built-in control panel to communicate with other indoor units and to the outdoor unit.

Unit shall have the following functions as standard:

a.Self-diagnostic function
b.Auto addressing
c.Auto restart function
d.Auto operation function
e.Auto clean function
f.Child lock function
g.Forced operation
h.Dual thermistor control
i.Sleep mode
j.Dual set point control
k.Filter life and power consumption display (with wired controller)

Unit shall be capable of refrigerant piping in 4 different directions.
Unit shall be capable of drain piping in 2 different directions.

EVAPORATOR UNIT

- The indoor evaporator unit shall be of the slim and compact design with a decorative cabinet. Units shall be suitable for flush-to-wall installation to conceal pipe connections.
- Where indicated on the drawings and schedules, evaporator units shall be of the horizontal concealed type with return air plenum suitable for ducted insulation. Unit casing shall be insulated with at least 12mm thick insulation of density 24 kg/M3.
- Evaporator coils shall be seamless copper tubing mechanically bonded to aluminium fins and shall be factory leak tested at a pressure of 28 kg/cm2. A drain pan made out of sheet steel shall extend throughout the length of the unit. The drain pan shall be coated with epoxy resin enamel and insulated prevent condensation.
- Evaporator fans shall be direct driven, forward curved, centrifugal type made of galvanized steel to deliver and accurate air flow at a low noise level. Fan driving components shall be mounted on rubber pad isolators to reduce noise required.
- A single phase squirrel cage motor shall be used to drive the evaporator fan. A fan speed control shall be provided to facilitate speed adjustment as required.

- Each unit shall be provided with a permanent washable filter and swing flow louvers to eliminate uncomfortable hot spots.

CONDENSING UNIT

- Casing of the unit shall be of sheet steel suitably treated to be rust resistant for a long life.
- Units shall have high efficiency, hermetic, line starting compressor. Each compressor shall be equipped with an internal thermostat, over current relay, a low pressure switch and a crankcase heater. Condenser coil shall be of seamless copper tubes mechanically bonded to aluminium plate fins with protection louvers. Each coil shall be factory leak tested at a pressure of 33 kg/cm2. Coils shall have here site or equal approved anti-corrosive coating.
- Condenser fans shall be direct driven by a weather proof squirrel cage induction motor. Fans shall be of the propeller type.

CONTROL

- The unit controls shall include main PCB, inverter PCB, inverter fan PCB, transformer, bridge diode, reactor. The noise filter shall also be incorporated to reduce the electrical noise.
- The main PCB shall consist of 4 digit, 7 segment LED display. Automatic addressing of the indoor unit shall be possible from the main PCB.
- The outdoor unit shall have Black Box Function which can record the full operation history of the units every 3 minutes which can help in precise analysis, decision and fast trouble shooting.
- The main PCB shall have a refrigerant checking device which charges the appropriate amount of refrigerant automatically through cycle operation, checks for any refrigerant leakage and over charging.
- In case of multiple compressors, each outdoor unit should have alternate cycling of the compressors in order to equalize the run time and increase the life time of the compressor.
- A serial connection port shall be available in the outdoor unit to display the cycle view, operating parameters like suction pressure, discharge pressure, voltage, current, ambient temperature, indoor unit refrigerant in/out temperature, room temperature, percentage loading of the invertors compressor, percentage opening of the EEV, and easy start up and Easy trouble shooting and fault alarm.
- Pump down and pump out operation of the refrigerant should be possible by changing the dip switch setting on the outdoor main PCB.
- The outdoor unit shall be able to connect laptop or mobile through Bluetooth during commissioning to view all the important cycle parameters.

ROOF MOUNTED PACKAGE UNITS:

- These shall be fully weatherproofed, suitable for external mounting and with acoustically lined, tropicalized steel casing on structural steel base frame. Centrifugal fan with belt drive and adjustable drive pulleys to permit adjustment of volume. Anti- vibration mountings and duct flexible connections.
- Hermetic compressors using refrigerant, coils to be copper tube with aluminium fins. Condenser fan to have vertical discharge.
- All controls to be integral to unit and pre-wired at factory. All components to be a accessible via easily removable panels. Remote controller circuit shall be provided.
- The units shall be used in high ambient temperature(52 degree centigrade)
- Units shall be provided with protective devices such as high and low pressure switch, over current relay, internal thermal protector and crank case heater etc. Freezing protection shall be provided. Capacities shall be provided on the drawings. However, Contractor responsibility to re-check the load and select the units accordingly without variation.
- Units should be with twin refrigeration circuits.
- Washable, removable, filter panels,MERV-13 filter, 95% efficiency.

- The units shall be used in high ambient temperature (52 degree centigrade)
- All units are self-contained and assembled on full perimeter base rail. The base rails have holes in the four corners for overhead rigging.
- Every unit is completely piped, wired, charged and tested at the factory to simplify the field installation and to provide years of dependable operation. Powder paint cabinets provide an exceptionally durable finish with the 750 hour salt spray process per ASTM- B117 test standard.
- All supply air blowers are equipped with a belt drive that can be adjusted to meet the exact requirements of the job.
- All compressors include crankcase heat and internal pressure relief. Every refrigerant circuit includes and expansion valve. A liquid line filter-drier, a discharge line high pressure switch and a suction line with freeze stat and low pressure loss of charge switch.
- Low ambient- an integrated low-ambient control allows all units to operate in the cooling mode down to 0 degree F outdoor ambient.
- Anti-short cycle protection- To aid compressor life, an anti-short cycle delay is incorporated into the standard controls. Compressor reliability is further ensured by programmable minimum run times. For testing, the anti short cycle delay can be temporarily overridden with the push of a button.

CONDENSER FANS

- The condenser fans are propeller type, aluminium alloy blades, directly driven by electric motors. Motors are totally Enclosed Air Over (TEAO) six pole with class “B” insulation and minimum IP55 protection to ensure long life. The motors are factory wired and suitable for high ambient operation.
- The condenser fans are statically and dynamically balanced at the factory. Complete fan assembly is provided with suitable acrylic coated fan guard.

EVAPORATOR

- Evaporator coils are manufactured from seamless copper tubes mechanically bonded to aluminium fins to ensure optimum heat transfer. All evaporator coils shall be tested against leakage by air pressure.
- Evaporator coil supplied with suitable size thermostatic expansion valve(s) and multi-circulated distributors providing capacity modulation to match the compressors.

EVAPORATOR FAN DRIVE

- Unit shall have dual fans mounted on a single heavy duty shaft driven by a single electric motor, Class B insulation IP55 protected and totally enclosed rated for continuous operations.
- The motors are fitted with an adjustable belt drive as standard.

EXTRACT FANS

IN-LINE DUCT MOUNTED

- Unit casing shall be manufactured in aluminium alloy and shall house fan assemblies, comprising Backward curved centrifugal or Axial impellers running in an individual scroll. Impellers may be either direct driven or belt driven depending upon the fan duty. Motors shall be manufactured to BS 5000, TEFC type with sealed for life bearing, class F insulation. Motors shall be prewired to fitted isolator accessible from outside.

WALL/WINDOW MOUNTED EXTRACT FAN

- Extract fans shall be of the propeller type suitable for window mounting. Fan construction shall be moulded plastic. Motor shall be of the shaded pole induction type enclosed in aluminium alloy cases and protected by a thermal overload cut-out. Motor bearing shall be self aligning oil impregnated porous bushes with an ample oil reservoir.
- Fans shall be provided with solenoid operated back draught shutters, opening and closing as the fan is switched On and Off.

TWIN ROOF MOUNTED FANS

- The fan unit shall comprise of a duplicate fan assembly complete with motors and drives mounted on a heavy0duty unit base plat by anti vibration mountings. The fan shall be direct/belt driven with double inlet forward curved centrifugal impellers running of two life-long lubricated bearings. The fan unit shall be fitted with automatic shutters to prevent air recirculation through standby fan and blow back when not in use.
- The fan unit electrical motors shall be manufactured in accordance with BS5999 and be on the shaded pole or capacitor start and run type suitable for an electrical supply of 220 volts, single phase, 50 Hz and capable of operating in ambient conditions as described earlier.
- The fan unit shall have air operated flow switched fitted, arranged to change over to be standby fan in the event of selected fan failing and also to proved a fault signal. The fan unit shall be prewired with switches and isolators to a single terminal box on the casing. The unit casing shall be constructed from galvanized mild steel / aluminium alloy with centrally positioned inlet spigot and access panel to allow ease of service.

VENTILATION AND AIR CONDITIONING

- The Contractor shall supply, install, test, commission and set to work a complete and fully operational ventilation and air conditioning system in accordance with the true intent of this particular specification, the standard specification and as indicated on the design tender drawings and equipment schedules.
- The HVAC systems serving the buildings shall be designed to provide sufficient conditioned air to maintain comfort conditions within the spaces and to provide sufficient outside air to satisfy indoor air quality requirements for occupants in accordance with ASHRAE standard 62.1.
- In addition, HVAC systems shall provide sufficient make-up air to replace that extracted from the toilets and kitchen areas whilst providing a slight positive pressure to minimize external air infiltration. Residential apartments fresh air system
- Fresh air flow rates shall be provided in accordance with the above mentioned ASHRAE 62.1 requirements but shall also be dictated by the extract rates from apartments (toilet and kitchen areas) to maintain the apartments under slight positive conditions and reduce air infiltration.
- Fresh air shall be treated and supplied by fresh air handling units located in mechanical plant room at the roof level to each residential apartment within the building.
- Fresh air from the roof level air handling units shall be vertically distributed up through the building in HVAC shafts located within the core area. The ducts shall branch from the riser at each floor and distribute horizontally via corridor services void to each of the apartments.
- The fresh air shall be supplied to main apartment areas (i.e. living rooms, bedrooms etc.) to maintain rooms positively pressurized relative to the toilet areas and kitchens.
- The outdoor air supply and extract ventilation shall operate on a continuous basis.
- The residential apartment fresh air handling units shall also provide fresh air for common corridor areas and other ancillary spaces on residential floors, as defined elsewhere.
- apartments fresh air handling units shall also provide fresh air to the restaurant, and retail spaces if any in the ground floor..

RESIDENTIAL TOILET VENTILATION

- All toilets and laundry rooms shall be provided with an exhaust in accordance with ASHRAE Standard 62.1.
- Make-up air to the toilet and laundry rooms shall be transferred from the adjacent spaces via door undercut.
- The toilet extract ductwork shall be routed back to air handling units for heat recovery via the thermal wheels. It is essential that cross contamination risk is minimized in these units and the thermal wheel must incorporate a purge section and guaranteed cross contamination performance as detailed elsewhere within the specification.

- Toilet extract duct from each apartment shall connect to a dedicated toilet extract riser within the core HVAC shaft. The toilet extract ducts shall join with residential kitchen extract ducts in mechanical plant room before connecting to air handling units for heat recovery.
- The ductwork distribution systems shall be complete with all ancillary equipment, such as (but not limited to) fire dampers, volume control dampers, etc. as indicated on the tender design drawings and detailed in the specification.

RESIDENTIAL KITCHEN VENTILATION

- All residential kitchens shall be provided with a separate ducted extract in accordance with ASHRAE Standard 62.1.
- Extract shall be also provided to all BOH kitchens in apartments.
- The extract will be provided via ceiling diffuser(s). Kitchen cooking areas shall also be equipped with recirculating carbon filter canopies as specified by the architect.
- Kitchen areas are open to dining and living spaces where the make-up air shall be transferred from.
- The kitchen extract ductwork shall be routed back to air handling units for heat recovery via the thermal wheels. It is essential that cross contamination risk is minimized in these units and the thermal wheel must incorporate a purge section and guaranteed cross contamination performance as detailed elsewhere within the specification.
- Kitchen extract duct from each apartment shall connect to a dedicated kitchen extract risers within the core HVAC shaft. The kitchen extract ducts shall join with toilet extract ducts at the roof level plant room before connecting to air handling units for heat recovery.
- The kitchen extract ductwork shall be provided with electrostatic filter prior to joining to the toilet extract ductwork
- The ductwork distribution systems shall be complete with all ancillary equipment, such as (but not limited to) fire dampers, volume control dampers, etc. as indicated on the tender design drawings and detailed in the specification.

RESIDENTIAL AIR-CONDITIONING SYSTEM

- Residential apartment spaces shall be comfort cooled using fan coil units.
- Typically, a dedicated fan coil unit shall be provided for each of the rooms listed below:
  a) Living room
  b) Master bedroom
  c) Each bedroom
  d) Bedroom bathrooms with external glazed walls
  e) Maid's room
- Residential apartments guest toilets, laundries and stores shall not be. directly cooled. These spaces shall be provided with extract ventilation, which shall draw. air from adjacent living room or bedroom space through door undercuts.
- The fan coil units shall be of ducted type located above false ceilings, strategically placed. over wet areas or kitchens. In some of the large apartments, fan coil units shall be placed. over circulation areas as indicated on the design drawings.
- When operating the chilled water fan coil units shall continuously recirculate air into the space. Air shall generally be drawn from return air grilles and in some cases from transfer ducts. Generally, the ceiling void shall be utilized as a return air plenum.
- The air shall be filtered and cooled before being returned to the space, typically via ceiling linear diffusers or ceiling louver face diffusers as specified on the design drawings.
- Metered chilled water connection shall be provided to each residential apartment. Chilled water connections complete with all accessories and pressure independent control valve shall be provided to each fan coil unit.
- Wall mounted temperature and fan speed controllers shall be provided in each residential and residential apartment space served by a FCU to allow the occupants to control the

internal temperature.

CORRIDORS AND LIFT LOBBIES

- Residential apartments floor corridors and lobbies shall be comfort cooled by recirculating fan coil units with fresh air supply from the AHU supplying the respective building as indicated on the design drawings.
- Fan coil units shall be of ducted type installed within the ceiling voids and supply and return air diffusers shall be provided in the ceiling. The fan coil units shall be located in the corridor outside the main front of house lobbies, where possible.
- Apart from cooling the common circulation areas, fan coil units shall also serve adjacent on-floor electrical and telecom riser rooms. Branches from the supply duct shall be taken and air shall be supplied / returned to the spaces via wall grilles complete with fire dampers.
- Chilled water connections complete with all accessories and pressure independent control valve shall be provided.
- The fan coil units shall be controlled by temperature sensors located within the rooms and shall be controlled & monitored by the BMS.

MAIN ENTRANCES AND LANDLORD AREAS

- Main entrances shall be cooled by recirculating fan coil units to meet the cooling requirements. The lobbies shall be positively pressurised to assist the buffer effect.
- Fan coil units shall be of ducted type installed within the ceiling voids and supply and return air diffusers shall be provided within the ceiling above the entrance lobbies areas.
- Refrigerant pipe connections complete with all accessories and controls shall be provided.
- The fan coil units shall be controlled by temperature sensors located within the rooms and shall be controlled & monitored by the BMS.
- The ventilation air to the main entrances and other landlord areas shall be served by the same air handling units serving the residential apartments.

GARBAGE ROOMS

- All on-floor garbage chute rooms shall be provided with continuous extract ventilation system. There shall be an extract air inlet in each of the rooms connected to a garbage extract riser distributed vertically throughout each building.
- Separate system shall be provided for each shaft and dedicated extract fans shall be located on the respective roof level.
- There shall be no dedicated cooling or air supply provided to the on-floor chute rooms Make-up air shall be transferred from adjacent corridor via door undercut.
- Refuse/waste collection rooms at the Ground level of each building shall be provided with continuous general extract ventilation and shall be connected to the extract fans located at the respective high-level ground and the air shall be discharged to atmosphere via louvres.
- The rooms shall be maintained at a slight negative pressure.
- Refuse/waste collection rooms shall be cooled by two pipe chilled water fan coils. Fan coil units shall be non-ducted decorative type wall-mounted inside the rooms. Chilled water connections complete with all accessories and pressure independent control valve shall be provided. The fan coil units shall be controlled by temperature sensors located within the rooms and shall be controlled & monitored by the BMS.

RETAIL/F&B SPACES

- Retail/F&B. spaces are proposed on the Ground levels of building. The units are designed with shell & core services provision as detailed below.
- capped-off connections will be provided to each retail space for further extension by the tenant's fit-out designer / contractor within the retail spaces. The connections shall be

complete with meters integrated with the sub-metering system managed by owner.
- Each retail unit will be provided with external louvre at the high level of the façade for the tenant to connect his supply and extract fans.
- The Contractor shall verify final requirements with the Engineer and Client prior to commencing the works associated.

CAR PARK VENTILATION SYSTEM
- Car park ventilation system is described and specified within the Smoke Management Systems section of this specification.

MAIN LV ROOMS
- Main LV equipment rooms for the building respectively shall be cooled by fan coil unit with a fresh air supply from the ventilation system served.
  from the associated building air handling unit as indicated on the drawings.
- Fan coil units shall be of ducted type installed outside the rooms in close proximity and shall supply and return air via wall mounted grilles.
- The fan coil units shall be controlled by temperature sensors located within the rooms and shall be controlled & monitored by the BMS.
- In case the temperature of the room rises above the pre-set value an alarm shall be raised.

MECHANICAL PLANT ROOMS
- All mechanical plant rooms located throughout the buildings shall be air-conditioned by fan coil units with a fresh air supply from the ventilation system served from the associated building air handling units as indicated on the drawings.
- All fan coil units within mechanical plant rooms shall be non-ducted decorative type wall mounted inside the rooms, as indicated on the design drawings and equipment schedules.
- complete with all accessories shall be provided.
- The fan coil units shall be controlled by temperature sensors located within the rooms and shall be controlled & monitored by the BMS.

TRANSFORMER ROOMS
- Transformer rooms' ventilation shall be in accordance with DEWA requirements.
- The rooms shall be provided with louvered doors.
- Additionally mechanical forced supply and extract ventilation shall be provided for the spaces. Dedicated supply and extract fans with associated ductwork and accessories will be provided and installed externally to the rooms in the parking areas. The fans shall provide air movement so that the temperature of the rooms does not rise above 55°C.
- Transformer room extract and supply fans shall be controlled via the BMS based on temperature readings within the transformer room.
- Motorised smoke and fire dampers shall be provided on all ventilation ducts connecting to the transformer rooms.
- There shall be no cooling provided to transformer rooms.

FIRE PUMP ROOM
- Fire pump room shall be ventilated in accordance with requirements of NFPA and DCD.
- Ventilation to the room shall be provided to deliver outdoor air rate required for normal operation of the room during non-fire condition and during electric fire pump operation
- During fire condition, in case of loss of power when the diesel standby fire pump is operating, the fire pump room shall be ventilated with un-tempered outside air supplied by a dedicated fan located in Lower Ground. The fan with supply ductwork shall be fire rated and shall be powered by supply from generator.

- The fan shall supply the outdoor air at the rate as required for the combustion demand of the diesel fire pumps. The final air flow rate requirement shall be coordinated by the Contractor with the final fire pump specification.
- Diesel fire pump shall be provided with fume exhaust system as detailed and specified in the Fire Protection design package.
- Cooling to the room shall be provided to maintain the room temperature below 49°C at fire operation mode. Cooling shall be provided by DX split units. The units shall be provided in a duty & standby arrangement (n+1) and shall be powered by supply from generator.
- The units shall be of recirculating non-ducted type and shall be installed inside the pump room.
- The DX split units shall be controlled by temperature sensors located within the rooms and shall be controlled & monitored by the BMS.

STANDBY GENERATOR

- This section covers the standby generator mechanical requirements. For other technical and operational details and specification of the generators Electrical Services Specification should be referred.
- System description The generators shall be located in dedicated rooms at Ground Level a) of each plot. The Contractor shall ensure that the selected generator manufacturer can accommodate the required equipment within the room allocated and that ventilation, complete with all associated ductwork, can be arranged to provide the ventilation necessary for engine combustion and heat rejection. The Contractor shall further develop the room
- arrangements with the selected generator manufacturer as part of the initial shop drawing process. The Contractor shall supply and install all ductwork, louvers, attenuators, etc. required to facilitate a complete installation appropriately coordinating with the building arrangements.
- The generator room shall be provided with a louvered air intake b) as shown on the MEP drawings. The active louver area shall be sufficient to ensure the air flow through the louvers is less than 2.5m/s face velocity and is in accordance with the approval of the Engineer and the selected generator manufacturer's requirements / recommendations. The discharge from the generator radiators shall be ducted to outside. The generator flue complete with silencers shall discharge at high level away from the air intake louver. Provide acoustic treatment to reduce the noise level to 65 dB at 10m from the generator room.

GENERATOR ENGINE EXHAUST

- Provide a complete system for removal of exhaust products of combustion from engine generators. System shall conform to requirements and recommendations of British Standards and specific requirements of engine generators manufacturer. Engine generator sets exhaust silencers and engine flexible piping connector shall be as per manufacturer recommendations.
- Provide pipe, fittings, flexible connectors, insulated roof thimble, tall cone flashing, anchors and guides, flapper caps and insulation. Piping shall be black steel pipe to BS 1387 medium grade, with welded fittings or equal and approved. Test system exhaust pipe at suitable test pressure in accordance with standby generator manufacturer's recommendations and system back pressure.

AIR HANDLING UNITS

- The Contractor shall supply, deliver, place in position and erect on prepared bases, the air handling plant and equipment as specified herein and elsewhere in the specification.
- The equipment shall be capable of meeting the duties as detailed in mechanical equipment schedule. The order of components shall be as shown on the drawings for each individual air

handing unit and detailed in the mechanical equipment schedule.

- Fresh air handling units shall typically consist of:
  a) Filters
  b) Heat recovery wheel (enthalpy)
  c) water cooling coil
  d) Re-heat thermal wheel (sensible)
  e) Supply fan
  f) Extract fan
  g) Motorized dampers
- The minimum efficiency for thermal wheel shall be 70%.
- Air handling unit cooling coils shall be selected with 20% extra capacity to allow for thermal wheel deterioration over exploitation period.
- Chilled water connections complete with all accessories and dedicated pressure independent control valve shall be provided to each air handling unit.
- All air handling unit motors shall be equipped with VFD’s for ease of commissioning.
- Each air handling unit shall be sectional, however the Contractor shall allow for assembly of the various sub-sections on site, the sections of the units being mounted on a continuous steel channel base which will also be used as a hoisting/cranage support.
- All air handling units shall be connected to and monitored & controlled by the BMS. (for more information refer to HVAC Controls and Building Management System sections, as well as BMS design schematic drawings).

FAN COIL UNITS

- The Contractor shall provide fan coil units of the type, size and capacity shown on the drawings and mechanical equipment schedules.
- The fan coil unit shall be complete with cabinet, high efficiency fan motor, chilled water connections, main and auxiliary drain pans, filters, control transformer, motor speed controller, valve package, drain cock and isolator switch. The valve package shall include but not be limited to: pressure independent control valve, isolating valves, flushing bypass valve, and strainer.
- A corrosion resistant drip pan shall be provided covering the entire length of the fan coil. Washable synthetic filters shall be provided with as part of each unit. All filters must be washed and cleaned before handover
- The FCUs shall be selected at medium speed.
- The FCUs shall be fitted with forward curved double width fan wheels. The wheels and scroll shall be constructed from galvanized steel or aluminium. Plastic is not acceptable.
- The motors shall be 3-speed, split capacitor type units and shall be designed for positive speed reduction with inherent overload protection and mounted on a resilient base. The motors shall typically be 220V, single phase, 50 Hz.
- The Contractor shall be fully responsible for selecting FCU's such that the specified noise limits are not exceeded with the indicated arrangement of ductwork and the architectural/structural construction shown on the Drawings. If necessary, the Contractor shall include for all necessary acoustic and vibration isolation measures to meet the specified noise criteria.
- All FCU’s shall be complete with a dedicated DDC controller for remote monitoring and control via the BMS.

DX SPLIT UNITS

- The Contractor shall supply and install DX split air-conditioning units in the positions indicated on the drawings or detailed elsewhere in the documents.
- For small capacity units of up to 3 TR nominal, the specified manufacturers construction standards shall generally apply, however, each system shall, as a minimum, be complete with liquid receiver complete with two stop valves, a sight glass, charging connection, a purge

valve, a suction line accumulator, a large capacity filter/drier and moisture indicating sight glass.

- The capacity of air-conditioning units shall be as detailed in the mechanical equipment schedule. Unless stated otherwise, the capacities mentioned are the actual requirements at specified operating conditions (i.e. ambient dry bulb of 46°C).
- The ducted split units shall consist of:
  a) Indoor units (i.e. evaporating units)
  b) Outdoor units (i.e. condensing units)
  c) Controls
- d) Interconnecting refrigerant piping, power and control wiring between indoor and outdoor units.
- Indoor units (fan coil units) should be of ducted type or decorative type as indicated on the design drawings and specified equipment schedules.
- It is a requirement to comply with Dubai Municipality Green Building Regulations & Specifications which require all refrigerants to have zero ozone depletion potential (ODP) or global warming potential (GWP) less than 100.
- All DX split units shall be connected to and be controlled via BMS and fire alarm.

HVAC CONTROLS

- The specialist contractor shall provide a fully programmed and functional control system for the HVAC operations as defined within this specifications section and elsewhere in sections referring to BMS and design drawings.
- Fan coil units – residential apartments
  a) Each FCU in all Residential apartments shall be provided with a dedicated controller installed on the wall in the services room/area.
  b) Each controller shall typically consist of:
  c) Digital display
  d) Room temperature sensor
  e) Fan speed set point
  f) Auto / manual selection
  g) On / off selection
  h) FCU's shall be controlled as per the controller unit settings.

FAN COIL UNITS – MAIN ELECTRICAL ROOMS, TELECOM ROOMS AND GSM ROOMS

- The FCU's shall normally be energised continuously and shall a) be controlled by the BMS. Local isolators shall be provided to enable the maintenance staff to switch the fan coils off as required.
- The controller shall be electronic with proportional or proportional plus integral output. During normal operation, the space temperature shall be maintained by modulating the flow of chilled water through the pressure independent control valve connected to the FCU cooling coil. The control valve shall be proportional type with control signal in proportion to the difference between set point and measured temperature.
- The BMS shall also control the FCU fan speed (low/medium/high).
- The BMS shall continually log the rooms temperature and shall raise a visual and audible alarm if the temperature in the room exceeds a pre-determined value, 26°C (adjustable).
- The BMS shall also record operative and inoperative hours for each room FCU.

FRESH AIR HANDLING UNITS – RESIDENTIAL APARTMENT

- During normal operation the AHU's shall operate under the control of the BMS with HO-A switch on the AHU control panel in Auto position.
- The control system shall be capable of resetting the "Off coil" temperature set point based on the ambient temperature and relative humidity. Relative humidity and

temperature sensors shall be installed to monitor the extract air condition upstream and downstream of heat recovery system and the supply air downstream of heat recovery system.

- A common set of sensors shall be provided to monitor ambient air conditions. Relative humidity and temperature sensors shall also be installed to monitor the air condition off the thermal wheels. These shall be used to monitor the efficiency of the thermal wheels and to assess the potential energy recovery of the units and to identify any failure/drop in performance.
- Control valves shall be of the globe type and the Contractor shall ensure that the valve closes when the AHU is switched off.
- The extract fan associated with each AHU shall also be controlled from the AHU control panel. A separate H-O-A switch shall be provided. In the normal course, the H-O-A switch shall be in Auto position and fan shall be operated by the BMS. The AHU extract fans and supply fans shall operate in tandem. In Auto position, Heat Recovery System control shall be carried out by the BMS. Power ON, Control power ON, Ammeters, Voltmeters shall be provided for each AHU control panel. Run, Trip indications shall also be provided for each fan, motor. Filter dirty indications shall be provided for each bank of filters.
- For residential fresh air handling units the fans shall operate at constant speed.
- Differential pressure switch shall be provided to monitor the condition of each bank of filters and annunciate on alarm when the pressure drop exceeds an adjustable value.
- Pressure sensors shall be located either side of the supply and extract fans to confirm airflow. In the event of no airflow the unit shall shutdown.
- Motorised low leakage dampers shall be provided on the outdoor air and exhaust air ductwork and in the event of a failure of the unit these shall close. At start-up these dampers shall fully open before the extract then supply fan start.
- In ‘Hand’ mode the fan shall run continuously, independent of the BMS, all necessary interlocks shall be provided by relays, auxiliary contacts etc.
- The AHU control panel shall include all necessary terminals for operation through
- Building Management Systems. The BMS outstation can be installed with the AHU control panel in a segregated controls section that can be accessed without switching off the power supply to the AHU.
- A smoke detection device shall be provided in the return air ductwork & Air intake duct. On sensing smoke the supply fan and extract fan shall be stopped and an alarm raised at the BMS central supervisor. The detector shall be manually reset.
- A manual switch at the Fireman’s Override Switch Panel n) (FOSP) will permit the department of Civil Defence to shut system down. Smoke detectors to be overridden from ECC room control panel.
- In the event of a fire in the building or in the event of fire or smoke being detected by
- the fire alarm system in the ductwork or the area being served by the AHU’s, the
- MCC’s shall receive the fire signal from the fire alarm system. The DDC shall then
- switch ‘Off’ the AHU and extract fans.
- A fireman’s override facility shall be provided for building to enable manual operation of
- each fan/ AHU to override any control function from the fire alarm system or BMS.
- Air handling units shall normally be energised continuously and shall be controlled by the BMS. Local isolators shall be provided to enable the maintenance staff to switch the fan coils off as required.
- Temperature sensors shall be installed within the served space. The temperature sensors shall relay the signal to a temperature controllers mounted along with the transformers, control fuses, and terminals, etc. in control boxes near to the FCU’s.
- The controllers shall be electronic with proportional or proportional plus integral output. During normal operation, the space temperature shall be maintained by modulating the flow

of chilled water through the pressure independent control valve connected to the AHU cooling coil. The control valve shall be proportional type with control signal in proportion to the difference between set point and measured temperature.

- The BMS shall also control the AHU fan speed (low/medium/high).
- A time clock shall also be provided as part of the BMS to facilitate programme scheduling of the fan 24 hours a day 7 days a week if necessary.
- Control valves shall be of the globe type and the Contractor shall ensure that the valve closes when the AHU is switched off.
- Differential pressure switch shall be provided to monitor the condition of filters and annunciate on alarm when the pressure drop exceeds an adjustable value.
- In 'Hand' mode the fan shall run continuously, independent of the BMS, all necessary interlocks shall be provided by relays, auxiliary contacts etc.
- The AHU control panel shall include all necessary terminals for operation through Building Management Systems. The BMS outstation can be installed with the AHU control panel in a segregated controls section that can be accessed without switching off the power supply to the AHU.
- In the event of a fire in the building or in the event of fire or smoke being detected by the fire alarm system in the ductwork or the area being served by the AHU's, the MCC's shall receive the fire signal from the fire alarm system. The DDC shall then switch 'Off' the AHU and extract fans.
- A fireman's override facility shall be provided for building to enable manual operation of each fan/ AHU to override any control function from the fire alarm system or BMS.
- The fire alarm shall be reset manually for the AHU's to run under normal mode again.
- The system shall comprise of duct mounted extract fan serving the kitchen. The fan shall be enabled at the dictates of an adjustable time programme demand from the BMS. The extract fan shall be enabled if the following conditions are satisfied: -
  a) The Hand/Off/Auto switches are in the Auto or Hand position
  b) The starter overloads are in healthy conditions.
  c) The kitchen supply air handling unit c) is running with flow
  d) A flow alarm override is not present
  e) The fire alarm override signal is not active
- A differential pressure switches installed across the fans shall monitor flow, and in event of flow failure, the fan shall be shut down, and an alarm shall be raised at the BMS.
- The system shall comprise of duct mounted extract fan serving the kitchen dishwasher. The fan shall be enabled at the dictates of an adjustable time programme demand from the BMS. The extract fan shall be enabled if the following conditions are satisfied:-
  a) The Hand/Off/Auto switches are in the Auto or Hand position
  b) The starter overloads are in healthy conditions.
  c) The kitchen supply air handling unit is running with flow
  d) A flow alarm override is not present
  e) The fire alarm override signal is not active
- A differential pressure switches installed across the fans shall monitor flow, and in event of flow failure, the fan shall be shut down, and an alarm shall be raised at the BMS.

CONDENSATE DRAIN FROM HVAC EQUIPMENT

- Condensate drain from all air handling units and most fan coil units within the building shall be connected in a condensate drain network and reclaimed to the condensate collection tank located in PH plant room at Lower Ground level for subsequent use, reducing the potable water requirement for irrigation.
- For the main condensate drain collection network distribution and specification refer to Public Health design package.
- Dedicated condensate drain risers shall be provided throughout the building as indicated in the drainage design drawings. All fan coil units within apartment floors as well as all air

handling units from mechanical plant rooms shall be connected to these risers.

- A number of fan coil units, particularly within the basement units may not be connected to the condensate drain network due to a remote location or installation level and the drains shall be connected to nearest floor drain or waste pipe as referred in the design drawings.
- These units however constitute a small percentage of overall cooling coils and shall not considerably impact the total condensate drain yield.

DUCT WORK

- This applies to duct work with mean velocities less than 2000 fpm (10 M/sec) and static pressures of 2 inches water gauge, (500 pascals) or less.
- Supply and install, where shown on the Drawings, galvanized sheet steel ductwork, complete with all necessary bracing and supports.
- All rectangular ducts shall be of the low pressure rating and all circular round ducts shall be of the high pressure rating.
- Galvanized sheet steel shall be fabricated, erected and installed in accordance with NFPA 90A and "SMACNA" sheet metal manuals.
- Galvanized sheet steel ducts shall be of G90 coating designation within ASTM A525- 75, standard specification for 'steel sheet zinc coated by the hot dip process'. The weight of coating on both sides of duct shall be 0.9 oz/ft2 (275g/m2) as a minimum check limit triple spot test.
- The ducts' gauges, thickness, type and method of jointing shall be as detailed and tabulated on the Drawings and/or in compliance with ASHRAE Standards and Handbooks.
- All ducts shall be constructed and erected so as to be rigid and free from sway, drumming and movement. Duct work shall be true to sizes indicated on Drawings, straight and smooth on the inside with neatly finished joints. Whenever internal acoustic lining is indicated on the Drawings, the duct sizes have to be increased to accommodate the lining.
- Ductwork joints shall be square with all sharp edges removed.
- The ducts shall be routed with a minimum of directional changes and abrupt transitions.
- Adequate space shall be provided around ducts to assure proper support and to allow the installation of the specified insulation.
- All connections between ductwork, including flexible connections, fittings and equipment, shall be made with gradually tapered transition fittings.
- Whenever a flexible duct is used to correct misalignment between the supply duct and the diffuser ceiling location, the misalignment (or offset) shall not exceed one-eighth (1/8) the length of the collar (or diffuser diameter).
- Changes in section of ductwork shall be effected by tempering in ducts with as long a taper as possible. All branches shall be taken off at not more than 45 degree angle from the axis of the main duct unless otherwise approved by the Engineer.
- The ducts shall be securely anchored to the building in an approved manner.
- The ducts shall be installed as to be completely free from vibration under all conditions of operation.
- The ducts and hangers shall be installed straight, plumb and level.
- Wherever ducts pass thru walls or floors, a sleeve of galvanized mild steel sheet shall be provided and the space between the insulated duct and the sleeve shall be caulked with lead wool and finished on each face with a mastic fill.
- Flexible ducts should be kept as short as possible and fully extended. All slip joints shall be made in the direction of flow.
- All elbows shall have a centreline radius equal to at least 1.5 times the width of the duct, otherwise turning vanes shall be installed in the elbows.
- Adjustable splitters and hinged volume dampers shall be provided at every duct junction on both supply and exhaust ductwork for adjusting air volumes.
- Where splitters and dampers are installed above suspended ceiling, flush-mounted controlling devices shall be used.

- Connection to diffusers, grilles and registers shall be made absolutely airtight.
- Equalizing grids or turning vanes shall be installed ahead of an air outlet whenever poor approach conditions, from the main duct to the outlet exist.
- In critical low noise level projects, poor approach conditions are not allowed. Where the duct is pierced for any reason, sealing compound shall be used. All joints and fittings concealed in vertical duct shafts shall be welded.
- All rectangular metal ducts shall be sealed in accordance with ASHRAE Standard 90 and SMACNA low pressure duct construction standards. All high pressure ducts shall be air tight.
- Kitchen exhaust ducts shall be constructed and installed in conformance with NFPA 96 and must:
- Be constructed from carbon steel (for concealed ducts) with a minimum thickness of
- 1.4 mm and from stainless (for exposed ducts) with a minimum thickness of 1.1 mm.
- Have all longitudinal seams and transverse joints continuously welded.
- Be installed without dips or traps which may collect residues, except where traps with continuous or automatic removal of residues were provided.
- Have provisions for expansion. Carbon steel ducts elongate 50 mm in 4 m. at 1100 deg. C temperature.
- The joints between the ducts will be completed with the installation of adhesive gasket to reduced air leakage.
- Ductwork shall be installed, using supports, as described in DW144 and according to manufacturer's requirements.
- Ductwork (non-fire-rated) materials shall be used as per the below schedule:

Supply / return / exhaust / outdoor air ductwork	**Hot dip galvanised steel**
FCU (concealed) ductwork	Hot dip galvanised steel or pre-insulated duct work
FCU (exposed) ductwork	Hot dip galvanised steel

- Sizes of ductwork on the drawings are internal clear airway dimensions. Where applicable, make allowance for any internal lining.
- Contractor shall prepare his shop drawings after careful coordination with other services. If necessary, Contractor can propose a change in dimension (subject to the cross-sectional area being maintained) if the specified duct size cannot be accommodated due to restricted space.
- Ductwork shall be classified as medium pressure to Table 1 of DW144 with air leakage as Class B.
- Ductwork shall be manufactured from zinc coated steel as DW144 Clause 7.2
- Provide access openings to DW144 and as required by the facilities maintenance engineers for regular inspection, servicing and cleaning of the systems.
- Ductwork support for insulated ductwork with a vapour barrier shall g) be as detailed in DW144. A rigid insulator shall insert between the support and ductwork to the thickness of the insulation. The insulation vapour seal shall be continuous through the insulator support and not taken over the support bracket.

FIRE RATED DUCTWORK

- Fire rated ductwork shall be provided for following ductwork systems and the ductwork shall be certified to conform to the below specified fire resistance:
- Smoke extracts ductwork – 2 Hours
- Stairs pressurisation ductwork – 2 Hours
- Lift shaft pressurization ductwork – 2 Hours
- Corridor smoke management ductwork – 2 Hours
- Car park extract ductwork – 2 Hours

- Commercial kitchen extract ductwork – 2 Hours
- Fire rated ductwork shall be constructed and protected to ensure that the stability,integrity and insulation of the ductwork system is maintained for the required duration. Supports and fixings shall be of sufficient strength and arranged to ensure that the ductwork is maintained in position throughout the period.
- Fire rated ductwork materials shall be used as per the below schedule:

Smoke Extract and smoke management systems	Hot dip galvanized steel with fire coating and fire rated Insulations
Kitchen Exhaust	MS or Stainless steel

- All fire rated ductwork utilised for smoke management systems shall be constructed from galvanizes steel, fabricated to a tested standard and coated with fire rated material.
- All fire rated ductwork for commercial kitchen extract system shall be constructed from 2 mm mild steel or 1.1 mm stainless steel sheets, fabricated to the specified standards and coated with fire rated material. Mild steel duct shall have a pre-treatment of minimum 100 micron of red oxide to prevent rusting.
- The duct shall be a composite fire duct manufactured to Method 3 of BS5588 Part 9, factory produced. Once erected, the duct shall be pressure tested to DW144 standard, in line with the pressure classification of the duct being installed.
- Fire rated ductwork construction utilized for kitchen extract systems or smoke exhaust applications shall be tested to BS476 Part 24 (1987) and ISO6944 (1985) for two hours stability and two hours integrity for both fire inside (Type B test) and fire outside (Type A test). The fire rated ductwork must be installed to UL and relevant DW standard for smoke and ventilation ducts.
- All kitchen exhaust ventilation systems and smoke extract systems shall comply with these fire rated ductwork specifications and shall also conform to DW143/144/172 standards as appropriate.
- Access doors must be installed at 3 meter intervals (where possible) for all kitchen extract ductwork, shall have same fire rating as of duct and to be installed outside the air stream including its frame or other supporting arrangements to have smooth internal surfaces to enable easy cleaning of the grease. Fire dampers must not be used in commercial kitchen extract system. Volume control dampers should be avoided however where necessary for balancing they must have stainless steel construction and access door shall be provided for regular cleaning.
- All fire rated ductwork must be manufactured and erected in line with the tested and assessed Certification to BS476 Part 24 and ISO6944. All fire rated ductwork must be treated in a factory controlled environment, away from the project location in order to maintain the highest quality of treatment.

CONTROL DAMPERS

- Install control dampers in the ductwork location a) shown on the drawings.
- Control dampers shall be suitable for systems having a maximum differential pressure of 1000Pa and shall have a maximum closed blade leakage of 25l/s/m2 at this pressure.
- Dampers to have marker labels for commissioning.

FIRE DAMPERS

- Whether shown on drawings or not, supply and install UL listed fire dampers at all positions where ducts/builders work holes or return air transfer openings penetrate fire rated walls, floors and partitions. Sizes to match duct sizes and ratings to match the fire rating of the fire separation.
- Fire dampers shall conform to the requirements of NFPA 90A or BS and the Dubai Civil Defence.
- Airtight, insulated, hinged access doors with latches shall be installed adjacent to all dampers and shall be sized for easy inspection and maintenance of dampers. Contractor shall not

obstruct access doors with piping, etc.
- Provide required ceiling access doors in areas with other than removable type ceiling.

SOUND ATTENUATORS

- The Contractor shall design, supply and install sound attenuators as indicated on the design drawings.
- After approval of the equipment and shop drawings the Contractor shall submit acoustic calculations demonstrating the specified noise levels will be achieved.
- The noise attenuators shall be sized to offer a maximum airflow resistance of 50 Pa per attenuator and maximum airway velocity of 15m/s.

GRILLES AND DIFFUSERS

- Grilles and diffusers shall be selected to achieve the appropriate throw and pressure drops based on the volume flow rates indicated on the drawings and equipment schedules. A maximum terminal velocity of 0.25 m/s in the occupied area shall be maintained.
- The overall noise level produced by all of the supply air outlets and return air inlets in various rooms shall not exceed specified limits.
- Design outlets shall distribute air in such a manner that the space temperature shall not vary more than 1.5°Cover the entire conditioned area. The conditioned area is defined as the area 0.6 meters above the floor to 2.1 meters above the floor, inclusive.
- Architectural Drawings and Specifications shall be referred for ceiling type and construction.

DUCTWORK SPECIFICATION:

- Duct work fabrication and hanging methods shall be as described in the 'Equipment' Volume of the American Society of Heating, Refrigerating and Air Conditioning Engineers Guide and Data Book, latest edition, and/or current edition of applicable manuals published by the Sheet Metal and Air Conditioning Contractors National Association Inc., where methods described in these volumes are not at variance with the requirements of any authority having jurisdiction, or do not conflict with methods described hereafter. Duct work shall not be shop fabricated until work has been job measured and interference situations coordinated.
- Any contractor who is more familiar with British Standards may refer to BS CP 352 and DW 142. Any discrepancy will be clarified by the consultant on request.
- In any event, the standard of work required is the best available, and the specifications are to be so interpreted.

DUCTWORK GENERAL:

- All dimensions shall be checked on site before duct work manufacturer is commenced.
- The whole of the duct work installation shall be carried out by an approved specialist in duct work manufacture and installation. No duct work shall touch the building structure or building finishes direct, but shall be isolated with insulating spacer.
- The fabrication shall be carried out in a neat and workmanlike manner with all duct work true in size and cross-section, braced and stiffened as specified and with all internal and external surfaces free from projections and sharp edges.
- Duct work shall be rigidly suspended or supported from building structure. Expansion type concrete inserts shall be placed so that the fastener is in shear rather than tension. Powder actuated fasteners placed by an explosive charge will not be accepted. Angel type trapeze hangers with rod supports spaced at 2 meters maximum shall be used. 'C' type beam clamps will not be accepted. Provide necessary steel angel iron required for bracing of duct work or equipment, and for supporting duct work from building structure.
- A layer of felt strip 12.5 mm compressed thickness shall be provided between any support member which is designed to clamp or grip the duct (e.g. circular duct band clip) or on which the duct is to rest. All supports shall be hot dip galvanized.
- Increase in duct size shall be gradual. Where width or largest dimension of a duct is over 450

mm, duct shall be stiffened by bending in a break across corners in both directions. Duct shall be self-supporting and complete in themselves. Single thickness partition between ducts will not to be accepted. Visible internal portions of duct outlets to grilles and registers shall be painted in dull back.

- All necessary allowances and provisions shall be made in the installation of the ducts for the structural framing of the building and when changes or offsets are necessary, the required cross-sectional areas shall be maintained. All of these changes however, shall be approved, and installed as directed at the time.
- During installation the open ends of ducts shall be protected with blank, flanged sheet metal baffles, securely attached to prevent debris and dirt from entering.
- Where ducts are shown connecting to masonry openings and/or along the edges of all plenums at floors, walls, etc., provide a continuous 30x30x40 mm galvanized angle iron which shall be bolted to the structure and made airtight to same by applying caulking compound on the angles before they are drawn down tight. The sheet metal at these locations shall be bolted to the angle iron framing.
- All air ducts, casings, plenums, etc., shall be constructed of lock forming quality prime galvanized steel sheets, which are free from blisters, silvers, pits, imperfectly coated spots, etc. No second quality sheet metal allowed.
- Where damaged (or rusting) has occurred on galvanized duct work the affected section shall be made good by painting with two coats of zinc-rich paint and approved finishing paint, or where the damage in the consultant's opinion cannot be made good, then a new section of duct work shall be provided at no cost to the contract.
- Duct shall be constructed using double of Pittsburgh lock corner seams. All seams shall be hammered down and made airtight by applying sealant before hammering down. For transverse joints, refer to current ASHRAE guide for low-pressure ductwork.
- Support the vertical ducts installed in the various shafts at each floor level with galvanized supporting irons of approved size. Install these angles across the width of the shaft, with their ends attached to angle irons securely anchored into the masonry walls of the shaft, or attached to the framing of the floor openings. The ducts shall be bolted to these supporting angle irons.
- Ensure that all openings required through floors, walls, partitions, etc., for the duct system are provided in the exact location required.
- Each piece of ductwork shall be wiped inside and out before installation and all open ends shall capped and sealed to prevent entrance of dirt during construction. Ensure that ductwork systems are clean and free from dirt, dust, grime, debris, etc., before initial operation of fans. Fans shall not be operated until the filters are installed and approval from the consultant has been obtained.
- The bottom joint and 150 mm of vertical joint on outside air intake ducts and mixing chamber ducts shall be soldered and made watertight. Provide drain connection and run copper drainpipe to nearest floor drain.
- All fixing devices including nuts, bolts, washers, etc, used in the construction or support of galvanized ductwork shall be Sherardised, or cadmium-plated where required by H.V.C.A/A.S.H.R.A.E. Specifications.
- All fastenings wholly or partly exposed inside ductwork shall be to H.V.C.A / A.S.H.R.A.E. Specifications.

FLEXIBLE DUCT WORK – LOW PRESSURE

- Flexible ductwork is shown diagrammatically on drawings and duct length shown may therefore bear no relation to actual length required. Allow sufficient length for connecting main duct to air distributing devices. Elbows shall have a central line radius of 2 times duct diameter. Length shall not exceed 3 meters per section.
- Flexible ductwork shall be manufacturer with a two ply Aluminium inner core, surrounded by 40 mm thickness of 12 Kg/M3 density fiberglass, all wrapped in a reinforced aluminium outer

jacket.
- Ductwork shall meet the standards of NFPA 90A, and be UL listed or to meet BS 476 and BS 413. Maximum flame spread rating – 12 (class 1). Maximum smoke developed rating – 50.
- Typical material, D.E.C. FLEXAL 40A or approved equal.
- Flexible ducts installed in an externally insulated duct system shall be factory insulated with glass Fiber insulation not less than 20 mm thick and a density not less than 24 kg/M3, re-covered with an acceptable vapour seal.
- Flexible ducts installed in internally (acoustic) insulated duct system, shall be factory insulated with glass fibre insulation not less than 20 mm thick and a density not less than 24 Kg/M3, faced on air side with PVC coated glass cloth having an open area not more than 25 %, and on room side with material specified above.
- Flexible duct installation shall be in accordance with manufacturer's instruction. Joint between factory insulated flexible ducts and field insulated ductwork shall be sealed and taped under this section.

FLEXIBLE CONNECTION

- Flexible connectors at inlet and discharged to air handling equipment shall be pre- assembled 24 gauge (0.7 mm) galvanized steel with minimum of 75 mm wide (exposed) fabric. For higher-pressure applications 25 mm of width of fabric shall be used for each 25 mm of static pressure.
- Flexible connectors attached to acoustically treated ductwork shall be insulated with 25 mm fiberglass insulation packed between flexible connector and 16 gauge (1.5 mm) galvanized steel housing. Housing shall be fastened to duct with sheet metal screws. At equipment collar caulk between collar and flange on housing with 10 mm thick permanently flexible sealant. Care shall be taken to ensure that ducts on both sides of the connection are independently supported and that no "bridging" occurs.
- All flexible connections will be by an approved manufacturer, with rating of fire spread, strength, etc., listed by a recognized testing authority.
- The fabric shall be rated for use up to 200 degrees F (93 degrees C) working temperature with a tensile strength of not less than 100 psi (690 kilopascals).
- The material shall also be impervious to moisture, dimensionally stable, and shall not rot.

ACCESS PANELS

- Duct access doors shall be minimum 450x300 mm. All access panels shall be constructed from double thickness 20 gauge (1 mm) galvanized steel sheets, 25 mm apart with necessary reinforcing inside for rigidly, with space filled with glass fiber insulation. Panels shall be made airtight with a continuous neoprene rubber gasket.
- Openings in ductwork shall be provided with continuous galvanized reinforcing bars, which on insulated ductwork, shall be extended to the face of the insulation. "Small" panels shall be provided with at least two brass window sash fasteners. All panels shall have brass drawer type handle.
- Apparatus access doors shall be minimum 600x1500 mm with angle or channel frame. Provide two 75 mm strap hinges with brass pins, and two handles minimum, operable from inside and outside. All access panels shall be constructed from double thickness
- 18 gauge (1.3 mm) galvanized steel, with 25 mm insulation between sides with necessary reinforcing for rigidly. Doors shall be made airtight with "tadpole" gaskets.
- Provide access panels where shown, required and directed and in the following locations:
- Bottom of all duct risers.
- Next to outside intakes and outlets. At each fire damper.
- Into apparatus casings to facilitate maintenance and cleaning of all components. Before each booster coil.

BALANCING DAMPERS

- Install volume dampers in accessible locations at all branch connections and wherever necessary to adjust the flow of air to secure correct, distribution. They shall be made of 20 gauge (1 mm) galvanized sheet metal and shall be equipped with an approved device for fastening in any desired position. This device shall such that the damper cannot move or rattle and pointer shall indicate the position of the damper from the outside of the finished duct insulation.
- Dampers shall be multi leaf opposed blade with blade height not more than 180 mm. PVC or similar blade seals shall be incorporated to the end of all blades.
- Dampers shall be non-leakage type with reference leakage at 500 Pascals pressure difference of not more than 35 M3/H. per M2.
- All blades shall be operated by a single operating quadrant with gears and links as necessary.
- Spindles shall be non-corrodible, passing through non-ferrous bushings or ball bearing supports with seals. The whole damper assembly shall be mounted in a galvanized frame with flanges.
- Seal material shall be rated up to 200 degrees F (93 degrees C) with low water absorption and excellent chemical resistance to acids, alkalis and oils.

BACKDRAFT DAMPERS (IF APPLICABLE)

- Back draft dampers shall be low leakage with parallel blades and neoprene edge seals.
- Dampers frames shall be constructed from galvanized sheet steel with aluminium blades. Bearing shafts shall be stainless steel, in brass bearings.
- All blades shall be coupled at the blade centres and shall be in width of not more than 1000 mm, with maximum blade size of 200 mm.
- Leakage shall not exceed 10 M3/H pre M2 at 1000 Pascals pressure differential.
- Pressure relief dampers shall be multi-parallel blade with weighted arm closing assist. The frame shall be anodized aluminium channel sections with formed aluminium blades. Maximum blade length shall be 100 mm, and polyester foam seating strips shall be incorporated on blade edges. Bearings shall be in PVC with non-corrodible shafts.

SPLITTER DAMPERS

- In each low-pressure system take off where opposed blade duct dampers are not specified, splitter dampers shall be provided. The only justification for not providing splitters is in a system, which can be demonstrated to be sized by a static regain programmer and which is run in high-pressure fittings.
- Splitters shall consist of hollow blades in a vane rail assembly, made from galvanized steel by a recognized manufacturer.
- A lockable quadrant adjustment lever shall be located outside the insulation, and marked clearly to show vane position.

FIRE DAMPERS

- Provide, where required by ordinances or codes, fire dampers made to British standards BS 476 or North American standards or NFPA 90A, and complete with angle iron frame, 160 degrees F (70 degrees C) fusible link, pivot rods, and spring catches. Fire damper housings shall be of non-ferrous metal. Enlarge duct sections around fire dampers, to allow 100% unrestricted duct area while in open position. Provide approved type access doors, with airtight gaskets, for inspection and servicing of fire dampers. Fire dampers shall be 2 hour rated and shall be labelled by manufacturer.
- Provided fire dampers in all ducts in the following locations, whether or not specifically required by ordinances and codes:
  - Duct entering and leaving fire shafts (duct shafts).
  - Ducts passing through designated fire walls.
  - Ducts through floors, and not encased in fire shaft.
  - Ducts leaving and entering plant area.

- Ducts entering and leaving storage areas.

- Where fire dampers are located remote from fire partition, duct between fire damper and partition shall be encased in double metal lathe and plaster or other fireproofing acceptable to authorities having jurisdiction. This shall be arranged and paid for by this trade unless otherwise noted on the drawings.
- Dampers shall be operated by two stainless steel springs with the blades being held in the open position by a fusible link and standard stainless steel strap. Blades shall be made from galvanized steel of 12 gauge (2.6 mm). blades shall overlap each other by 12 mm and shall close against a 25x25x3 mm angle iron stop fixed at all four sides.
- Fire dampers shall NOT be installed in kitchen exhaust ductwork. This shall comply in al respects with NFPA 95.

SILENCERS

- Silencers are not fully indicated on drawings since actual requirements for them depend on the choice of equipment to be installed and the base noise levels generated.
- The contractor is to maintain the noise levels as mentioned in clause 2.2.1.
- At shop drawing stage the contractor will undertake calculation for every unit and every area and determine the insertion loss required to meet the stated noise criterion. Supply and return duct noise must be considered as well as duct breakout noise.
- All the materials used in the silencer shall be inorganic, inert, rot, odour, and moisture proof. All metal shall be galvanized steel. Minimum acceptable thickness shall be 22 gauge (0.85 mm) for outer shells, with 24 gauge (0.7 mm) for inner shells, 75 mm slip joints will be possible at either and without modification. Inlets and outlets shall have a rounded profile acceptable to the consultant.
- Silencers shall be factory made, and shall have available certified test date concerning insertion loss. This must be available when required, in advance of ordering units. Where a silencer is made under licence, or part assembled locally, then tests must also be made after assembly and witnessed by an independent authority.
- The static pressure loss of any required silencer must be considered in air handling equipment original selection.

VIBRATION ISOLATION

- All plant and equipment shall be isolated form the building structure in such a manner that noise and vibration is not transmitted through the structure.
- Anit-vibration equipment will be manufactured by a specialist company acceptable to the consultant.

AIR DISTRIBUTION DEVICES

- Diffusers, registers and grilles shall be arranged for flush mounting in lay-in type ceilings and over lap mounting in plaster, mineral tile and similar ceilings, with concealed fixings unless otherwise directed.
- Grills, register and diffusers locations shall be adjusted to suit reflected ceiling drawings, or consultant's site instructions.
- Provide plaster frame for grills, and diffusers installed in plaster ceilings.
- All diffusers, grills and registers shall be powder coated aluminium, colour to architect's approval.
- All supply and extract grilles and diffusers will have opposed blade balancing dampers. All will have foam rubber sealing band around the edge to seal to the structure. All pivots will be round section, not of formed sheet, and not relying on a spring steel locking wire.
- Basic grills and diffusers materials shall be aluminium extruded sections. Sections in the airstreams shall be carefully selected to minimize turbulence.
- Diffusers with removable core shall have fixed blades for horizontal air supply and shall be sized for reasonable throw and velocity at maximum flow conditions, but with consideration to avoid 'dumping' at minimum flow conditions.

- Wall grills shall be double deflection with front bar across the shorter dimension, and shall have fully adjustable independent blades.
- Return grilles shall be single deflector.
- Linear grills shall be complete with plenum in short sections such that inlet velocities can be maintained at less than 4.5 M/sec., internal velocities at less than 3 M/sec. Plenums to be fully sound-lined.
- Slot diffusers shall be coordinated with the ceiling system and shall be complete with plenum in short sections. The diffuser shall be complete with air deflection blades, integral air straightener and hit and miss damper. On multi slot diffuser, each slot shall be adjustable. Slot diffuser construction shall be of extended aluminium sections. Plenum bases shall be acoustically lined.

INSULATION GENERAL

- All insulations shall be in accordance with British Standard specifications and codes of practice in all respects. Insulation meeting ASTM standards will be considered if properly submitted.
- In addition to complying with the relevant standards, all insulating materials shall be free from asbestos.
- Pipes and ducts passing through outside or unconditioned spaces must be insulated with the minimum insulation thickness specified in table 502.11(1) and (2) , section 5, green building regulations and specification –Dubai municipality
- Insulating materials shall be acceptable only if they are equal to or better than the grades or classes of fire resistance as follows:
    - BS 4735, class Q, (Burning rate nil, and not producing melted droplets).
    - BS 476, Part 4, Non-combustible grad).
    - BS 476, Part 5 Class P, (Not easily ignitable).
    - BS 476 Part 6, (Fire Propagation index a maximum of 12.6).
    - BS 476 Part 7, Class 1, (Surface spread of lame).
    - BS 476 part 9,
- (Production of emitted smoke shall not give more than 35% obscuration of the light beam).
- All insulation finishes and coverings shall be classified as Class 1 (one) surface spread when tested in accordance with B.S.476, Part 7.
- All adhesives, mastics, coatings, sealers and primers shall be classified as fire resistant class O as per building regulations section E-15.
- They shall not in any way attack the insulation or the surface to which the insulation is being applied and shall be suitable for the working temperatures.
- Any rough, irregular and badly finished surfaces shall be stripped down and re-insulated to the Consultant's satisfaction.
- All systems are to have been tested and approved by the Consultant prior to installation of insulation.
- Insulating materials shall have thermal conductivity values not more than those listed as follows:

MATERIAL	TYPE	THERMAL CONDUCTIVITY W/M CO
Mineral Wool	Sectional	0.036
Mineral Wool	Slabs	0.041
Fibre Glass	All	0.036
Rubber	Sectional	0.038
Polyurethane	Sectional	0.035
Styrofoam	Rigid	0.036

- All conductivity figures are rated at an average temperature of 24 deg C.
- All material delivered to site shall be new and fully dried out and so maintained throughout the progress of the works.
- All insulating materials shall be stored in storage shads, and in accordance with the

Manufacturer's recommendations.

- Particular attention shall be paid to the finished appearance of all thermal insulation, which must present a neat and symmetrical appearance running true in line, which pipe layouts, etc.
- Any rough, irregular and badly finished surfaces shall be stripped down and re-insulated to the Consultant's satisfaction.
- All systems are to have been tested and approved by the Consultant prior to installation of insulation.

CONCEALED COLD AIR DUCTS

- Insulate rectangular supply, return and extract air GI ductwork with 25 mm thick, 32 kg/M3 density rigid fibreglass board duct insulation with integral Aluminium foil vapour barrier. Fasten the insulation with adhesive on 200 to 250 mm centres. But all joints tightly and seal all breaks and joints by adhering a 75 mm wide Aluminium foil vapour barrier tape or sheet with a fire retardant adhesive.
- Apply on all insulations a 6 oz canvas adhered between a coat of fire resistance lagging, adhesive, as Benjamin Fosters ' Sealfas ' or approved Equal and a coat of vapour barrier over it.

EXPOSED COLD AIR DUCTS

- For GI ducts (supply, return and extract) exposed inside conditioned spaces, insulate as described above for concealed air ducts.
- Where exposed cols air ductwork runs through occupied or public areas, but is not exposed to foot traffic, accidental or intentional damage, the finished insulation is to be painted to an approved colour with a recommended paint. Where ducts are exposed to foot traffic, or damage, the exposed duct shall be clad with sheet Aluminium, of 22 gauge, (0.9 mm) or thicker.
- Pipes and ducts passing through outside or unconditioned spaces must be insulated with the minimum insulation thickness specified in table 502.11(1) and (2) , section 5, green building regulations & specification –Dubai municipality
- For ducts exposed in non air conditioned areas ,insulate using the method described for concealed ducts, but using insulation with a minimum thickness of 50 mm ,48 kg/M3 density rigid fibreglass board insulation .if necessary due to market availability , this may be installed in two layers ,but with each layer properly finished.
- Then apply a 6 oz canvas cover adhered between a coat of fire resistant lagging adhesive and a coat of vapour barrier over it.
- Then cover with sheet Aluminium, of 22 gauges, (0.9 mm) or thicker all ductwork exposed externally to the building is to be clad with aluminium of 22 U.S gauge (0.9 mm) or thicker.
- Where ducts penetrate the building shell, the duct shall be flashed waterproofed before any insulation is applied.

ACOUSTIC INSULATION

- Allow pressure duct work shall be lined with acoustic insulation up to the first take off point, or 3 meters from the fan/unit outlet.
- Duct acoustic lining shall be 25 mm thick ,fiberglass ,24kg/M3 density with a thermal conductivity of 0.035 Watts/M/ deg C with approved coating on inside.
- Liner will be attached by a fire resistant adhesive such as Benjamin Foster 81-99, or equivalent. in addition mechanical fasteners shall be used on 400 mm centres on top and side sections .these fasteners must be spot-welded to the inside of the duct and must not perforate the duct.
- All abutting edges must be caulked and at the extremities of lining shall have a sheet metal nosing of at least 40 mm length.

REFRIGERANT AND CONDENSATE DRAIN PIPEWORK

- Insulate suction lines of refrigerant piping and the condensate drain pipework with 19 mm thick proprietary, closed cell rubber insulation.
- For piping on roof or other areas subject to mechanical damage, provide 60z canvas cloth with two coats of adhesive and apply aluminium cladding of 22 US gauge (0.9 mm) or thicker over insulation described above.

SMOKE MANAGEMENT SYSTEMS

- The Contractor shall supply, install, test, commission and set to work a complete and fully operational smoke management system in accordance with the true intent of this particular specification, the standard specification and as indicated on the design tender drawings an equipment schedules.
- The smoke management systems shall be in accordance with the Fire & Life Safety requirements
- and the Fire Strategy document approved by Civil Defence.

STAIRCASE PRESSURISATION

- In the event of a fire situation all staircases shall be mechanically pressurized in accordance with NFPA standards, Dubai Civil Defence and UAE Fire and Life Safety Code of Practice requirements.
- In general the staircases shall be pressurized with un-tempered outdoor air supplied to each staircase to maintain a positive pressure relative to the adjacent areas, with all doors closed. The pressurization system shall also be capable of maintaining efflux velocities through the 3 No open doors during operation. The pressure differential shall be 12.5Pa as per a sprinklered buildings requirement.
- The pressurization fan sets shall be located within each of the staircases at the high level of the top floor. Outdoor air louvers to serve the fans intake shall be provided within the staircase wall at the top floor.
- The fans shall be in line axial flow, encased in circular casing. The fans shall be capable of handling high temperature smoke up to 400°C for at least 2 hours. The fans shall operate at variable speed to maintain the required pressure differential. Two fans shall be installed as duty and standby where shown. These fans shall be hard-wire interlocked with the associated fire alarm system. The pressurization fan sets and system shall be designed to satisfy the requirements of NFPA and Dubai Civil Defence requirements.
- The secondary power supply to the duty & standby fan arrangement shall be capable of providing the power supply within 15 seconds of failure of the primary power. The standby generator shall be capable of providing power for at least 3 hours without the replenishment of fuel. The primary and secondary power supply cables shall be fire rated MICC and shall terminate in the at the control panel. This panel shall include a changeover device to automatically effect the transition from primary to secondary power supply if the primary supply fails. Similarly changeover of fans shall be automatic upon failure of the duty fan.
- Outdoor air shall be distributed via builders work shafts adjacent to each staircase and be fed into the staircase via supply grille provided every second floor as detailed in the design drawings and equipment schedules. The contractor shall ensure the air-tightness of the shafts.
- To maintain the required pressure within the stair, air pressure relief dampers shall be provided as a failsafe system with instant reaction time. The dampers shall relief excess volumes of air from protected stair cases. The dampers shall be positioned as indicated in the drawings discharging directly to outside. The damper shall consist of brush polished stainless steel blades pivoted on sealed for life ball bearing, each with its own balance weight ssembly, fixed into aluminum powder coated casing. The damper shall be factory tested for pre-set specified operating pressure and shall be able to maintain the operating pressure to an

accuracy of ± 2 Pascals. Dampers shall be easily adjustable on site should the need arise. The size of the damper shall be as indicated on the drawings or as necessary to maintain the specified pressure.

- 8. Smoke sensors shall be installed in the outdoor air intake plenum / duct to ensure that only clean outside air is supplied to the staircases.

CONTROLS

- The stair pressurization controls shall be hard-wired to a) the Fire Alarm system and operate via signals from the Fire Alarm Panel.
- If a fire is sensed in the building the fire alarm panel shall provide signals to switch on the pressurization through approved fire rated cables as described elsewhere Gravity operated pressure relief dampers shall operate if pressure in the staircase exceeds a set predetermined (to be confirmed on site) level, say 50Pa.
- Key operated "H.O.A." switches shall be provided for each fan on the control panel. The Hand position is only for testing purposes. In the Auto position, the fan(s) shall be started 'ON' on receipt of a fire signal from the fire alarm panel.
- In case the duty fan fails to start, trips or the duty fan "H-O-A" switch is not in the Auto position the standby fan shall start automatically.
- Fireman's switch(s) shall be provided at the ECC room at ground floor level for manual operation of staircase pressurization by fire authorities. This switch shall enable the fireman to switch each staircase pressurization fan ON/OFF.
- These switches shall override all controls referred above (except overload and H-O-A in OFF position). The final location of the switches shall be in accordance with the regulations and requirements of the local authority having jurisdiction.
- A motorized low leakage inlet damper shall be installed for each set of pressurization fans as indicated on the drawings. The damper actuator shall be spring return type and the damper shall be held in closed position during normal operation. In case of a signal from the fire alarm system or loss of power, the damper shall open. It shall close automatically if the power is restored or the fire alarm signal is reset.
- Smoke sensors mounted in pressurization air intake paths shall give an alarm and trip the fans if smoke is sensed at the inlet of pressurization fans.

LIFT SHAFT PRESSURISATION

- In the event of a fire situation all passenger lift shafts opening to residential corridors shall be mechanically pressurized in accordance with NFPA standards, DCD and UAE Fire and Life Safety Code of Practice requirements.
- In general the lift shafts shall be pressurized with un-tempered outdoor air supplied to each shaft to maintain a positive pressure relative to the adjacent areas. The pressure differential shall be 12.5Pa as per a sprinklered buildings requirement.
- Outdoor air shall be supplied into each lift shaft via single injection point. Each lift shaft shall be served by a separate dedicated pressurization fan set located within mechanical plant areas on top levels of each core, in the vicinity of the lift shaft. For locations of fans for each of the lift shafts refer to mechanical floor layout drawings.
- The fans shall be in line axial flow, encased in circular casing. The fans shall be capable of handling high temperature smoke up to 400°C for at least 2 hours. The fans shall operate at variable speed to maintain the required pressure differential. Each set shall comprise two fans installed to operate in a duty and standby arrangement. These fans shall be hardwire interlocked with the associated fire alarm system. The pressurization fan sets and system shall be designed to satisfy the requirements of BS or NFPA and Dubai Civil Defence requirements.
- The secondary power supply to the duty & standby fan arrangement shall be capable of providing the power supply within 15 seconds of failure of the primary power. The standby

generator shall be capable of providing power for at least 3 hours without the replenishment of fuel. The primary and secondary power supply cables shall be fire rated MICC and shall terminate in the at the control panel. This panel shall include a changeover device to automatically effect the transition from primary to secondary power supply if the primary supply fails. Similarly changeover of fans shall be automatic upon failure of the duty fan.

- To maintain the required pressure within the lift shaft, air pressure relief dampers shall be
- provided as a failsafe system with instant reaction time. The dampers shall relief excess volumes of air from protected lift shafts. The dampers shall be positioned as indicated in the drawings discharging directly to outside. The damper shall consist of brush polished stainless steel blades pivoted on sealed for life ball bearing, each with its own balance weight assembly, fixed into aluminum powder coated casing. The damper shall be factory tested for pre-set specified operating pressure and shall be able to maintain the operating pressure to an accuracy of ± 2 Pascals. Dampers shall be easily adjustable on site should the need arise. The size of the damper shall be as indicated on the drawings or as necessary to maintain the specified pressure.
- Smoke sensors shall be installed in the outdoor air intake plenum / duct to ensure that only clean outside air is supplied to the lift shaft.

CONTROLS

- The lift pressurization controls shall be hard-wired a) to the Fire Alarm system and operate via signals from the Fire Alarm Panel.
- If a fire is sensed in the building the fire alarm panel shall provide signals to switch on the pressurization through approved fire rated cables as described else where
- Gravity operated pressure relief dampers shall operate if pressure in the lift shaft exceeds a set predetermined (to be confirmed on site) level, say 50Pa.
- Key operated “H.O.A.” switches shall be provided for each fan on the control panel. The Hand position is only for testing purposes. In the Auto position, the fan(s) shall be started ‘ON’ on receipt of a fire signal from the fire alarm panel.
- In case the duty fan fails to start, trips or the duty fan “H-O-A” switch is not in
- the Auto position the standby fan shall start automatically.
- Fireman’s switch(s) shall be provided at the ECC room at ground floor level for manual operation of lift shaft pressurization by fire authorities. This switch shall enable the fireman to switch each lift shaft pressurization fan ON/OFF.
- These switches shall override all controls referred above (except overload and H-O-A in OFF position). The final location of the switches shall be in accordance with the regulations and requirements of the local authority having jurisdiction.
- f) A motorized low leakage inlet damper shall be installed for each set of pressurization fans as indicated on the drawings. The damper actuator shall be spring return type and the damper shall be held in closed position during normal operation. In case of a signal from the fire alarm system or loss of power, the damper shall open. It shall close automatically if the power is restored or the fire alarm signal is reset.
- Smoke sensors mounted in pressurization air intake paths shall give an alarm and trip the fans if smoke is sensed at the inlet of pressurization fans.

CORRIDOR SMOKE MANAGEMENT SYSTEM – RESIDENTIAL FLOOR CORRIDORS

- The corridor smoke extracts system for residential floors shall be provided as per DCD requirement. The Contractor shall supply, install, test and commission the corridor smoke extracts system as indicated on the drawings, equipment schedules and specified herein. The complete installation shall comply with NFPA and UAE Fire Code requirements.
- The fresh air and extract air systems are dedicated to provide to extract smoke and supply

make up air to the residential floors corridor in event of fire.

- Each building shall be provided with a dedicated system with duty and standby extract/supply fans. The fans shall be located on the roof level as indicated on the drawings.
- Dedicated ductwork risers shall be provided for each of the systems to distribute vertically and provide extract/supply connections at each residential level.
- Extract points shall be located at high level in form of a bell mouth extracting from the ceiling void plenum. Air make-up point shall be located at low level within the riser wall adjoining to the corridor. Extract and make-up points shall be interchanging and separated by maximum 15m.
- The fresh air duct / extract air duct connection to the residential corridor is provided with motorised smoke & fire damper that will open & close in the event of fire.
- The system shall operate to provide a negative pressure in the corridor of the fire affected floor only.
- The secondary power supply from generator shall be provided to all corridor smoke management fans. The primary and secondary power supply cables shall terminate in the plant room at the wall mounted control panel. This shall include a changeover device to automatically effect the transition from primary to secondary power supply if the primary supply fails. Smoke sensors shall be installed in the outdoor air intake duct so that only clean outside air is supplied.

CONTROLS

- The corridor smoke extracts system fans shall be a) variable speed type. A motorised low leakage inlet damper shall be installed for each fan as indicated on the drawings. The damper actuator shall be spring return type and the damper shall be held in closed position during normal operation. In case of a signal from the fire alarm system or loss of power, the damper shall open. It shall close automatically if the power is restored or the fire alarm signal is reset.
- Key operated “H.O.A.” switches shall be provided for each fan on the control panel. The Hand position is only for testing purposes. In the Auto position, the fan(s) shall be started ‘ON’ on receipt of a fire signal from the fire alarm panel.
- In case the duty fan fails to start, trips, fails to register airflow (from an airflow switch) after a predetermined time delay or the duty fan “H-O-A” switch is not in the Auto position the standby fan shall start automatically.
- The following indications shall be provided on the panel:
  1. Power on
  2. Fan Run - each fan
  3. Fan Trip - each fan
  4. Fan airflow – each fan
- Smoke sensors mounted in air intake path shall give an alarm and trip the fans if the smoke is sensed at the inlet of fans.
- The MD/ MSD (Motorized smoke damper) on the fresh air duct/ extract air duct connection to each floor shall be normally closed and powered to open position. All MD/MSDs (Motorized smoke dampers) shall have spring return facility and shall be UL listed for such use. In case of fire, a signal from fire alarm system shall open the branch dampers and close the dampers on normal ventilation system
- Fireman’s switch(s) shall be provided at the ECC room at ground floor level for manual operation fan by fire authorities. This switch shall enable the fireman to switch fan ON/OFF. These switches shall override all controls referred above (except overload and H-O-A in OFF position). The final location of the switches shall be in accordance with the regulations and requirements of the local authority having jurisdiction

SMOKE EXTRACT SYSTEM

- Smoke extract systems shall be provided in areas where required by DCD and the Fire and Life Safety Consultant. The served areas shall include: Fire Pump room at Basement a) level of each plot
- The smoke extract fans shall be located within mechanical plant rooms at the basement level and discharge smoke directly to outside via an exhaust louver.
- The system shall be capable of smoke clearance mode at 9 air changes per hour from from the fire pump room. On detection of fire the normal ventilation systems will stop, including the adjacent areas and the car park ventilation, and motorized dampers will be used to extract the smoke from the zone on fire mode.
- The fans shall be in line axial/mixed flow. The fans shall be capable of handling high temperature smoke up to 400°C for at least 2 hours. The fan motor shall be sized to overcome the resistance of the duct network. The fans shall operate at variable speed. Two fans shall be installed as duty and standby where shown. These fans shall be hardwire interlocked with the associated fire alarm system. The smoke extract fan sets and system shall be designed to satisfy the requirements of BS or NFPA and Dubai Civil Defence requirements.
- The secondary power supply to the duty & standby fan arrangement shall be capable of providing the power supply within 15 seconds of failure of the primary power. The standby generator shall be capable of providing power for at least 3 hours without the replenishment of fuel. The primary and secondary power supply cables shall be fire rated MICC and shall terminate in the at the control panel. This panel shall include a changeover device to automatically effect the transition from primary to secondary power supply if the primary supply fails. Similarly changeover of fans shall be automatic upon failure of the duty fan.

CONTROLS

The smoke extract controls shall be hard-wired to the Fire Alarm system and operate via signals from the Fire Alarm Panel.

- If a fire is sensed in the smoke zone the fire alarm panel shall provide signals to switch on the smoke extract through approved fire rated cables as described elsewhere
- In the event of a fire being detected the motorized smoke and fire damper on the fire zone only shall open. All other motorized smoke and fire dampers shall remain closed.
- The damper actuator shall be spring return type and the damper shall be held in closed position during normal operation. In case of a signal from the fire alarm system or loss of power the damper shall open. It shall close automatically if the power is restored or fire alarm signal is reset.
- The duty fan shall be selected at the control panel using duty selector switch. In case the selected duty fan is not available for any reason, the controls will automatically start the standby fan. A key operated Hand-Off-Auto switch shall be provided for each fan. The Hand position is only for testing the fan. In the Auto position, the fan shall be started "ON" on receipt of a fire signal from the fire alarm panel.
- Fireman’s switch(s) shall be provided at the ECC room at Ground Level for manual operation of smoke extract fans by fire authorities. This switch shall enable the fireman to switch each smoke extract fan ON/OFF. These switches shall override all controls referred above (except overload and H-O-A in OFF position). The final location of the switches shall be in accordance with the regulations and requirements of the local authority having jurisdiction
- A motorized low leakage inlet damper shall be installed for each set of smoke extract fans as indicated on the drawings. The damper actuator shall be spring return type and the damper shall be held in closed position during normal operation. In case of a signal from the fire alarm system or loss of power, the damper shall open. It shall close automatically if the

power is restored or the fire alarm signal is reset.

CAR PARK VENTILATION AND SMOKE EXTRACT SYSTEM

- The enclosed car park area on podium levels Plots shall be ventilated in accordance with Dubai Civil Defence requirements.
- Each car park floor shall be a fire zone.. Each of the extract points shall be inclusive of a set of fire rated extract fans in a duty & standby arrangement which shall be located in dedicated fan rooms. Associated control panels shall be positioned within the rooms.
- The fan rooms shall be positioned directly below plenums of external exhaust louvers located above ground where the air/smoke shall be discharged.
- The extract fans shall be installed as N duty + 1 standby arrangement as shown on the design.
- Extract ducts shall be distributed throughout the car park levels and terminate with a number of extract air/smoke inlets. 50% of the inlets shall be positioned at high level and 50% of the inlets at low level. Each inlet shall be provided with a grille complete with a VCD for balancing.
- Extract ducts from each floor shall be provided with a motorized smoke damper when connecting to the main extract shaft/plenum.
- All car park extract ductwork shall be fire rated.
- Make-up air to the car park areas shall be provided mechanically via dedicated fan sets.
- The fan rooms shall be positioned directly below plenums of external intake air louvers above ground and shall supply air via dedicated shafts. The make-up air be supplied to the parking areas at low velocity directly via louvers on intake shaft walls / fan room walls.
- The supply fans shall be installed as duty only. There shall be 2 fans per each set/room with each of the fans having 50% of the total capacity.
- The mechanical make-up air intake will be supplemented by natural air movement via the car park entrance/exit ramp.
- All car park fans shall be in line axial/mixed flow, encased in square/circular casing. The fans shall be capable of handling high temperature smoke up to 400°C for at least 2 hours. The fan motor shall be sized to overcome the resistance of the duct network and the sound attenuators. The fans shall operate at variable speed.
- The fans shall be hard-wire interlocked with the associated fire alarm system. The extract fan sets and system shall be designed to satisfy the requirements of NFPA and Dubai Civil Defence requirements.
- The secondary power supply to the duty & standby fan arrangement shall be capable of providing the power supply within 15 seconds of failure of the primary power. The standby generator shall be capable of providing power for at least 3 hours without the replenishment of fuel. The primary and secondary power supply cables shall be fire rated MICC and shall terminate in the at the control panel. This panel shall include a changeover device to automatically effect the transition from primary to secondary power supply if the primary supply fails. Similarly changeover of fans shall be automatic upon failure of the duty fan.
- Each of the fan sets shall be provided with inlet and outlet sound attenuators to achieve the noise levels specified elsewhere.
- The car park ventilation system shall be controlled by a CO-monitoring system to optimize efficiency. During normal conditions the variable speed fans shall adjust their operation speed the during normal car park operation according to CO concentration within the basement.
- CO-monitoring devices shall be mounted remotely throughout the car parks. The devices shall be mounted in accordance with the manufacturer’s guidelines but no greater than 1.5m above FFL. Protective guards shall be provided to detectors to prevent damage.
- The contractor should refer to the specific requirements of the extract fan supplier for

quantities of co-detectors.

CONTROLS

- The car park extract fan controls shall be hard-wired a) to the Fire Alarm system and operated via signals from the Fire Alarm Panel and monitored by the BMS.
- All fans shall have the facility to be controlled with reference to the interfaces provided to the fire alarm system and CO detection systems.
- For each fan a mode selection switch shall be provided on the MCC i.e. "Hand, Off, Auto" switch. In 'Auto' mode the complete operation of the fans shall be carried out by the fire alarm system, i.e. control set points, hours run, auto changeover (twin fan only) etc.
- The duty fan shall be selected at the control panel using duty selector switch. In case the selected duty fan is not available for any reason, the controls will automatically start the standby fan. A key operated Hand-Off-Auto switch shall be provided for each fan. The Hand position is only for testing the fan. In the Auto position, the fan shall be started "ON" on receipt of a fire signal from the fire alarm panel.
- Fireman's switch(s) shall be provided at the ECC room e) at Ground Level for manual operation of car park extract fans by fire authorities. This switch shall enable the fireman to switch each smoke extract fan ON/OFF. These switches shall override all controls referred above (except overload and H-O-A in OFF position). The final location of the switches shall be in accordance with the regulations and requirements of the local authority having jurisdiction
- A motorized low leakage inlet damper shall be installed for each extract and intake fan as indicated on the drawings. The damper actuator shall be spring return type and in case of a signal from the fire alarm system or loss of power the damper shall open. It shall remain open until the power is restored and/or the fire alarm signal is reset.
- During normal mode, the car park extract fans shall operate at low speed to maintain the CO concentration below 25ppm and to avoid still air throughout the entire car park. In general, this shall result in about 3 air changes per hour.
- In the event the pollution CO levels rise to between 30-75ppm, the car park extract fans shall increase in speed & shall operate to achieve approximately 6 air changes per hour per underground car park level that has increased pollution. Once the CO concentration drops below 25ppm the system shall revert to the normal mode of operation.
- If a fire is sensed in the car park the fire alarm panel shall provide signals to switch on the smoke clearance mode. The motorized smoke dampers on extract and intake points shall remain open for the floor on fire mode only and shall close for the other floor. The fan associated with the zone on fire shall continue to run and adjust the speed to achieve extract rate of 9 air changes per hour. The system shall operate in this condition until reset manually.
- The make-up air fans shall be controlled in conjunction with the car park extract fans.

FIREMAN'S OVERRIDE SWITCH PANEL

- A fireman's override facility shall be provided for building to enable manual operation of each fan/ AHU as noted below to override any control function from the fire alarm system or BMS.
- The fireman's switch shall be located in close priority to the fire alarm panel within ECC room on Ground Level. The override switch shall enable each of the following to be switched "on" or "off", overriding any other on or off control input from the BMS or fire alarm: -
- a) Staircase pressurization fans
- b) Lift shaft pressurization fans
- c) Corridor smoke management system fans
- d) Fire pump room smoke extract fans

- e) All fresh air handling units
- f) All extract fans
- g) Kitchen extract fans
- The fireman's switches shall be incorporated in a wall amounted panel and shall include

the following

- a) Lockable glass cover, inhibiting access to switches.
- b) Run and fault indicator lamps / LED's for each fan/ AHU as noted in above.
- c) Three position switch to control each fan or AHU. Each switch shall have "auto"
- central position (for control by BMS or fire alarm), 'on' position and "off" position.
- The panel shall include a clear display by mimic/ schematic or equivalent to clearly indicate the location and function of each fan or AHU.
  I. Motorized smoke and fire dampers
- The motorised smoke and fire dampers (MSFDs) shall be controlled by the fire alarm system, and monitored by the BMS, as indicated on the drawings and elsewhere in the Specifications. The Contractor shall use fire rated cabling (i.e. FP200) for operation and monitoring of these Dampers. The MSFDs shall automatically revert to their normal position after the fire alarm has been reset or power restored. In case the MSFDs are not in their normal operating positions, an alarm shall be raised at the BMS.
- Motorised smoke and fire control dampers shall be installed in locations as indicated on the contract drawings and as required by NFPA and Dubai Civil Defence. They shall also be provided on all HVAC ductwork to clean agent protected rooms. The damper control mode shall operate on 24 volt and is to be located external to the ductwork, and protected within a thermally insulated enclosure. All control mode cables shall be halogen free low smoke and fume cables. Two separate volt free contacts shall be provided for external damper status indication. The damper operation shall be a motor closed spring open type and shall Interface with the fire alarm system. The Contractor shall allow for interfacing all smoke dampers with the fire alarm system, including all necessary interface modules, wiring, panels, etc.

SECTION 3

DUCTWORK AND AIRSIDE EQUIPMENT

SECTION 3 – DUCTWORK AND AIRSIDE EQUIPMENT

INDEX

DUCTWORK MATERIAL.

- All ductwork material except that for Commercial Kitchens and smoke extract ducts shall be manufactured from strip mill cold reduced sheet, continuously hot dip galvanized in accordance with ASTM A 653 and 653M (lock forming quality) with zinc coating grade of G90 (1.25 oz/sqft nominal).
- All flanges and stiffeners used in the construction of galvanized mild steel ductwork shall be galvanized steel section.
- This Section includes insulation for ducts, semi rigid and flexible duct, plenum, field-applied jackets, accessories, and attachments. The insulation material must be Flexible Elastomeric Foam made of NBR with closed cell technology with 18 microns Aluglass facing that is factory laminated. The insulation shall be manufactured as per ASTM C534/C534M: Grade 1 Standard specification for performed flexible elastomeric cellular thermal insulation in sheet and tubular form.

- Smoke extract ductwork shall conform to the requirements of BS 5588. Ductwork shall be tested to BS 476 for stability, integrity and insulation for a period of two hours for both inside and outside fire condition.

- All commercial kitchens extract duct work shall be of 16g black steel to BS 2989. Ductwork shall be air and water tight welded construction. Access panels shall be provided throughout the length at 10 ft intervals. In exposed conditions within kitchens ductwork shall be 20g polished stainless-steel construction. All black steel ductwork shall be provided with two coats of heat resistant paint for protection against corrosion.

DUCTWORK SPECIFICATION.

- Ductwork fabrication and hanging methods shall be as described in the "Equipment" Volume of the American Society of Heating, Refrigerating and Air Conditioning Engineers Guide and Data Book, latest edition, and / or current edition of applicable manuals published by the Sheet Metal and Air Conditioning Contractors National Association Inc., where methods described in these volumes are not at variance with the requirements of any authority having jurisdiction, or do not conflict with methods described hereafter. Ductwork shall not be shop fabricated until work has been job measured and interference situations coordinated.

- Any Contractor who is more familiar with British Standards may refer to B.S. CP. 352 and DW. 144. Any discrepancy will be clarified by the Consultant on request.
- In any event, the standard of work required is the best available, and the specifications are to be so interpreted.

DUCTWORK GENERAL.

- All dimensions shall be checked on site before ductwork manufacture is commenced.

- The whole of the ductwork installation shall be carried out by an approved specialist in ductwork manufacture and installation. No ductwork shall touch the building structure or building finishes direct but shall be isolated with insulating spacer.
- The fabrication shall be carried out in a neat and workmanlike manner with all ductwork true in size and cross-section, braced and stiffened as specified and with all internal and external surfaces free from projections and sharp edges.

- At each main branch in ductwork and at each fan discharge and suction, provide sufficient number of Pitot tube test holes for balancing systems. Also provide test holes for traverse fan discharge and at all equipment. Test holes shall be located within easy reach of catwalks or ladder. Each test hole shall have 20 mm clear opening, provided with a metal ring plate with a threaded hole in the boss, and matching screwed head plug. Where these plugs are installed in insulated ductwork, provide an extension collar against which the insulation can be finished.

- Reinforced holes shall be provided where thermometers, manometers, thermostats, gauges, damper rods, etc., occur in ductwork. Extended collars shall be provided for the reinforced holes where these occur on insulated ductwork. Where copper tubing passes through ductwork, or casing, provide a rubber grommet to prevent damage to copper tubing.

- Ductwork shall be rigidly suspended or supported from building structure. Expansion type concrete inserts shall be placed so that the fastener is in shear rather than tension. Powder actuated fasteners placed by an explosive charge will not be accepted. Angle type trapeze hangers with rod supports spaced at 2 meters maximum shall be used "C" type beam clamps will not be accepted. Provide necessary steel angle iron required for bracing of ductwork or equipment, and for supporting ductwork from building structure. A layer of felt strip 12.5 mm compressed thickness shall be provided between any support member which is designed to clamp or grip the duct (e.g. circular duct band clip) or on which the duct is to rest. All supports shall be hot dip galvanized.

- Increase in duct size shall be gradual. Where width or largest dimension of a duct is over 450 mm, duct shall be stiffened by bending in a break across corners in both directions. Ducts shall be self-supporting and complete in themselves. Single thickness partition between ducts will not be accepted. Visible internal portions of duct outlets to grilles and registers shall be painted in dull black.

- All necessary allowances and provisions shall be made in the installation of the ducts for the structural framing of the building and when changes or offsets are necessary, the required cross-sectional areas shall be maintained. All of these changes, however, shall be approved, and installed as directed at the time.

- During installation the open ends of ducts shall be protected with blank, flanged sheet metal baffles, securely attached to prevent debris and dirt from entering.

- Where ducts are shown connecting to masonry openings and / or along the edges of all plenums at floors, walls, etc., provide a continuous 30 x 30 x 4 mm galvanized angle iron which shall be bolted to the structure and made airtight to same by applying caulking compound on the angles before they are drawn down tight. The sheet metal at these locations shall be bolted to the angle iron framing.

- All air ducts, casings, plenums, etc., shall be constructed of lock forming quality prime galvanized steel sheets, which are free from blisters, slivers, pits, imperfectly coated spots, etc. No second quality sheet metal allowed. Where damage (or rusting) has occurred on galvanized duct work the affected section shall be made good by painting with two coats of zinc-rich paint and approved finishing paint, or where the damage in the Consultant's opinion cannot be made good, then a new section of ductwork shall be provided at no cost to the contract.
- Duct shall be constructed using double or Pittsburgh lock corner seams. All seams shall be

hammered down and made airtight by applying sealant before hammering down. For transverse joints, refer to current ASHRAE guide for low pressure ductwork.

- Support the vertical ducts installed in the various shafts at each floor level with galvanized supporting irons of approved size. Install these angles across the width of the shaft, with their ends attached to the angle irons securely anchored into the masonry walls of the shaft, or attached to the framing of the floor openings. The ducts shall be bolted to these supporting angle irons.

- Ensure that all openings required through floors, walls, partitions, etc., for the duct system are provided in the exact location required.

- Each piece of ductwork shall be wiped inside and out before installation and all open ends shall be capped and sealed to prevent entrance of dirt during construction. Ensure that ductwork systems are clean and free from dirt, dust, grime, debris etc., before initial operation of fans. Fans shall not be operated until the filters are installed and approval from the Consultant has been obtained.

- The bottom joint and 150 mm of vertical joint on outside air intake ducts and mixing chamber ducts shall be soldered and made watertight. Provide drain connection and run copper drain pipe to nearest floor drain.

- All fixing devices including nuts, bolts, washers, etc., used in the construction or support of galvanized ductwork shall be sherardized, or cadmium-plated.

- Final connections to diffusers shall be carried out using flexible ductwork.

LOW PRESSURE DUCTWORK.

- This applies to ductwork with mean velocities less than 2,000 fpm (10 M/Sec) and static pressures of 2 inches water gauge, (500 Pascal's) or less.

- Rectangular low-pressure ductwork shall be fabricated from prime quality; re-squared, tight coat galvanized steel sheets of the following gauges.

Size of Duct (Millimetres)	Gauge of Sheet Metal
300 mm less in width/depth	No.26 US Std. Gauge (0.6 mm) 300
to 750 mm width/depth	No.24 US Std. Gauge (0.7 mm)
750 to 1350 mm width/depth	No.22 US Std. Gauge (0.85 mm)
1350 to 2100 mm width/depth	No.20 US Std. Gauge (1.00 mm)
2100 mm and over	No.18 US Std. Gauge (1.3 mm)

- Low pressure suction and discharge plenum chambers shall be fabricated from 1.3 mm (18 gauges) galvanized steel with galvanized angle iron framework and bracing.

- In square elbows and in elbows where radius is less than 1-1/2 x width of duct, sheet metal deflector vanes shall be installed the full height of the duct, being securely riveted in place. All vanes shall be double thickness vanes of two gauges heavier than the duct in which they are installed, and shall be factory made, not site fabricated. Vanes shall be tack welded to vane rail. For vane lengths over 1000 mm, tack weld vanes to 10 mm tie- rod at mid-span.

- Circular low-pressure ductwork shall be fabricated from prime quality as per DW 144 or SMACNA Standards.

- Exposed circular low-pressure ductwork shall be double wall duct with 30% perforated inner shell and with 25mm thick, 25kg/m3 density fiberglass insulation

FLEXIBLE DUCT WORK - LOW PRESSURE.

- Flexible ductwork shall be manufactured with a two-layer aluminized polyester inner core, surrounded by 25 mm thickness of 25kg/m3 density fiberglass, all wrapped in a reinforced multilayer aluminium polyester outer jacket.
- Ductwork shall meet the standards of NFPA 90A and be U.L. listed or to meet BS 476 and BS 413. Maximum flame spread rating - 12 (Class 1). Maximum smoke developed rating - 50.
- Duct clamps shall be made of an 8MM wide stainless-steel band with 'Lifted' edges to avoid damage to the duct. The clamp shall be provided with a 'flip up' and 'quick lock'. Tightening head for ease of firing and speed of application.
- Flexible duct installation shall be in accordance with manufacture's instruction. Joints between factory insulated flexible ducts and field insulated ductwork shall be sealed and taped under this section.

FLEXIBLE CONNECTION

- Flexible connectors at inlet and discharge to air handling equipment (AHU / FCU) shall be pre-assembled 24-gauge (0.7 mm) galvanized steel with minimum of 75 mm wide (exposed) fabric. For higher pressure applications 25 mm of width of fabric shall be used for each 25 mm of static pressure.
- Flexible connectors attached to acoustically treated ductwork shall be insulated with 25 mm fibreglass insulation packed between flexible connector and 16 gauges (1.5 mm) galvanized steel housing. Housing shall be fastened to duct with sheet metal screws. At equipment collar caulk between collar and flange on housing with 10 mm thick permanently flexible sealant. Care shall be taken to ensure that ducts on both sides of the connection are independently supported and that no "bridging" occurs.
- All flexible connection will be by an approved manufacturer, with ratings of fire spread, strength, etc., listed by a recognized Testing Authority. All flexible connectors shall be UL certified.
- The fabric shall be rated for use up to 200 degrees F (93 degrees C) working temperature with a tensile strength of not less than 100 psi (690 Kilopascals).
- The material shall also be impervious to moisture, dimensionally stable, and shall not rot.

ACCESS PANELS

- Duct access panels shall be minimum 450 x 300 mm. unless restricted by duct dimensions. All access panels shall be constructed from galvanized steel sheets, 25 mm apart with necessary reinforcing inside for rigidity, with space filled with glass fibre insulation. Panels shall be made airtight with a continuous neoprene rubber gasket.

- Openings in ductwork shall be provided with continuous galvanized reinforcing bars, which on insulated ductwork, shall be extended to the face of the insulation. 'Small' panels shall be provided with at least two self-tightening camlocks. All panels shall have drawer type handle.

- Apparatus access doors shall be minimum 600 x 1500 mm with angle or channel frame. Provide two 75 mm strap hinges with brass pins; and two handles minimum, operable from inside and outside. All access panels shall be constructed from double thickness
- 18 gauge (1.3 mm) galvanized steel, with 25 mm insulation between sides with necessary reinforcing for rigidity. Doors shall be made airtight with "tadpole" gaskets.

- Provide access panels where shown, required and directed and in the following locations:
    - Bottom of all duct risers.
    - Next to outside air intakes and outlets.
    - At each fire damper.
    - Into apparatus casings to facilitate maintenance and cleaning of all components.

LOCAL GUIDELINES

- ASTM E84 (ANSI/UL 723): Test method for surface burning characteristics of building materials.
- DCD/ ADCD: Civil Defense certificate with listing of product being proposed.
- DCL: For the emirate of Dubai, 2020 Al Sa'fat Green Building System certification from Dubai Central Laboratory Department.

FACTOY APPROVALS

- ISO 9001-2015: Manufacturing and Trading Thermal and Acoustic Insulation Materials in the Shapes of Pipes, Sheets, Rolls, Tapes and their Accessories and trading of fire Stopping Materials.

INTERNATION ACCREDITIONS

- REACH Directive (EC) 1907/2006
- Singapore Green Building Council Product Certificate
- FM 4924: Material classification and full-scale test requirements for insulation material used on the exterior of non-combustible pipes and ducts.
- Environmental Product Declaration in accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for the insulation.

PRODUCT DATA

- For flexible elastomeric foam (FEF), including conductivity, thickness, jacketing if needed, flame spread, and smoke development indices.

DETAIL INSTALLATION OF INSULATION FOR THE FOLLOWING

- Installation for air ducts, including outer insulation and duct lining when applicable.
- Installation for protective shields, saddles and duct hangers.
- Insulation for elbows, conics, splitters, and specialties.

SAMPLES

Provide samples of all thermal insulation of same material indicated for final work andother accessories sets.

MATERIAL TEST REPORTS, INCLUDING BUT NOT LIMITED TO

- Thermal Conductivity
- Water Absorption
- Fire Behavior
- Resistance to Fungal and Mold Formation
- Permeability
- Corrosion Prevention
- Ecological Data

BALANCING DAMPERS.

- Install volume control dampers in accessible locations at all branch connections and wherever necessary to adjust the flow of air to secure correct distribution. Casing and blades shall be made of galvanized sheet metal and shall be equipped with an approved device for fastening in any desired position. This device shall be such that the damper cannot move, or rattle and pointer shall indicate the position of the damper from the outside of the finished duct insulation.

- Dampers shall be aerofoil multi-leaf opposed blade with blade height not more than 180 mm.
- All blades shall be operated by a single operating quadrant with gears and links as necessary.
- Spindles shall be non-corrodible, passing through bearings made of special plastic temperature resistant to 100 deg. C. Capped bearings shall be provided on non-drive side.

- The whole damper assembly shall be mounted in a galvanized frame with flanges.

BACKDRAFT DAMPERS.

- Back draft dampers shall be low leakage with parallel blades and neoprene edge seals.
- Dampers frames shall be constructed from galvanized sheet steel with Aluminium blades. Blade stub shafts shall be brass with PVC bearings. Sealing strips on blades shall be polyester foam.
- Pressure relief dampers shall be multi-parallel blade with weighted arm closing assist. The frame shall be anodized Aluminium channel sections with formed Aluminium blades. Maximum blade length shall be 100 mm, and polyester foam seating strips shall be incorporated on blade edges. Bearings shall be in PVC with non-corrodible shafts.

SPLITTER DAMPERS.

- In each low-pressure system take off where opposed blade duct dampers are not specified, splitter dampers shall be provided. The only justification for not providing splitters is in a system which can be demonstrated to be sized by a static regain programme and which is run-in high-pressure fittings.

- Splitters shall consist of hollow blades in a vane rail assembly, made from galvanized steel by a recognized manufacturer.

- A lockable quadrant adjustment lever shall be located outside the insulation and marked clearly to show vane position.

FIRE DAMPERS.

- Provide, where required by ordinances or codes, fire dampers made to British standard B.S. 476 or North American Standards or N.F.P.A. 90A, and complete with angle iron frame, 72 deg. C fusible link, pivot rods, and spring catches. Enlarge duct sections around fire dampers, to allow 100% unrestricted duct area while in open position. Provide approved type access doors, with airtight gaskets, for inspection and servicing of fire dampers. Fire dampers shall be 2 hours rated and shall be labelled by manufacturer.

- Provide fire dampers in all ducts over 125sq.cm in area, in the following locations, whether or not specifically required by ordinances and codes: -

- Duct entering and leaving fire shafts (duct shafts). Ducts passing through designated fire walls.
- Ducts through floors, and not encased in fire shaft. Ducts leaving and entering plant area.
- Ducts entering and leaving storage areas.

- Where fire dampers are located remote from fire partition, duct between fire damper and partition shall be encased in double metal lathe and plaster or other fireproofing acceptable to Authorities having jurisdiction.

- Fire damper construction shall be of galvanized steel casing, galvanized steel blades, fitted with catch plates, stainless steel closure springs and stainless-steel seals. Blades shall be out of the air stream. Fusible links shall be rated for a fusing temperature of 72 deg. C unless otherwise specified.

- Damper installation frame shall be supplied by the fire damper manufacturer.
- Dampers shall be UL listed or approved by an equal certification authority.

SILENCERS

- Silencers shall be installed as shown on drawings and when required to achieve the specified noise levels.

- Silencers shall have moisture protected mineral wool insulation suitable for hospital application. No material shall be used that encourages bacteria and fungus in the presence of moisture.

- At shop drawing stage the contractor will undertake calculation for every unit and every area and determine the insertion loss required to meet the stated noise criterion. Supply and return duct noise must be considered as well as duct breakout noise.
- Materials of construction shall be galvanized sheet metal and mineral fiber acoustic fill which is inorganic, inert, moisture and vermin resistant. Silencers shall be so constructed as to prevent erosion and acoustic fill.

- Silencers shall be factory made and shall have available certified test data concerning insertion loss. This must be available when required, in advance of ordering units. Where a silencer is made under license, or part assembled locally, then tests must also be made after assembly and witnessed by an independent authority.

- The static pressure loss of any required silencer must be considered in air handling equipment original selection.

VIBRATION ISOLATION (FOR ALL SECTIONS)

- All plant and equipment shall be isolated from the building structure in such a manner that noise, and vibration is not transmitted through the structure.

- Anti-vibration equipment will be manufactured by a specialist company acceptable to the Consultant, and with standard ratings information available.

- Anti-vibration equipment shall comply with Seismic code for zone 2A.

- The following requirements are minimum requirements and must be confirmed by the specialist manufacturer's representative as being adequate for the particular equipment proposed by the Contractor.

PUMPS.

- Water pumps are to have open type spring isolators with equal stiffness in horizontal and vertical planes. They shall be supported on 10 mm thick ribbed neoprene Vibropad and will be color coded to aid identification.
- They will also have level adjustment screws and be rated for 75% more than the anticipated design load.
- The spring shall be located between the housing pad and the concrete inertia base. The inertia base for each pump will be designed to give a static deflection of 50 mm at rated design load conditions and shall be fully independent of any adjacent pad.

CONDENSING UNITS / AIR HANDLING UNIT, FLOOR MOUNTED.

- Floor mounted air handling units shall be mounted on open spring type isolators with 25 mm static deflection, equal stiffness in horizontal and vertical planes and a 10 mm neoprene

Vibropad base.

- They shall have built in levelling screws and limit stops, and color coding to aid in identification.

CENTRIFUGAL FANS, FLOOR MOUNTED.

- Centrifugal fans shall have open type spring isolation with equal stiffness in horizontal and vertical planes. They shall be supported on 10 mm thick neoprene Vibropad and will be color coded to aid identification.
- They will have level adjusting screws and will be rated at 50% more than the design load and shall have a static deflection of 50 mm.
- The springs will be located between the fan and the secondary support steel.

AXIAL FANS, FLOOR MOUNTED.

- Axial fans shall have open type spring isolation with equal stiffness in horizontal and vertical planes. They shall be supported on 10 mm thick Vibropad and will be color coded to aid identification. They will have level adjusting screws, and will be rated at 50% more than the design load, and shall have a static deflection of 50 mm.

PIPING.

- All piping connected to isolated equipment shall be supported with isolation hangers for at least the first three points of support on all sides of the equipment.
- The first point shall have a static deflection of 2 x the equipment deflection, except that the deflection of this point shall not exceed 50 mm (with pipe work full).
- The second and third supports shall have deflections of 25 mm each.
- For suspended piping, springs will be box type spring hangers with snubber, level adjustments, and colour coding.
- For low level piping mounted on roller systems, the isolators will be open type springs with equal stiffness in horizontal and vertical planes.
- In addition, Flexible connection shall be included at inlet and outlet to all items of equipment.
- Flexible connections will be of full line size and will be flanged with gaskets and mating flanges.
- Connectors shall be twin sphere manufactured with multiple plies of nylon tire cord fabric and neoprene in hydraulic rubber presses and vulcanized.

DUCTWORK.

DUCT DRAINS AND DRIP PANS.

- Provide drip pans under cooling coils mounted in ducts in built up plenums.
- Drip pan shall be constructed of not less than 18-gauge steel (1.2 mm) hot dip galvanized after fabrication, with all joints welded or soldered. Sides of pans shall be not less than 100 mm upstream and 300 mm downstream of coil. For Hygiene AHUs the drain pan shall be stainless steel Grade SS – 316.
- Pan width shall extend 50 mm outside coil headers and/or return bends to collect any condensation released from these surfaces.
- Pan shall be equipped with one or more 25 mm drain which must be run to drainage system as indicated by the Consultant.

- Drain pans for fan coil units shall be factory manufactured and supplied with the fan coil unit. Drain pan must be factory insulated on the sides in addition to on the bottom, with at least 20 mm rigid insulation.
- Drain lines will have traps and falls as specified in the plumbing section.

DIFFUSERS AND REGISTERS

- Diffusers, registers and grilles shall be arranged for flush mounting in lay-in type ceilings and overlap mounting in plaster, mineral tile and similar ceilings, with concealed fixings unless otherwise directed.

- Grilles, register and diffuser locations shall be adjusted to suit reflected ceiling drawings or Consultant's site instructions. All grilles, registers, diffusers, louvers shall be from one manufacturer.

- Provide plaster frame for grilles, and diffusers installed in plaster ceilings.

- All diffusers, grilles and registers shall be supplied completely factory powder coated. Finish color shall be to the approval of the Architect. The interior of all grilles and diffusers is to be factory painted matt black.

- All grilles and diffusers will have opposed blade balancing dampers. All will have foam rubber sealing band around the edge to seal to the structure. All pivots will be round section, not of formed sheet, and not relying on a spring steel locking wire.

- Unless otherwise specified basic grilles and diffuser materials shall be Aluminium extruded sections. Sections in the air stream shall be carefully selected to minimize turbulence.

- All grilles and diffusers supplied on this project shall be tested and rated in accordance with ASHRAE Standard 70-72, ADC Test Code 1062-GRD and ISO 3741.
- Linear bar grilles shall be fabricated from aluminium, with 6.4mm wide bars on 12.5mm centre’s pressed into a notched steel retaining bar. The core can be either welded into the cuter frame, or, where the grille is used in a sill application, held in the outer frame by spring clips fixed to the core retaining bar.
- The outer frame shall be 35mm deep and shall have a visible flange 25.4 mm wide. Mitred end caps shall be welded to give a near invisible joint.
- The grill shall be complete with an opposed blade damper painted matt black and shall be fixed with universal mounting brackets. Both the damper and the fixing brackets shall be accessible through the face of the grille.
- Continuous grilles shall be provided with positive alignment strips, which fit into special keyways extruded into the frame of the grille to ensure clean unbroken lines.
- Ceiling Diffusers shall be multitone giving 4-way horizontal discharge.
- The three centre cones of the diffuser shall be manufactured from pressed aluminium, with the remaining cones and the outer frame fabricated from extruded aluminium welded at the corners to give near invisible joints.
- One, two, and three-way pattern cores shall be used as indicated on schedules. All cores shall be interchangeable.
- The core shall be removable without the use of special tools, but for safety, shall be fixed to the outer frame by a small length of chain.
- The diffuser shall be complete with an opposed blade damper painted matt black.
- Wall registers shall be double deflection fabricated from aluminium, the front vanes being horizontal, the rear vanes vertical. This grille shall be complete with an opposed blade damper

painted matt black and adjustable from the face of the diffuser.

- Both sets of vanes shall be fully adjustable without the use of special tools.
- Egg crate return or extract grille shall be provided with a steel lattice core of 12.7mm x 12.7mm openings, giving a free area of 90%.
- The core shall be fixed into an extruded aluminium frame, with welded corners and a 25mm face flange.
- The grille is complete with an opposed blade damper painted matt black and adjustable through the face of the diffuser.
- Circular ceiling diffuser shall be of aluminium construction with two concentric inner spinning's.
- The diffuser core shall be fully adjustable for vertical or horizontal air discharge and shall be removable without the use of special tools.
- A flap damper shall be provided in the neck of the diffuser which is adjustable from the diffuser face.
- Vision proof door transfer grilles shall consist of a steel core with inverted-vee type blades, and an extruded aluminium frame with matching rear flange.
- The frame shall be adjustable from 28mm to 60mm to suit the door width.
- Transfer grilles for 'LIGHT TIGHT' applications such as dark rooms shall be with two vision proof cores back to back.
- Transfer grilles on fire rated doors shall include a block of intumescent material as a fire stop.
- Linear Slot diffusers shall provide unobtrusive continuous air diffusion with a pleasing aesthetic appearance. Hairline butt joints shall ensure clean unbroken linear runs for active and dummy sections.
- The diffusers shall be complete with pattern control blades, fully adjustable from face of diffuser through 180 degrees and shall be fitted with end caps at each end.
- The diffuser members shall be constructed from high quality aluminium extrusions to BS 1474 while the pattern control blades shall be of black rigid PVC.

- Exhaust valves shall be manufactured from high quality sheet steel spinning protected by a stove enamelled or powder coated paint finish. Flanges shall be fitted with sealing gaskets.

- The valves shall be installed with the aid of a mounting ring and air flow adjusted by rotating the central disc. Finish of valves shall be to Architect's choice.

EXTRACT AIR LOUVRES

- Louvers shall be extruded Aluminium frame with Aluminium blades of not less than 2 mm thickness and shall be firmly fixed so as not to vibrate. Unsupported blade width shall not exceed 1800 mm
-
- Behind each louver shall be an insect mesh screen 76 x 6 mm made from 2 mm diameter wire. The screen will be clamped by a 20 mm frame and will be firmly fixed to the outer edges of the louver. The screen and frame shall be hot dip galvanized after fabrication.
-
- The connection to the louver shall be flexible and shall ensure no duct load is transmitted to the louver.
-
- Louvers shall be provided with powder coated finish to the approval of the Architect.

SAND TRAP LOUVRE FOR AIR INTAKE

- Sand trap louvers shall have a double deflection inlet passage to separate sand from incoming air by means of centrifugal forces.

- Separation efficiency particle size 350-700 shall not be less than 90% at a face velocity of 1M/Sec and not less than 70% at a face velocity of 2M/sec.
- Sand trap louver shall be of aluminium construction, self-cleaning and maintenance free. The base of the louver shall have self-emptying sand holes.
- Pressure drop at 2 M/Sec average face velocity shall not exceed 120 Pascal's.
- Insect mesh shall be included.
- Sand louvers shall be provided with powder coated finish to the approval of the Architect.

MOTORIZED SHUT OFF DAMPERS

- The Contractor shall furnish and install, control dampers as required for the proper functioning of the system.
- All control dampers shall be opposed blade.
- Casing shall be in galvanized sheet steel to BS 2989 Z2 G275 M.
- Blades shall be of natural anodized extruded aluminium to BS 1474.
- Blade tip seals silicon rubber, temperature resistant up to 300 deg. C.
- Side seals shall be stainless steel to BS 1449 Grade 302 – 525.
- External blade linkage shall be stainless steel to BS 970 304-S 15.
- Blade spindles shall be in stainless steel to BS 970 304 S11 supported in plain brass bearings.
- Shut off dampers shall have a leakage rate not exceeding 200 M3 / hr / M2 at 1000 Pa.
- Motorized dampers used in smoke extract ducts shall be low leak type confirming to UL 5555. The actuator shall be fixed in the factory and certified for proper operation.

SUBMISSIONS

- **TECHNICAL SUBMISSIONS**
  - For any minor revisions in architectural layouts, contractor shall be responsible to carry out part redesign for such area and submit necessary calculations for Engineer's approval before proceeding with the shop drawings.
  - The Contractor will prepare full shop drawings, including sections of distribution systems and equipment.
- **HARDWARE SUBMISSIONS.**
  - The Contractor will submit catalogue information for all distribution equipment including, but not limited to:
  - Duct work - Low pressure
  - Flexible duct work - Low pressure
  - Flexible connections
  - Access panels
  - Balancing dampers.
  - Motorized dampers
  - Back draft dampers. Splitter dampers
  - Turning vanes
  - Fire dampers
  - Silencers
  - Vibration isolators
  - Diffusers, grilles and registers
  - Louvers, Sand louvers

- Filters
- The Contractor will submit samples for any or all of the above as requested by the Consultant after receipt of the catalogues.
- Samples will definitely be required of these items which are exposed, such as diffusers, grilles, louvers. The samples must be of size, specification and finish to be relevant to the project.
- Where a Country of Origin is given, this refers to the Head Office in the case of International Corporations. However, for each product not manufactured in that Country, separate approval must be obtained from the Consultant.

DUCT INSULATION

GENERAL

- This Section includes insulation for ducts, semi rigid and flexible duct, plenum, field-applied jackets, accessories, and attachments. The insulation material must be Flexible Elastomeric Foam made of NBR with closed cell technology with 18 microns Aluglass facing that is factory laminated. The insulation shall be manufactured as per ASTM C534/C534M: Grade 1 Standard specification for performed flexible elastomeric cellular thermal insulation in sheet and tubular form.

REFERENCES

LOCAL GUIDELINES

- ASTM E84 (ANSI/UL 723): Test method for surface burning characteristics of building materials.
- DCD/ ADCD: Civil Defense certificate with listing of product being proposed.
- DCL: For the emirate of Dubai, 2020 Al Sa’fat Green Building System certification from Dubai Central Laboratory Department.

FACTOY APPROVALS

- ISO 9001-2015: Manufacturing and Trading Thermal and Acoustic Insulation Materials in the Shapes of Pipes, Sheets, Rolls, Tapes and their Accessories and trading of fire Stopping Materials.

INTERNATION ACCREDITIONS

- REACH Directive (EC) 1907/2006
- Singapore Green Building Council Product Certificate
- FM 4924: Material classification and full-scale test requirements for insulation material used on the exterior of non-combustible pipes and ducts.
- Environmental Product Declaration in accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for the insulation.

SUBMITTALS

Product Data:

- For flexible elastomeric foam (FEF), including conductivity, thickness, jacketing if needed, flame spread, and smoke development indices.

Detail installation of insulation for the following:

- Installation for air ducts, including outer insulation and duct lining when applicable.
- Installation for protective shields, saddles and duct hangers.
- Insulation for elbows, conics, splitters, and specialties.

SAMPLES

- Provide samples of all thermal insulation of same material indicated for final work and other accessories sets.

Material Test Reports, including but not limited to:

- Thermal Conductivity
- Water Absorption
- Fire Behavior

- Resistance to Fungal and Mold Formation
- Permeability
- Corrosion Prevention
- Ecological Data

QUALITY ASSURANCE

Regulatory Requirements:

- Flame and Smoke Ratings: Installed composite flame spread not to exceed 25 and smoke developed not to exceed 50 as tested by ASTM E84 (ANSI/UL 723).

Protection

- Exterior insulation shall be covered with a protective material to protect against mechanical damage, equipment maintenance, and dirt. In case of outdoor installation, protection or cladding should be UV and scratch resistant 400 µm (0.4 + .03 mm) thickness.

Source Quality Control:

- Service: Use insulation specifically manufactured for service specified.
- Labeling: Insulation labeled or marked with brand name and number.
- Insulation and accessories shall not provide any nutritional or bodily use to fungi, bacteria, insects, rats, or other vermin, shall not react corrosively with equipment, piping, or ductwork, and shall be asbestos free.
- Any material or composite risk causing a hazard shall not be used for its allergic and dangerous reactions to humans. All fibrous insulation materials cannot be used for health reasons

PRODUCT

DUCT INSULATION (Closed Cell Flexible Elastomeric Foam Insulation)

- Flexible elastomeric foam (FEF) with a closed cellular structure with factory applied water-based self-adhesive and factory laminated Aluglass facing.
- Thermal Conductivity (ASTM C518-17) at 35°C: max 0.034 W/mK
- Water Absorption (ASTM C1763-20) ≤ 0.2 volume %
- Water Vapor Permeability (ASTM E96): 0.018 Perm.in
- Density: 55-70 kg/m3
- ASTM E84 (ANSI/UL 723): Test method for surface burning characteristics of building materials.
- DCD/ ADCD: Civil Defense certificate with listing of product being proposed.
- DCL: For the emirate of Dubai, 2020 Al Sa'fat Green Building System certification from Dubai Central Laboratory Department.
- Flame Spread Index ≤ 25
- Smoke developed ≤ 50
- Material Resistance to Fungi and Mold (ASTM G21): Rating of 1 (or less)
- Service Temperature: As per ASTM C534 Grade I
- Minimum Service Temperature: -183°C
- Maximum Service Temperature: 104°C
- Adhesive: Factory applied water-based self-adhesive.
- ASTM E96: Water Vapor Transmission of Insulation with value '0'
- ASTM C518-17: Thermal Transmission Properties by Heat Flow Meter with value not more than 0.038 W/m.K at mean temperature 35 deg C.
- ASTM D 5116: CDPPH standard method V1.2 2017: Volatile Organic Compound Emission complying to GEV classification criteria (EMICODE EC 1) and are free from hazardous volatile compounds.
- ASTM C209-2015: Determination of Water Absorption of Insulation with less than 0.2% water absorption by Volume.
- ASTM C871: 2008: Chemical Analysis of Thermal Insulation Materials for Leachable Chloride, Fluoride, Silicate, and Sodium Ions should be below the detection level.

- ASM C1338-14: Fungi Resistance of Insulation Materials and Facings resulting 'no fungal growth' after 4 weeks.
- ASTM G154-16: UV Weathering should have no changes for 300 hours.
- ASTM D1149-18: Ozone Resistance for 300 hours resulting no cracks or no visible changes on the insulation.
- IEC 62321-6 Chemical Analysis complying to 2011-65/eu regulations.
- CDPH Standard Method V1.2 2017
- CFC and HCFC to be less than 1 ppm.
- EPA-600/R-93/116: Insulation should be free from Asbestos, dust, Fiber Free.
- ISO 22196: The material should have inbuilt Antibacterial properties
- Aluglass facing shall be with 18 Microns Aluminium, 200 GSM Glass Fiber Cloth withstanding Tensile Strength of 600 N/25 mm when tested against ASTM D828.
- Cladding material if factory laminated should be equal to or better than 400 Microns made of 20 Micron Aluminium and 430 GSM Glass Fiber Cloth with Weave Satin.
- The product, while manufactured, the GWP total shall be no more than 0.18 Kg CO2 e. This can be referred in the EPD certification.

DUCT SYSTEM APPLICATION:

- Insulation materials and thicknesses are specified in the belowschedule:

THICKNESS SCHEDULE

Service	Insulation Thickness (in mm)	Jacketin g
Indoor Ducts	19	Aluglass
Unconditioned Ducts	25	Aluglass
Outdoor Ducts	32	Factory Applied UV Jacket
Risers	32	Plain

DUCT INTERNAL LINING (Closed Cell Flexible Elastomeric Foam Insulation)

- Flexible elastomeric foam (FEF) with a closed cellular structure with factory applied water-based self-adhesive.
- Sound Absorption according to DIN EN ISO 11654:1997: Weighted sound absorption coefficient ≥ 0.7
- Sound absorber class D or higher
- Thermal Conductivity (ASTM C518-17) at 35°C: max 0.034 W/m K
- Water Absorption (ASTM C1763-20) ≤ 0.1 volume %
- Water Vapor Permeability (ASTM E96): 0.019 Perm.in
- Density: 55-70 kg/m3
- Fire Behavior (ASTM E84 or ANSI/ UL 723):
- Flame Spread Index ≤ 25
- smoke developed ≤ 50
- Material Resistance to Fungi and Mold (ASTM G21): Rating of 1 (or less)

- Service Temperature:
- Minimum Service Temperature: -183°C
- Maximum Service Temperature: 105°C
- Adhesive: Factory applied solvent-based self-adhesive.

THICKNESS SCHEDULE:

Service	Insulation Thickness (in mm)	Jacketing
Duct Liner	13 - 19 - 25	None

- The insulation shall have an NRC value of 0.4 when tested as per ASTM C423-23- Standard test method for sound absorption and sound absorption coefficients by the Reverberation Room Method.
- Insulation supplied shall be seld-adhesive type, preferably with scrim if the insulation is for PID.

ACCESSORIES

Glue: Polychloroprene Adhesive Suitable for insulation in heating and refrigeration system.
- Viscosity at 20°C should be 700 Pa.S. (±100 Pa.S Tolerance)
- Solid Content should be 20,5% (±1% Tolerance)
- Density should be 850kg/m3 (±50kg/m3 Tolerance)

Anticondensation Rubber Tape: (For CDP/ plain insulation)
- Highly flexible closed cell elastomeric foam (FEF) suitable for hot and cold thermal insulation.
- Water Absorption (ASTM C1763-20) ≤ 0.2 volume %
- Water Vapor Permeability (ASTM E96): 0.019 Perm.in
- Density: 55-70 kg/m3
- Fire Behavior (ASTM E84 or ANSI/ UL 723):
- Flame Spread Index ≤ 25
- smoke developed ≤ 50

Aluglass/Clad Tape:
- AG / Clad tapes can be used for sealing the joints at frequent interval. The AG tapes must be of the same finish of the insulation
- All the accessories in order to complete the installation of insulation shall be recommended by the insulation supplied with a conformity document.

EXECUTION
GENERAL
- Applicators: Applicators shall be employed by a firm that specializes in insulation work.
- Preparation: Surfaces of equipment, and ductwork shall be clean, free of oil or dirt, and dry before insulation is applied.
- Apply insulation materials, accessories, and finishes according to the manufacturer's written instructions; with smooth, straight, and even surfaces; and free of voids throughout the length of ducts and fittings.
- Labels: ASTM, FM, and similar markings and labels shall not be covered.

- Any insulation that becomes damaged, chemical soaked, or stained shall be replaced.
- All joints and edges should be sealed by manufacturer approved glue.

DUCTWORK INSTALLATION
General:

- Install in accordance with the manufacturer's instructions.
- Insulation at access panels shall be removable.
- All surfaces to be insulated must be thoroughly cleaned.

External Duct Insulation:
- All corners shall be made with joints or seams; bending insulation around corners will not be allowed.
- For self-adhesive sheets position the uncovered edge at the starting point and pull the backing paper off gradually, pressing the material down as you go.
- All edges must be glued together.
- All joints and seams shall be butted tight together.
- Butt joints with 2-inch-wide Foam tape for plain insulation/ Cladding tape for Clad facing.
- Finish corners with 2-inch-wide foam tape of similar finishing.

Verification:
- The manufacturer and/or an authorized supplier shall observe the installation of insulation at regular intervals, ensuring that it occurs monthly. A comprehensive report, including photographs, details of the observed areas, and confirmation of adherence to recommended installation practices, must be provided. These reports, along with the manufacturer's or supplier's warranty letter, shall be submitted by the MEP contractor at the conclusion of the project

PIPES INSULATION
GENERAL
DESCRIPTION
- This Section includes insulation for piping and equipment. The insulation material must be Flexible Elastomeric Foam made of NBR with closed cell technology with 18 microns Aluglass facing that is factory laminated. The insulation shall be manufactured as per ASTM C534/C534M: Grade 1 Standard specification for performed flexible elastomeric cellular thermal insulation in sheet and tubular form.

REFERENCES
LOCAL GUIDELINES
- ASTM E84 (ANSI/UL 723): Test method for surface burning characteristics of building materials.
- DCD/ ADCD: Civil Defense certificate with listing of product being proposed.
- DCL: For the emirate of Dubai, 2020 Al Sa'fat Green Building System certification from Dubai Central Laboratory Department.

FACTOY APPROVALS
- ISO 9001-2015: Manufacturing and Trading Thermal and Acoustic Insulation Materials in the Shapes of Pipes, Sheets, Rolls, Tapes and their Accessories and trading of fire Stopping Materials.

INTERNATION ACCREDITIONS
- REACH Directive (EC) 1907/2006
- Singapore Green Building Council Product Certificate
- FM 4924: Material classification and full-scale test requirements for insulation material used on the exterior of non-combustible pipes and ducts.
- Environmental Product Declaration in accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for the insulation.

SUBMITTALS
- Product Data: For flexible elastomeric foam (FEF), including conductivity, thickness, jacketing if needed, flame spread, and smoke development indices.

- Detail installation of insulation for the following:
- Outer installation for pipe, linear lengths, elbows, and singularities.
- Removable or immovable preformed solutions for valves, actuators, and special fittings.
- SAMPLES:
- Provide samples of all thermal insulation of same material indicated for final work and other accessories sets.
- Material Test Reports, including but not limited to:
- Thermal Conductivity
- Water Absorption
- Fire Behavior
- Resistance to Fungal and Mold Formation
- Permeability
- Corrosion Prevention
- Ecological Data
- QUALITY ASSURANCE

Regulatory Requirements:

- Flame and Smoke Ratings: Installed composite flame spread not to exceed 25 and smoke developed not to exceed 50 as tested by ASTM E84 (ANSI/UL 723).

Protection:

- Exterior insulation shall be covered with a protective material to protect against mechanical damage, equipment maintenance, and dirt. In case of outdoor installation, protection or cladding should be UV and scratch resistant 400 µm (0.4 + .03 mm) thickness.

Source Quality Control:

- Service: Use insulation specifically manufactured for service specified.
- Labeling: Insulation labeled or marked with brand name and number.
- Insulation and accessories shall not provide any nutritional or bodily use to fungi, bacteria, insects, rats, or other vermin, shall not react corrosively with equipment, piping, or ductwork, and shall be asbestos free.
- Any material or composite risk causing a hazard shall not be used for its allergic and dangerous reactions to humans. All fibrous insulation materials cannot be used for health reasons.

PRODUCT

PIPE INSULATION (Closed Cell Flexible Elastomeric Foam Insulation)

- Flexible elastomeric foam (FEF) with a closed cellular structure with factory applied Aluglass facing.
- Thermal Conductivity (ASTM C518-17) at 35°C: max 0.034 W/mK
- Water Absorption (ASTM C1763-20) ≤ 0.2 volume %
- Water Vapor Permeability (ASTM E96): 0.018 Perm.in
- Density: 55-70 kg/m3
- ASTM E84 (ANSI/UL 723): Test method for surface burning characteristics of building materials.
- DCD/ ADCD: Civil Defense certificate with listing of product being proposed.
- DCL: For the emirate of Dubai, 2020 Al Sa’fat Green Building System certification from Dubai Central Laboratory Department.
- Flame Spread Index ≤ 25
- Smoke developed ≤ 50
- Material Resistance to Fungi and Mold (ASTM G21): Rating of 1 (or less)
- Service Temperature: As per ASTM C534 Grade I
- Minimum Service Temperature: -183°C
- Maximum Service Temperature: 104°C
- Adhesive: Factory applied water-based self-adhesive.
- ASTM E96: Water Vapor Transmission of Insulation with value ‘0’

- ASTM C518-17: Thermal Transmission Properties by Heat Flow Meter with value not more than 0.038 W/m.K at mean temperature 35 deg C.
- ASTM D 5116: CDPPH standard method V1.2 2017: Volatile Organic Compound Emission complying to GEV classification criteria (EMICODE EC 1) and are free from hazardous volatile compounds.
- ASTM C209-2015: Determination of Water Absorption of Insulation with less than 0.2% water absorption by Volume.
- ASTM C871: 2008: Chemical Analysis of Thermal Insulation Materials for Leachable Chloride, Fluoride, Silicate, and Sodium Ions should be below the detection level.
- ASM C1338-14: Fungi Resistance of Insulation Materials and Facings resulting 'no fungal growth' after 4 weeks.
- ASTM G154-16: UV Weathering should have no changes for 300 hours.
- ASTM D1149-18: Ozone Resistance for 300 hours resulting no cracks or no visible changes on the insulation.
- IEC 62321-6 Chemical Analysis complying to 2011-65/eu regulations.
- CDPH Standard Method V1.2 2017
- CFC and HCFC to be less than 1 ppm.
- EPA-600/R-93/116: Insulation should be free from Asbestos, dust, Fiber Free.
- ISO 22196: The material should have inbuilt Antibacterial properties
- Aluglass facing shall be with 18 Microns Aluminium, 200 GSM Glass Fiber Cloth withstanding Tensile Strength of 600 N/25 mm when tested against ASTM D828.
- Cladding material if factory laminated should be equal to or better than 400 Microns made of 20 Micron Aluminium and 430 GSM Glass Fiber Cloth with Weave Satin.
- The product, while manufactured, the GWP total shall be no more than 0.18 Kg CO2 e. This can be referred in the EPD certification.

PIPING SYSTEM APPLICATION:

- Insulation materials and thicknesses are specified in the below schedule:

ACCESSORIES

- Glue: Polychloroprene Adhesive Suitable for insulation in heating and refrigeration system.
- Viscosity at 20°C should be 700 Pa.S. (±100 Pa.S Tolerance)
- Solid Content should be 20,5% (±1% Tolerance)
- Density should be 850kg/m3 (±50kg/m3 Tolerance)
- Anticondensation Rubber Tape:
- Highly flexible closed cell elastomeric foam (FEF) suitable for hot and cold thermal insulation.
- Water Absorption (ASTM C1763-20) ≤ 0.2 volume %
- Water Vapor Permeability (ASTM E96): 0.019 Perm.in
- Density: 55-70 Kg/m3
- Fire Behavior (ASTM E84 or ANSI/ UL 723):
- Flame Spread Index ≤ 25
- smoke developed ≤ 50
- Aluglass/ Clad Tape:
- AG/ Clad tapes can be used for sealing the joints at frequent interval. The tapes must be of the same finish of the insulation
- All the accessories in order to complete the installation of insulation shall be recommended by the insulation supplied with a conformity document.

EXECUTION

GENERAL

- Applicators: Applicators shall be employed by a firm that specializes in insulation work.
- Preparation: Surfaces of equipment, and pipework shall be clean, free of oil or dirt, and dry before insulation is applied.

- Apply insulation materials, accessories, and finishes according to the manufacturer's written instructions; with smooth, straight, and even surfaces; and free of voids throughout the length of ducts and fittings.
- Labels: ASTM, UL, and similar markings and labels shall not be covered.
- Any insulation that becomes damaged, chemical soaked, or stained shall be replaced.
- All joints and edges should be sealed by manufacturer approved glue.

PIPING INSTALLATION

General:

- Joints: Coat both sides of the complete joining area with applicable adhesive or glue. Pipe inner sides should be connected and well glued to avoid any heat leakage or thermal bridges.
- Longitudinal Joints: Make joints on top or back of pipe to minimize visibility. Seal with closure system of 2-inch-wide rubber tape. In case of the use of
- aluminium glass cloth or aluminium cladding pipes, longitudinal joints can be covered by a factory applied overlap or 2-inch-wide tape of the same covering material.
- Butt Joints: Butt tightly together, and seal with 2-inch-wide rubber tape.
- Multiple Layered Insulation: Joints shall be staggered.
- Access: Strainer and other items requiring service or maintenance with easily removable and replaceable section of insulation to provide access.
- Voids: Fill all voids, chipped corners, and other openings with the same insulating material.
- Seal joints, seams, and fittings of metal watertight jackets at exterior locations.
- Longitudinal and butt joints should be covered as previously mentioned ONLY after the consultant’s inspection and approval.
- Elastomeric Tubular Insulation:
- Slit full length and snap around pipe.
- Make cuts perpendicular to the insulating surface leaving no cut section exposed.
- Do not stretch insulation to cover joints or fittings.
- Glue the edges of insulating tube cut along its length and wait till it dries properly before attempting to stick the edges.
- Seal joints with adhesive. Covering joints with tape only will not be allowed.
- Finish seams with 2-inch-wide rubber tape or other covering material tapes where applicable.
- Butt joints with 2-inch-wide Foam tape for plain insulation/ Cladding tape for Clad facing.
- Finish corners with 2-inch-wide tape of similar finishing

Verification:

- The manufacturer and/or an authorized supplier shall observe the installation of insulation at regular intervals, ensuring that it occurs monthly. A comprehensive report, including photographs, details of the observed areas, and confirmation of adherence to recommended installation practices, must be provided. These reports, along with the manufacturer's or supplier's warranty letter, shall be submitted by the MEP contractor at the conclusion of the project.

END OF SECTION

SECTION 4

WATER SUPPLY SYSTEM

SECTION 4

WATER SUPPLY SYSTEM

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GENERAL:

- This specification shall be read in conjunction with other related documents, including contract conditions,
- All systems shall comply with the requirements of the Authority having jurisdiction and shall comply with the requirements given within this Specification and the applicable documents referred to therein.

PLUMBING SYSTEMS

- Plumbing systems shall be installed with the minimum pipework and fittings required to satisfy Plumbing systems requirements, as per Division 22 specification.
- All plumbing systems shall be clearly identified, as per Engineer specifications.
- Plumbing systems shall provide safe operation for the end user under site conditions.

SCOPE OF WORK:

- The work shall include design, check, supply, installation, testing, commissioning, adjustment and setting to work of the water services installation.
- By virtue of carrying out a check on the complete design the contractor is to assume the full responsibility for the correct functioning of the system and to carry any liability or guarantee as may be necessary to protect all parties in this regard.
- The plumbing services sub contractor shall be responsible for coordination with other trades and services and shall provide all materials, labor and supervision, equipment, tools, appliance, service etc, for carrying out the following items of woks.
- The Contractor shall supply, install, test and commission the complete Public Health Engineering Services, as described herein and indicated on the tender drawings and equipment schedules.
- The Contactor shall energize all systems on completion of tenants' installation.
- This particular specification includes the Public Health services associated with this building only.

- The Public Health services associated with this package shall include but not be limited to the following systems:
- The Prime objective of the Public Health services installations shall be to provide a fully functional environment via discrete efficient and fully maintainable systems and equipment.
- The main objectives shall be to:
- Ensure that the complete system is protected against Legionella disease.
- Ensure that all water supplies are clear, odourless, tasteless and wholesome for use.
- Provide potable water in the amount and at the pressure required by the building occupancy and type of plumbing fixtures utilized in the building.
- Prevent contamination from non-potable liquids, solids or gases.
- Maintain a velocity of 1.0 - 1.5 m/s in the water supply piping to prevent noise and decrease the danger of surge pressure shock.
- Ensure that hot water is stored and supplied to appliances at safe water temperatures.
- Ensure flow rates to fixtures are limited to Green Building and Regulations & Specifications.
- Prevent the transmission of foul air in to the building.
- Minimise the frequency of any blockage and provide access to enable the clearance of any such blockage.
- Provide efficient conveyance of discharge from sanitary, kitchen and wash-down facilities to enable the correct function of each appliance.
- The system operates without blockage.
- The flooding frequencies shall be limited to a minimum.
- Coordination with existing services prior to starting of works.

DOMESTIC HOT AND COLD WATER INSTALLATIONS:

- Thermal and protective installation of pipe work plant and equipment installed within the plumbing and drainage contract.
- Noise suppression measures for plant and equipment installed within this section of the contract.
- Protective painting of all materials installed within this section of the contract.
- Identification of all services.
- Electrical power and control panels for power supply to equipment including cabling from panels to respective equipment. Power supply terminating in an isolating switch terminating within 3 meters of each equipment shall be provided by the electrical sub-contractor.
- All builder's work including breaking and making of good walls, floors, etc, equipment foundation pads, sleeves, etc.
- Hoisting of heavy equipment.

SERVICES DESCRIPTION:

DESIGN STANDARDS

- The design, specification and installation of the water systems shall be in accordance with good engineering practice and shall conform to the following, but not limited to, Standards, Codes of Practice and Specifications:
- Dubai Electricity and Water Authority (DEWA) – Guidelines for new Development Projects 2017
- Dubai Green Building Regulations & Specifications
- Dubai Municipality – Requirements and Regulations
- Dubai Technology and Media Free Zone Authority
- Dubai Civil Defence Standards
- UAE Fire and Life Safety Code of Practice

- Dubai Municipality Regulations on Building Conditions and Specifications LEED
- British Standards
- Chartered Institute of Building Services Engineers (CIBSE) Guidelines
- Plumbing Engineering Services Design Guide (IOP)
- Plumbing Engineering Services Design Guide (CIPHE)
- NFPA 54 National Fuel Gas Code
- NFPA 58 Liquefied Petroleum Gas Code
- ASHRAE Standards

RECORD DRAWINGS

- Retain on-site a marked-up set of drawings showing the "as installed" routes, sizes, gradient, invert levels, access locations etc. as a minimum of the foul drainage above ground system.
- Prior to completion transfer this information to "record drawings" as detailed within section A of the preliminaries.

SYSTEM DESIGN:

- The plumbing system shall be designed in accordance with the recommendations of the Chartered Institution of Building Services Engineers and to the requirements of the Load Authorities.

SYSTEM DESCRIPTION:

- From the DEWA main, an incoming supply shall be connected to UNDER GROUND water storage tank of the capacity as indicated on the drawings. This tank shall have enough capacity to meet the daily storage requirements of the building along with the minimum water storage requirements as recommended by Civil Defence for fire fighting purpose.
- A low cutout installed in this tank shall ensure the availability at all times of the minimum quantity of water for fire services.
- Water from the ground tank shall be pumped out by transfer pump to roof mounted water tank where to be distributed to all flats by either pressure booster pump set or by gravity. Hot water supply shall be through individual water heaters in the ceiling void of kitchens and toilets.

PIPE WORK INSTALLATIONS:

GENERAL

- Pipe materials shall be as specified elsewhere herein. The contractor shall install unions or pairs of flanges in the pipe work system at the following locations:
- At intervals not greater than 15m.
- To enable the disconnection of all items of plants, equipment, pumps and valves for maintenance purpose.
- To enable sections of pipe work to be subsequently dismantled and refixed without disturbing other pipe work, plant, equipment and building fabric.
- Where indicated on the drawings.
- Pipework shall be run neatly, parallel to walls or grid lines on plan and shall, where necessary, set around columns and projections. All vertical pipe shall be plumb. Direction changes in pipe groups shall be set out from a common center point.
- All screwed and flanged joints shall be easily accessible.
- Where pipes are fitted around columns or pillars, the direction change shall be made using short sets and shall be supported from the floor on the centerline of the column.
- It shall be the responsibility of the contractor to ensure that the pipe work systems are at least 150mm from any electrical conduit, cable or wiring.

- Pipes entering or leaving trenches or ducts, shall be square to them. All pipe work shall be fixed in such manner as to be readily accessible for tools, welding operations or removable without dismantling adjacent pipe work.
- Where pipe work passes through walls, etc the contractor shall ensure that these pipes are not subsequently bedded-in by making good by other trades.
- Pipe work passing through brick or concrete shall be enclosed in a steel pipe sleeves of such diameter that the pipe may be removed without cutting or deforming the pipe work. In particular where plastic pipes pass through a fire rated wall, fire-arresting sleeves shall be used with a packing of in tumescent strip between the pipe and the sleeve.
- Pipe work passing through internal or external dry sheet construction walls shall pass through formed holes of such size that the pipe work may be removed without cutting or deforming the pipe.
- All holes through such walls shall be closed after the erection and testing of the pipe work by means of removable split closing plates in both sides of the wall. All closing plates shall be manufactured from galvanized sheet steel of not less than 16G thickness in the case of brick or concrete walls. All closing plates shall be painted or finished in accordance with the requirements of this specification.

PIPE WORK SUPPORTS:

GENERAL

- All pipe supports shall be manufactured to B.S.3974 PART1:1974.
- Supports shall allow free movement for expansion or contraction of pipe work.
- Vertical rising pipes shall be supported at the base and the support shall withstand the total weight of the pipe and fluid contained.
- All supports shall be specifically designed for outside diameter of the pipe including specified packing.
- Pipe work shall be supported as recommended by the manufacturer.
- Pipe work shall be supported at intervals not exceeding the following:

Steel, Light Gauge Copper		Plastic water pipe work	
Diameter	**Length**	**Diameter**	**Length**
15 mm	1.2 m	15 to 50 mm	1.0 m
20 mm	1.5 m	65 mm	1.15 m
25 mm	1.8 m	80 mm	1.30 m
32 mm	2.2 m		
40 mm	2.5 m		

PIPE WORK MATERIAL:

INTERNAL COLD AND HOT WATER SUPPLY PIPES:

- Shall be as following:
- Cold and hot water running above false ceiling as a ring main shall be made of PPR, complete with compression fittings, suitable for 10 bar working pressure. Hot water pipes shall be preinsulated to protec-2000.
- Vertical, connected cold and hot water pipes shall be made of cross linked polyethylene-line (80- 85 degree) of 16 mm dia and 2.2 mm wall thickness (ie Pex pipes) complete with 28 mm dia blue/red

pvc conduits, pvc boxes. All fittings and accessories are made of zinc-leach out resistant bass (DZR), with compression type fittings.

EXTERNAL COLD WATER SUPPLY PIPES (ON ROOFS, INSIDE OTS OR SHAFTS)

- On roof shall be UPVC CLASS-E.
- Exposed pipes shall be thermally insulated with polyisocyanate insulation of 25 mm thickness and 50 kg/m3 density covered with 8 oz canvas cloth applied with approved vapour barrier and aluminum cladded.

PVC PIPES - TYPE 2(DOMESTIC COLD-WATER SUPPLY UNDERGROUND OUTSIDE THE BUILDING)

- PVC pipes -Type 2 (Un-plasticized Polyvinyl Chloride) pipes shall be to BS 3505 latest edition Class E 130 psi (900 Kpa) working pressure at 68 deg. F (20 deg. C) fluid temperature and 103 deg. F (40 deg. C) ambient temperature or ASTM specification D 1785.
- The pipe shall be homogenous throughout and free from visible cracks, holes, foreign inclusions or other defects. The pipe shall be as uniform as commercially practicable in color, opacity, density and other physical properties.
- All fittings and accessories shall be of same material and quality as the pipe and jointing up to 2 1/2" diameter shall be of the spigot and socket cemented type where solvent cement is applied to both parts all in compliance with B.S. 4346: Part I: 1969 joints and fittings for use with unplaticized PVC pressure pipes, Part I: Injection molded PVC fittings for solvent welding for use with pressure pipes. After pushing the pipe into the socket, the joint shall be allowed to set for at least 10 hrs.
- Jointing for all pipes buried underground outside buildings and for all pipes above 3" diameter shall be of the rubber ring integral socket type to BS 4346 Part 2.
- Expansion joints with guides as recommended by manufacturer shall be installed on long run solvent cemented pipes every 30 meters of length, and wherever shown in the specification.
- Bending PVC pipes is only allowed in non-critical application at room temperature and after the approval of the Engineer.
- Before bending, the pipe should be heated at the section to be bent to a temperature of about 275-300 deg.F (135-150 deg.C).
- The bore should be supported by packing with sand or by insertion of thick rubber pipe, the heating being carried out in a hot air oven or by immersion in hot oil or glycerine. Overheating should be avoided, and the pipe should not be held at the bending temperature too long.
- PVC pipes -Type 2 are allowed to be used for domestic and potable cold water and swimming pool piping, and / or as specifically mentioned in the schedule of pipe materials.

WATER SUPPLY PIPE SUPPORTS:

- Pipe supports shall be of electrolytically zinc plated clip with factory rubber lining complete with 8 mm threaded G.I. rod, fixed with drilled plugs.

VALVES:

GLOBE VALVES:

- One ¾" size valve with blue marking to be provided in each bathroom, toilet and kitchen. Location to be approved by the Consultant. Valves shall be bronze consulted to BS:2060;1964 with non rising stem.
- From: Grohe – Germany.

CHROME PLATED ANGLE VALVES:

- Bath in blue and red marking ½" inch size each 2 Nos. to be provided for every wash basin, bidet, water heater and kitchen sink. One no to be provided for every WC flush tank, dish washer, washing machine and washing hose combination.

GATE VALVES:

- Should be bronze construction, screwed ends to BS:21:1973 with cast aluminum handles to BS:3464.

FLOAT VALVES:

- Should be bronze construction with rubber seating, copper/gun metal float arm and plastic ball float to BS:1212 part 1:1953 or 2:1970.

NON RETURN VALVES:

- Should be bronze construction, swing type, screwed ends to BS:5154.

PRESSURE REDUCING VALVES:

- Wherever shown in the drawing press reducing valves shall be provided. Valve shall be of brass construction with direct acting diaphragm, adjusting screw and pressure gauge.

FLEXIBLE CONNECTORS:

- Shall be of twin sphere type with multilayered tire cord fabric and neoprene body completes with
- G.I female unions.

Note: All screwed ends valves are to be provided with two unions.

STRAINERS:

- Strainers up to 50mm dia shall be bronze body with threaded ends and open mesh monel metal screen.
- Strainers size 65mm dia and more shall be of CI straight Y-Type with standard perforated st steel screen.

SOLAR WATER HEATER:

- Water heater capacity as shown in the drawings to be complete with solar panels, back up electrical power connections safety, non-return and gate valves etc…..according to Dubai Municipality regulations and approval.
- The design and installation shall be carried out by a supplier approved from Dubai Municipality. Solar water heater is to be guaranteed for 5 years from the date of installation
- The system shall be designed as a fully integrated, packaged heating a) system incorporating the high performance solar collectors, drain back thermal storage tanks, primary circulating pumps, and all other necessary controls for safe and efficient operation.
- The boilers, SS tanks and master control panel shall have a warranty of 2 years.
- The solar circuit shall be an 'active' pumped system that utilizes drain back technology.
- When the unit is at the target temperature with the solar pump stopped the heat transfer fluid is able to drain back from the solar collectors into the Heat Store. This technology shall provide over temperature protections in times of high solar gain with little or no potable water use.
- The closed circuit is controlled by a temperature differential controller that gives preference to any effective energy available from solar rather than the back-up boost.
- Fully-flooded systems utilising additional heat exchangers, intercooler and chemical makeup system on the primary side to prevent over heating in summers shall not be acceptable.
- The solar system selection and design shall ensure the system functions as listed below:-

- The issue of summer overheating is to be addressed; a drain back type solution is
- preferable based on previous experiences in the region and the climatic conditions
- The roof top collectors shall be specified to suit the climate conditions and have a
- high collector efficiency
- Solar hot water systems shall comply with Australian Standard AS2712 or equivalent to American/ DIN standard or DM requirement, and manufactured under a Quality Assurance System complying with ISO 9001. Technical design, construction quality and measurement of performance (accelerated ageing, continuous control) all shall be done as per AS 2712. This is very important as the Australian solar standard has testing procedures to the higher temperature experienced in the Middle East.
- The solar system shall be equipped with all necessary valves and safety features, including provision for automatic drain in case of emergency/overheating.

CODES AND STANDARDS:

- Solar hot water systems shall comply with European standard, and manufactured under a Quality Assurance System complying with ISO 9001, ISO 9002. The solar collector is to be in compliance with EN 12975 and Solar Keymark certificate, while the system is in compliance with EN 12976 and Solar Keymark certificate.
- Typical Solar Thermal System involves:
- Solar Panels and
- Storage Tank
- Solar hot water systems shall comply with Australian Standard AS2712 or regulations/codes/standard applicable to Dubai or equivalent to American, and manufactured under a Quality Assurance System complying to ISO 9001.
- Technical design, construction quality and measurement of performance (accelerate ageing, continuous control) all shall be done as per AS 2712.
- The manufacturer shall have a minimum experience of 25 years.

SOLAR PANELS

- Each collector shall have a harp design with a nominal absorption surface area of 2m2.
- Collectors shall be accredited to EN 12975, constructed of copper waterways with 13 risers mechanically bonded to the copper absorber sheet capable of taking maximum working pressure of 1400kPa.
- Insulations shall be 38mm and 12kg/m3 glass wool insulation in the base and 28mm,32Kg/m3 polyster insulation on the sides, 0.7mm marine grade aluminium outer casing and 0.2mm copper absorber plate.
- The absorber plate shall be copper, Tinox sputtered Titanium Oxide surface having a minimum absorbtance factor of 0.95+/-2 and maximum emissivity factor of 0.04+/-2.
- The collector shall be glazed with low iron toughened safety matt glass of thickness 3.2mm with a maximum iron oxide content of 0.04%.
- The net weight of each collector shall not exceed 33.5 Kgs to keep the roof load within limits.
- The collectors shall have a stagnation temperature greater than 210oC.
- The collectors shall be manifolded in parallel into the appropriate number of rows with a maximum of eight collectors per bank.
- The flow and return lines from the collectors to the storage tank shall be of minimum required diameter and insulated with a minimum of 13 mm thick UV stabilized insulation

STORAGE TANK

- The Storage tank shall operate with neutral treated water at a pH level of 8.5 to 9.0.
- The Storage tank shall contain the appropriately sized internal thermal expansion volume to suit the maximum number of solar collectors with a storage temperature rise of 75ºC and be equipped with all necessary pressure relief valves.
- The storage tank shall be fully insulated with a minimum of 50 mm high density glass fibre bats and cased within a totally weatherproof outer enclosure made from Color bond steel.
- It shall come completely attached to a support frame for locating on a floor or concrete plinth.

POTABLE WATER HEAT TRANSFER COIL

- The heat transfer coil shall be constructed from eight copper coils connected in parallel which and have a pressure drop across the coil of no more than 35kPa.
- It shall be constructed with a ring main circulation return connection at the centre to facilitate re heating.
- It shall be fully brazed and hydrostatically pressure tested in manufacture to 2200 kPa for operation on a potable water circuit of up to 1200 kPa.
- Only Anti scaling, Anti-corrosion and Anti-freeze liquid shall be used on the primary closed circuit. Water shall not be used in the primary circuit.
- Working Pressure:-
- Closed Circuit = 50kPa
- Potable Circuit = 1200 kPa

CIRCULATING PUMP

- A SS 316 circulating pump, rated for high solar fluid temperatures, shall be supplied along with the heat store to circulate the fluid between the heat store and the collectors.
- The working of this pump shall be controlled through a delta T controller.
- The pump shall circulate the fluid only if the temperature in the system falls below the set point and if there is enough radiation to heat the fluid.
- The solar pump shall be of a centrifugal design and circulate (as required) L/s at a total pressure head of (as required) kPa. It shall have a stainless steel 316 body, impeller and shaft.
- Pumps shall be electrically-driven, single-stage, centrifugal type circulating pumps. Support pumps on a concrete foundation or mounting intended for the purpose, or by the piping on which installed if appropriate to the size. Construct the pump shaft of corrosion resistant alloy steel with a mechanical seal. Provide stainless steel impellers and casings of bronze.
- Pump motor start stop shall be controlled by the solar thermal temperature control system that is compatible with open communication protocol and meets CSI thermal incentive program requirements complete with manual override (Hand-Off-Automatic). Pumps shall be installed with isolation valves so the pump can be serviced without draining the system.

SOLAR COLLECTORS:

- The collector surface area of 2.2 m2. The absorber plate should be with blue selective treatment and laser welding. The solar collectors should be suitable for flat or inclined roof installation. The absorber plate is to be with highly selective titanium oxide treatment coating (BLUETEC ETA PLUS) to ensure high absorption of solar radiation and low emission of solar radiation (95% absorption and 5% reflection) according IEA-SHCP. Copper hydraulic circuit. The collector pipes are to be harp geometry with 16 pipes (6 mm diameter) and 2 manifolds (18 mm dia.) Zn- Mg anticorrosion chassis. It ensures high performances and subsequently high temperature of the domestic hot water. Special glass with low iron content (< 0.015% Fe_2O_3), according EN12150-1, highly transparent, Tempered. Silicon bi-component seal, UV and Ozone resistant, runs around the entire perimeter of the glass panel Lower insulation in glass wool (30 mm) heat- resistant and low-emission. The solar collectors are to be highly resistant against weather conditions (wind, hail and humidity). Rubber seal in extruded EPDM runs around the entire perimeter of the glass panel. The solar collectors to have anti-condensation holes corresponding to the four hydraulic attachments and on the lower part of the collector and Solar KeyMark certification and test report according EN 12975-2.

STORAGE TANK:

- The steel boiler is to be with exclusive titanium-based enamel treatment. It should have homogeneous surface tank, elastic, and therefore more resistant to high temperatures and corrosion. The enameling protection at 850°C of tank resistant to thermal shocking treatment to be ensured. The magnesium anti-corrosion anode to be in accordance to DIN 4753/3. The polyurethane insulation free of CFC & HCFC (35 mm for 150 and 200 litres, 45 mm for 300 litres) is required for the storage tank. The external tank is to be in pre-painted zinc-steel. The maximum working pressure is to be 8 bar. There should be flange for inspection. The working pressure 8 bar (sanitary side). There should be available heating element kit.

WASHING HOSE COMBINATION:

- To be provided in each bathroom & toilet near WC unit, complete with 90cm flexible metal tube, angle valve, hand spray and wall support.
- All flexible hoses to be covered with factory applied transparent polyethylene casing.

INDIVIDUAL WATER METERS: (TENANTS/USERS)

- As shown on the drawing, and as per local water department specifications and requirements. Meters should be clearly labeled according to the flat number and other areas served. Metal type labels should be as per the water department requirements as well as pipe supports.

MAIN INLET WATER METER:

- As shown in the drawing, and as per local water department specifications and requirements a main water meter with two gate valves is to be housed inside suitable size recessed type box with door.

WATER SUPPLY SYSTEM

- The water system shall be cleaned, disinfected and left operational with lines well vented to prevent the entrapment of air.
- All safety valves shall be manually operated to ensure reliability.
- All gate valves in the system shall be opened completely. However, valves in bypass lines shall be completely closed.
- All services crossing building expansion joints shall be adequately installed to ensure maintenance free operation.

WATER TANKS:

ROOF WATER TANKS:

- Shall be of hot press moulded GRP panel sandwiched type complete with stainless steel Gr 316 nuts, bolts and internal brackets, external and internal PVC ladders, air vent, float valve, PVC inlet/outlet/drain/over flow connections flanged to the tank panel, with PVC flanges, concrete foundation and steel footings. Panels shall be sealed by EPDM rubber gasket. Panels to be thermally insulated i.e sandwitched type for the roof tanks only.

WATER TANK SIZE: (AS SHOWN ON THE DRAWINGS).

- Tank is to be guaranteed for 10 years from the date of installation.

- Underground Water Tank to be made of concrete with a capacity as per mentioned in he drawings and in conjunction with the Civil Works complete with inlet pipe from DEWA, outlet, overflow Drain pipe and Valve, (the value to be Water type), Warning Pipes, Fire Fighting Reservation Switch, Dry Running Protection, 60x60 cover to be installed at the top side corner, Float Valve, Isolation Valves, all the necessary works and foundation blinthes, PVC ladder for maintenance.

DOMESTIC WATER SUPPLY PUMPS:

- In general either booster or transfer sets as shown in the tender drawing the pumps shall be vertical, multi-stage, centrifugal type complete with stainless steel impellers, shaft and casing. Each pump shall be driven by totally enclosed, fen cooled, squirrel cage, star-delta electric motor with class "F" insulation, IP54 protection suitable for 380V/50Hz power supply automatically controlled and protected by approved control panel. Each pump suction/discharge line will incorporate a gate valve plus quick closed spring actuated check valve (in discharge line only).
- Suction/discharge and manifold pipes shall be made of hard copper to BS 2871 part 1 suitable for working press 10 bar.

TRANSFER PUMP SET:

- 3x water pumps (2 duty, 1 standby). The pumps shall be running alternatively by means of auto cyclic operation facility. Duty pump shall start / stop by means of level switch located in elevated water tank. (For duty refer to drawings).

BOOSTER PUMP SET:

- 3 pumps (2 mains, 1 jockey) the main pumps shall be one duty, one stand-by. The 2 main pumps shall be running alternatively by means of auto. Cyclic operation facility. The jockey pump shall start running only in case of low demand, and cut off when main pump starts. The pump set shall be complete with press tank, press switches, press gauges, control panel & dry running protection. The pump set is to be kept in
- Aluminum, louvered penthouse of suitable size complete with door and lock. (for duty refer to drawings).

CONTROL PANEL:

- One of common control panel comprising of:
- Star delta starters (size according to motor KW).
- Selector switches (Auto / off / hand) each pump.
- Auto cyclic operation switch.
- H/L voltage cut-out.
- Transformer to give 24V out-put for float switches.
- Dry running protection.
- Phase fail protection.
- Run, trip, power indication lights.
- TP MCB + ELCB back up protection.
- Ammeter / each pump.
- 1 x interlocked isolator door.
- Enameled steel enclosure with IP: 54 protection made by Simmons-Germany or equivalent.

TESTING AND COMMISSIONING AT SITE:

- The Contractor shall thoroughly clean, lubricate as necessary, and ensure free operation of all plant and installations prior to testing and commissioning.

- After each pump unit has been set, anchored and fixed in place and prior to testing, a final check of lubrication, alignment between pump and motor, electrical connections, discharge and suction gauge readings and pump balance shall be made under the supervision of a representative of the manufacturer.
- The Contractor shall secure a letter of certification from the manufacturer stating that the check items meet their recommendations and shall submit the letter to the Engineer.
- Horizontally mounted pumps will be supplied correctly aligned on a bed before shipment.
- The Contractor shall ensure the direction of rotation of the driver matches that of the pump.
- Each building zone will consist of 1 set of central solar heating system supplying hot water through a common manifold to all the hot water outlets.
- The bottom of the collectors shall be a minimum of 1 Mts above the top of the heat store to enable the Natural Drain Back of close circuit fluid to take place.
- Hot water circulating pump set to be installed in the return line to ensure continuous hot water availability.
- The solar collectors shall face south with an inclination of 25 deg and shall be mounted on a stand or plinth. Note, a minimum of 10 deg shall be provided for drain back systems.
- A 2-core cable from the hot sensors in the panels to each heat store to be conducted.

SUBMITTALS

- General: Comply with the requirements of Section 013300 Submittal Procedures and submit the following.
- Technical Submittals
- Shop Drawings.
- Product Data: Provide product description, thermal characteristics, list of materials and
- components for use.
- Supplementary Product Literature: Include a statement from the manufacturer for the
- design life of the system.
- List of tests included.
- Certified test data.
- The Solar heating systems shall have certificate from Dubai Central Laboratory Department, and copy of the certificate has to be kept at the construction site
  Outline technical specifications reflecting proposed materials and systems.
- A list of proposed suppliers and Subcontractors intended to be used.
- Preliminary Method Statement.
- Preliminary Quality Plan.
- Summary of deviations from the Specification.
- Manufacturer's Instructions: Indicate installation procedures that ensure acceptable workmanship and installation standards will be achieved.
- Reports of factory witness tests as called for in each section and requested by the Engineer.
- Valve Tag Charts.
- Field Test Reports.
- System Verification Certificates.
- Coordination Construction Drawings: Prepare and submit to the Engineer for acceptance a dimensioned layout with required sections of piping for each floor. These drawings shall be used as coordination drawings and be distributed to all Divisions whose Work is affected. Drawings shall be in a larger scale, usually 1:20 and shall include piping sections in accordance with planned fabrication, all pipe and conduit fittings, valves and cleanouts with all dimensions for erection.

TECHNICAL SAMPLES

- 1. Submit 1 solar panel sample c/w connection fittings, support, insulation, etc.

CLOSEOUT SUBMITTALS

- A. General: Comply with the requirements of Section 017000 Execution and Closeout Requirement, and submit the following:
- 1. Warranties.
- 2. Operation and Maintenance (O+M) Manuals: Include component list with manufacturer's reference numbers, descriptions of materials and procedures for repairing and cleaning of finishes and cleaning frequency.

HYDROSTATIC TESTING OF WATER PIPING SYSTEMS:

- All water piping systems and swimming pool piping shall be hydrostatically tested for ensuring complete tightness under the test pressure and for the duration of time as specified under the respective system concerned.
- Systems may be tested as a whole or in sections to facilitate the progress of the work.
- No part of any piping system shall be tested to pressure less than the specified test pressure measured at the highest point of the system.
- Care shall be taken not to subject any equipment, apparatus or device to a pressure exceeding its prescribed test pressure as obtained from its name plate data or from manufacturer's published data. Pressure tests shall be applied before connecting piping to equipment. Relief valves instruments, automatic air vents and all devices that might be damaged by the test pressure shall be removed, disconnected or blanked off.
- No pressure shall be applied against the closed gate of gate valves. All valves shall be in the open position but not completely back seated during testing. End valves shall be capped.
- In testing flanged piping, temporary blank flanges shall be installed and firmly anchored to accommodate all developed end thrust.
- All piping that can be damaged by end thrust developing from hydrostatic testing shall be properly anchored during testing especially at changes of direction.
- The piping system to be tested shall be closed by plugging and blanking all openings in the system in an approved manner and filled slowly with water making sure to vent all entrapped air. Plugs shall be released temporarily to ensure that water has reached all parts of the system.
- Pressure shall be applied to the system by means of a hand pump drawing from a water container. The pump discharge shall be connected to the system through a globe valve, check valve and recently calibrated pressure gauge of suitable range to have test pressure ready in the middle of the range.
- After the test pressure is reached, the pump shall be blocked off by closing the globe valve and the variations of pressure in the system monitored on the pressure gauge.
- The test pressure shall be maintained on the system for 24 hours during which time the system shall have no noticeable drop in pressure.
- While the system is under pressure, a careful inspection shall be made of all pipes and joints and if any leaks in joints or evidence of defective pipe or fitting is disclosed the defective work shall be corrected immediately by replacing defective parts with new joints and materials. No make shift repairs or application of any repair compound will be permitted.
- After the correction is made the pressure test shall be repeated until a completely tight system is ensured.
- The test pressure shall be released slowly so as not to produce shocks and sudden contractions that might damage the piping.

DOMESTIC WATER PIPING STERILIZATION AFTER COMMISSIONING:

- On completion of the potable cold water supply services system, including any parts of the existing service they shall be sterilized by the application of chlorine. The minimum requirements for chlorination shall be to flush the pipework thoroughly to remove dirt. Chlorine shall then be added to the storage tanks and water drawn from all draw-off points until chlorine is detected at each outlet point.
- The mains are then to be allowed to stand for a period of 24 hours, after which the outlets are to be tested for chlorine. If chlorine is not found to be present, all the above shall be repeated until residual chlorine is detected. After this, the system is to be completely flushed with clean water and precautions taken to avoid subsequent contamination of the installation.
- The final requirements for chlorination will be advised by the Engineer during the course of the Contract. The Contractor shall however allow in his tender for the requirements as outlined above.

WATER AMI (ADVANCED METERING INFRASTRUCTURE) PROJECT

GUIDELINES FOR DOMESTIC WATER METER INSTALLATION IN METER ROOM FOR BUILDINGS

TABLE OF CONTENTS

GUIDELINES FOR DOMESTIC WATER METER INSTALLATION IN METER ROOM

- All Domestic water meters in high-rise residential and commercial buildings shall be arranged in groups and housed inside the water meter rooms as shown in standard drawing PEW-STD-AMI-003.Rev3.
- The meter rooms shall be used solely for water meter installation to protect them against exposure to undue external interferences. No other service installations will be allowed in the water meter room. The meter should not be installed in the basement of the building, pump room or in underground meter chambers.

A. WATER METER ROOM STANDARD:

WATER METER ROOM SHALL COMPLY WITH THE FOLLOWING REQUIREMENTS:

- Meter Room in high-rise buildings shall be located on each floor of the building and adjacent to the Electricity Meter Room and Proper water meter room shall be considered for Retail shops meters, if any.
- The size and shape of the water meter room shall be determined based on the number of water meter to be installed and minimum size of the water meter room shall be 1500mm length x 1500mm depth x 3000mm height.
- The meter room door shall have a minimum width of 800mm and a height of 2000mm with an appropriate handle.

- The water meter room shall be designed to arrange the water meters into a single layer with sufficient clearance to install and maintain the meters.
- Meter room shall be located within the common area for safe, free and uninterrupted access.
- Install a 25mm dia. PVC conduit along a pulling spring between the water meter rooms from the highest floor to the ground floor to lay interconnecting cable for meter communication and Install junction boxes, Trunking and associated fittings in every floor as described in Section C.
- An adequate drainage system shall be provided within the meter room to remove flushed water during installation and maintenance of the water meters.
- The meter room flooring shall be clear from obstacles and shall be even, rigid and not slippery.
- The meter room level shall be lower than (min.100mm) the passage or other floor areas such that water from the meter room should not flow into the passageway, other floor areas and elevators/lifts.
- There shall be sufficient ventilation inside the meter room.
- Light with IP 65 switches shall be provided inside the water meter room for sufficient illumination.
- A waterproof 13 Amp. Power socket shall be provided in the meter room.
- All electrical fittings in the water meter room shall be waterproof and minimum 2mtr. clearance shall be provided from all electrical services.
- A permanent Notice/Warning plate “Landlord/Customer is not allowed to install any other devices in the water meter room and also not allowed to do any modification to meter connection, pipes and fittings without written permission from DEWA” in Arabic and English language shall be affixed inside the meter room.

B. METER INSTALLATION AND ASSOCIATED PLUMBING WORKS STANDARD:

- All plumbing work shall be performed in accordance with all local codes, ordinances, laws and regulations; and as recommended by BS EN 12056-2:2000 and BS 8000, Part 13, 14 and 15. Piping shall be installed not to obstruct the room entrance or passageways and rigidly anchored to walls with suitable wall brackets. Meter and associated piping works shall be in line with the dimension mentioned in the typical installation Drawing PEW-STD-AMI-003.Rev3.
- The following practice shall be adopted in plumbing work for meter room: -
- The fittings at the meter position should facilitate the installation and removal of the water meter without having work on other pipes. All-time maintenance of the meters shall be taken into account while installing new meters.
- Pipes, Valves, PRV, NRV and other fittings used for meter connection shall be high quality, heavy duty, non-toxic, non-corrosive material approved by DEWA and these shall be located at the easily accessible places for future maintenance.
- Pipe size for ½” meter installation shall be ½” to 1” maximum and for 1” meter installation shall be 1” to 2” maximum.
- ½” meter can deliver up to 10,000 gallons water in 24 hours and 1” meter can deliver up to 20,000 gallons in 24 hours, approximately.
- Consumer must have a water storage tank equivalent to 24 hours of consumption for residential premises and storage equivalent to 48 hours of consumption for labor accommodation and other high consumption premises.
- The pipework at the meter position should be securely fixed to support the weight of the water meter and to resist any torsion, bending and tension during the installation and removal of the meter.
- The meter shall be installed on a straight length of pipe the same diameter as the meter and equivalent in length to 3 times the meter dia. at the inlet and 2 times dia. at the outlet.

- Isolation valves shall be installed upstream and downstream the meter to stop water flow from both directions and a stop cock shall be installed prior to the meter for locking/ disconnection of supply to the meter. An isolation valve shall be installed on the main pipeline branch in the meter room to isolate the supply in case of pipe breakages to prevent flooding.
- The main isolation valve shall be provided for every group of five meters in order to sectionalize the group of meters during maintenance without affecting any other group of meters.
- Both valves shall be fully open while the meter is in service and no flow control shall be conducted by adjusting the inlet and outlet valves.
- A threaded joint/union shall be provided after the meter to make length adjustments for meter connection in accordance with the meter length.
- Meters shall be installed at a height of 1200mm from the finished floor/ground level. The clearance between the back plate of the meter and the wall plane shall be 200 mm and the entire meter group must be adjusted in a single layer. Multiple layers and different elevations of meters are not permitted in the water meter room.
- Suitable size single header supply lines are to be considered in the plumping design for each meter room, multiple or parallel header lines shall not be allowed in a single room.
- Sufficient space (at least 200mm.) shall be provided between and around the water meters to allow for installation, reading, servicing and removal of the water meters.
- The water meter and its associated fittings/pipes shall not be part of electrical earthing.
- No water pump shall be installed upstream or downstream the meter and any sudden change in flow/pressure shall be avoided.
- Maximum pressure at the meter inlet shall not exceed 2 bar, PRV and Pressure Gauge shall be installed, at least 1 mtr. before the meter, to achieve the required pressure.
- The water meter shall be easily accessible for reading without the use of mirror, ladder etc.
- The meter shall be installed in accordance with the arrows indicated on the body of the meter and the register shall be located in the most convenient position for reading.
- The meter room shall be suitable for the Technician to stand upright and remove/re-install the meter easily. The instructions for the installation of the water meter must be strictly followed.
- Prepare As-built drawings showing the pipelines to the meter and after meter to the customer pipe connection to their storage tank and the PVC Conduit from meter to the building's LV room for communication cabling and submit to DEWA.
- A/c. No. & Flat No. plates in engraved label shall be affixed on wall immediately adjacent to the meter (not on the meter) for identification.
- DEWA is responsible for the supply and installation of water meters for new connections. Meters are installed, either by DEWA staff or Contractors acting on behalf of DEWA. Customers and property developers are responsible for the supply and installation of all ancillary fittings and pipework beyond the meter in accordance with relevant DEWA specifications and standard drawings.

C. M-BUS CABLE CONTAINMENT INSTALLATION STANDARD:

- Install a 25mm dia. PVC conduit vertically between the water meter rooms from the highest floor to lowest floor for interconnecting communication cable for Smart water meters. Install PVC junction boxes at 25m interval (if the length exceeds 25m) and corner portions with pulling spring so as to provide ease of access to cable circuits throughout the route. The JB shall be a two way PVC JB for easiness of cable pulling.

- In the water meter room located in each floor, install a 50X50 mm PVC trunk of sufficient length horizontally on the wall surface behind the back plate of the meter by utilizing the achieved clearance of 165mm. The trunk shall be made of a single piece securely fixed and aligned at an elevation of 1600 mm from the finished floor level and installed with proper end caps.
- The PVC trunk shall be installed with PVC PG-7 glands equal to the number of Water Meters in the Water Meter room.
- The PVC trunk shall be of high quality conforming to BS4607 and rectangular cross-section with a removable side.
- All PVC conduits, trunks, glands and coupling accessories must be conforming to BS 4607 and be suitable for the ambient conditions expected. It shall be corrosion resistant.
- The external and internal surface of pipes shall be clean, smooth and virtually free from grooves or other indentations or projections. The smoothness of the internal surface of the pipe shall be such that the pulling through of the cables long lengths shall be facilitated without risk of damage to the exterior surface of the cable. The ends of the conduit, trunk and cable tray shall be provided with bushes or other finished ends such that the cables do not sustain damage during installation or throughout the life of the Installation and must be suitably sealed against the ingress of water.
- PVC conduits installed vertically shall be clamped at every one-meter interval and installed horizontally shall be clamped every half meter using suitable and safe fixtures. The connection of conduit to JB shall be done with proper couplings/adaptors.
- Provide an IP65, PVC Junction Box of dimension150X150X100mm for up to 10 Water Meters and 200X150X100mm for up to 25 Water Meters in water meter room on each floor with DIN rail for fixing the water meter communication interfacing units.
- The Junction Boxes shall be mounted on the wall planes inside the meter room at an elevation of 1800mm from the finished floor level and minimum 500mm away from the nearest water meters.
- PVC trunk behind the meters shall be connected to the Junction Box through 25mm PVC conduit by using proper couplings/adaptors. The PVC conduit shall have soft bends and proper mounting. There shall be one connection for up to 10 water meters and two connections for up to 25 water meters.
- A heavy duty GI conduit of 25 mm dia. shall be provided between the ground floor water meter room and LV room (or the water meter room on the floor nearest to LV room) with pulling rope for interconnecting meter communication cable with data concentrator. Install pulling junction boxes in every 25 meters interval and corner portions. The connection of GI conduit shall be done with proper GI couplings.
- Install an IP56 GI Junction Box of dimension150X150X50mm at suitable position on the wall inside LV room at an elevation of 1600mm from the finished floor level. This shall be connected with the GI conduit mentioned above.
- As built drawing, continuity test and IR test reports of the communication installation and cable shall be submitted for approval. Cabling should be done through DEWA approved contractor.

D. M-BUS CABLE INSTALLATION STANDARD:

- Water meter communication cable shall be properly routed and secured along with the water pipe line using suitable nylon cable ties to reach the PVC trunk as shown in Picture Reference # 1, Section-E. These works shall be performed by an approved DEWA contractor.
- M-bus cable from each meter shall be passed in to the PVC trunk through PVC PG-7 gland inside water meter room. All the cable shall be labelled with suitable size of PVC ferrule sleeve and affixed near to the PG-7 glands. For example, the ferrule marking for Shop No.XX shall be marked as S-XX

and Flat No.XX as F-XX. Moreover, all the cables inside the trunking shall be neatly dressed and secured using cable ties throughout the PVC trunk. Refer Picture Reference # 2, Section-E.

- Extend and label each M-bus cable of meter from the PVC trunk to adjacent PVC junction box for termination. In order to achieve sufficient length, approved M-bus Communication cable can be joined together using suitable Splicing connectors inside the PVC trunk. Special care should be taken when connecting the cables with exact polarity/color of wires. The brand of M-bus cables and Splicing connectors shall be approved by DEWA Engineer prior to installation.
- PVC trunk shall be properly closed and PG-7 glands securely fixed to ensure the water proofing of the cable containment. Labelling of cable inside the trunk shall be similar to the one outside the trunk as mentioned in Picture Reference # 2, Section-E.
- Sufficient quantity of Terminal connectors, End plates and DIN Rail stoppers shall be fixed inside the PVC junction box to achieve a pair for dual polarity connection. The cables inside the junction boxes and to the terminal connectors should be neatly dressed in such way that all the ferrules are easily visible, Refer Picture Reference # 3, Section-E.
- All Red/Positive wires from the water meter cables shall be terminated and properly labelled to one part of pair of terminals and Black/Negative to the other part of pair of terminal inside the Junction Box. Suitable size of PVC ferrule sleeve labels shall be affixed on each wire of each cable. For example, the ferrule marking for Shop No.XX shall be marked as S-XX and Flat No.XX as F-XX. Refer Picture Reference# 3, Section-E
- Each pair terminals inside the junction box shall be shortened by using terminal short links to make a star connection. Refer sample picture # 3, Section-E, White colored strip in middle of the terminals.
- The junction boxes located inside each water meter room of all the floors shall be looped together using M-bus cable through the conduits to build a local M-bus communication network.
- In case of multiple loops inside a high-rise building, Water meters in the entire premise shall be grouped in such a way that a loop contain a maximum number of either 245, 120, 60 or 28 Nos. of water meters. For example, if the total number of meters are 245-250, then all the meters shall be grouped in one loop. Likewise, if the total number of meters are 300, then first loop will contain 240-245 Nos. of meters and second loop will contain 55-60 Nos. of meters. Moreover, if there are 490 Nos. of meters, then it will be divided in two groups of 245 Nos. connected together to maintain two loops.
- Junction boxes shall be properly closed to maintain its water proofing capability and to be labelled with PVC engraved label. For Example, the Label for junction box on the floor No-AA shall me marked as JB WMR FAA and on the Ground floor shall be as JB WMR FG. This label shall be affixed below the one already provided on the junction box and conduits as DEWA W AMI.
- The communication cables between the junction boxes located in the adjacent floors and from there to LV rooms shall be properly labelled with origin and destination names at both ends. The ferrule label for the cables shall be in similar fashion as mentioned earlier.
- In case of multiple loops, all loop cables shall be routed together through containments in parallel up to the LV room junction box and properly labelled. For example, the cable from a junction box in Floor No-AA water meter room to junction box in LV room in Basement Floor-1 shall be marked as JB WMR FAA–JB LVR B1 using suitable size of ferrule sleeve. Each loop cable extended up to the LV room should be left with minimum 10 meters of extra cable length and properly coiled for further extension. These cables can be placed on the wall of LV room where there is enough space available to install DEWA communication related devices as shown in the Picture Reference# 4, Section-E.
- Continuity Test, IR test with reports shall be conducted on each loop cable before terminating to water meters and other devices. Insulation Resistance test to be performed at 100VDC for one

minute with a minimum insulation resistance value of 1MΩ. Test reports shall be submitted to DEWA for approval.

- All the materials used for above works shall be approved by DEWA Engineer prior to installation. Preferred M-bus cable, (2x18AWG, Screened, Polypropylene, FRNC/LSZH sheath any standard brand) and Terminal Blocks/ Splicing connectors: Wago or similar standard subject to DEWA approval. Cable installation, termination and testing should be done through DEWA approved contractor.
- As built drawing of the entire cabling and termination, specifying the physical route and locations shall be prepared and submitted for final acceptance. Sample form of As-built schematic drawing is shown in the Section-F, M-Bus Cable Schematic Sample Drawings.
- Exact GIS coordinates (lat. Long. WGS84) of the cable route from water Meter Chamber to the Retaining/Boundary wall, risers, concealed spaces in corridors, parkings etc. shall be mentioned in As-built drawing for DEWA Geographic Information System (GIS) update.
- A copy of the as-built drawing shall be framed in the LV room.
- All the above requirements should be strictly complied by Developers /Contractors before submitting application for the New Connection.

HEAT EXCHANGER

- Heat Exchanger (if required by system design): For system designs a) requiring a heat exchanger, provide a minimum design pressure rating of 150 psi. Construct heat exchanger of 316 stainless steel, titanium, copper-nickel, or brass. Furnish heat exchanger with a capability of withstanding Temperatures of at least 100 °C. Tube-in-tube copper side-arm heat exchangers are acceptable for appropriate system types.

PIPING SYSTEM

- Provide a piping system complete with pipe, pipe fittings, valves, strainers, expansion loops,pipe hangers, inserts, supports, anchors, guides, sleeves, and accessories with this specification and the drawings. Pipe shall be designed to observe limits on flow velocity, pressure drop and gauge pressure associated with the pipe type and characteristics.
- Piping shall be Type L or Type M copper tubing, ASTM B88, with 95-5 tin-antimony soldered joints.

WATER METER INSTALLATION AND ASSOCIATED PLUMBING WORKS STANDARD AS PER DEWA

REGULATION: AMS APPROVAL WIRE SYSTEM .

DELIVERY, STORAGE AND HANDLING CONNECTED TO A SUBSTANTIAL EARTH TERMINAL.

A. DELIVERY

- Manufacturer's Preparation for Shipping:
- Clean flanges and exposed machined metal surfaces and treat with anticorrosion compound after assembly and testing. Protect flanges, pipe openings and nozzles with wooden flange covers or with screwed-in plugs.
- Comply with solar hot water system manufacturer's written rigging instructions.

B. STORAGE

- Store solar panels, water heaters, solar control panels and pumps in dry location.
- Retain protective covers for solar panels, water heaters, pumps, flanges and any protective coatings during storage.

C. HANDLING

- Protect solar panels, pumps bearings and couplings against damage from sand, grit, and other foreign matter.
- A. Protect solar panels and solar water vessel from weather and construction traffic, dirt, water, chemical, and mechanical damage, by storing in original wrapping.

A.PP-R Pipes & Fittings:

- PPR Pipes shall be manufactured according to DIN 8077, 8078. PPR Fittings shall be manufactured according to DIN 16962, DIN EN ISO 15874, ASTM F2389 and shall be injection molded, no fabricated fittings shall be used.
- Domestic cold water system supply PPR pipes and fittings inside the building distribution pipes shall be SDR 6 and hot water system should be SDR 7.4 with Fiber layer. The fiber layer PP-R pipes shall be more rigid and have thermal expansion factor 0,035 mm/m.K .
- System shall have a MOP of 8.1 bar at 70 Deg for 50 operating years according to DIN 8077. The pipe shall be long term pressure tested at 11 bar at 95 Deg C.
- Pipes and fittings shall be joined by Poly-fusion welding, Electro-fusion or Butt welding depending on requirements and comply with the International Standard requirements.
- Welding machine shall be selected as per pipe manufacturer's recommendation and suitable for their Piping system. Welders to be trained and certified by manufacturer.
- Metal inserts for PP-R threaded fittings shall be made of brass by using dezincification resistance type CW617N (CuZn40Pb2). Metal insert shall NOT be nickel coated as per DVGW hygiene guidelines.
- The polymer used for the manufacturing of the products must be approved by Worldwide Known Quality Body Certifier such as,
- DVGW System 1+, W 270 & W 534
- NSF Standard 61 (Cold & Hot 180 F/ 82 C)
- SKZ
- Pipe and fittings should be covered by WRAS and from same manufacturer. The certification should be for full PPR system i.e., both pipes & fittings (including Metal inserts) together as a system approval, material certification of just the PPR pipe is not acceptable.
- Piping materials shall bear label, stamp, or other markings of specified testing agency.
- PPR Pipes, Fittings, and system to be manufactured in an environment which operates a Quality Assurance System assessed to ISO 9001 & 14001
- All PPR pipes and fittings shall be made using virgin raw material from a single source.
- All certificates shall be under the manufacturer's name, any certificate under the local agent other than manufacturer is not acceptable.
- The pipes & fittings used for potable water shall not constitute a toxic hazard, shall not support microbial growth, and shall not give rise to unpleasant taste/ odor /cloudiness /discoloration of water.
- PPR shall be virgin (no mix batching), pre-pigmented black color. The polymer used for the manufacturing of the products must contain metal deactivators to reduce the iron ions in the water to prevent corrosions.

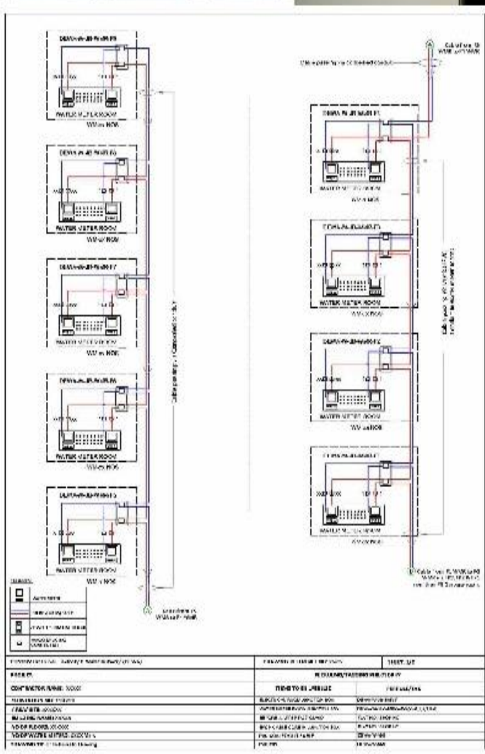
- Provide necessary expansion compensators like bends and loops and follow proper bracketing with the appropriate use of fixed bracket and guiding bracket to avoid effects of liner expansion of pipes.
- All pipes on roof or outdoor shall be insulated and cladded with Aluminum sheet.
- Pipes and fittings shall be stored under cover in a flat horizontal position to prevent sagging or bending and protected from direct sunlight and harmful soil elements until ready for installation.
- PPR & PEX should be from same manufacturer.

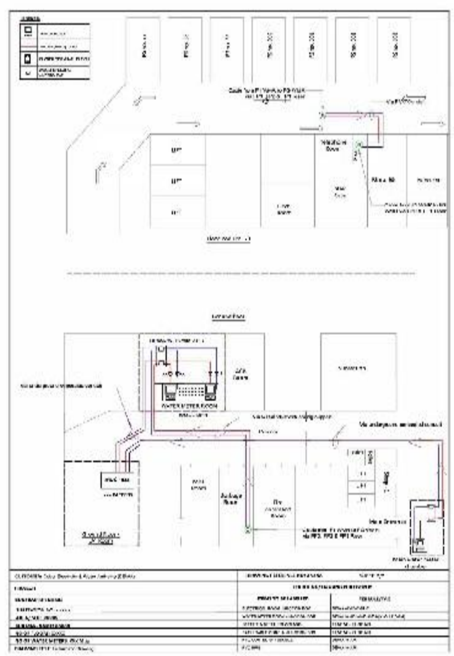
B.PEX

- PE- XC pipes are produced of a high-density polyethylene and are subjected to crosslinking with an electron beam („c" method – a physical method, without using chemical agents)
- The inner pipe shall be a PEX pipe withstanding an operating temperature up to 70 Deg.C at a max. pressure of 10 bar without deformation or damage for a service life expectancy of 50 years as per application class 2. Type of installation and application class (acc. ISO 10508).
- PE-XC pipes are equipped with an EVOH anti-diffusion layer, therefore they can be used in both heating and drinking water systems.
- The following Type Tests for PEX or PERT pipe & fittings should be tested mandatorily by the certification laboratories, the certification should be for the full PERT/PEX system i.e., both pipe & fittings (inlc. Metal inserts) together as a system approval, Material Certification of just the PEX/PERT pipe is Not Acceptable.
- DVGW W 534-(P)
- UBA Metalle
- UBA KTW
- DVGW W 270
- PEXC fittings be either made of made of PPSU (Polyphenylsulfone) or Dezincification Resistant Brass (DZR) certified by DVGW for all product ranges (16 – 25mm).
- The complete system (i.e. both pipe & fittings) should have DVGW & WRAS certification.
- Jointing between PEXC pipe & fittings are to done with Ultraline method with plastic (PVDF) sliding sleeves without the need for compression type fittings, torches, glues, solder, flux or gauges etc+

APPROVALS AND FIXING

All dewa, dm is scope of contractor only for dewa water





ALL DEWA, DM IS SCOPE OF CONTRACTOR ONLY FOR DEWA WATER

- Any item which is not mentioned in the vender list Contractor shall submit RFI.

SECTION 5

DRAINAGE WORKS SYSTEM

(SOIL, WASTE AND VENT PIPE SYSTEMS)

SECTION 5

DRAINAGE WORKS- SYSTEM -

(SOIL, WASTE AND VENT PIPE SYSTEMS)

INDEX

- **GULLY GRATES**
- **TESTING OF DRAINAGE AND VENT PIPE SYSTEMS.**

DRAINAGE SYSTEMS

- All drainage pipelines shall be cleaned, free from deposits or debris.
- Drainage systems shall be left operational with adequate venting, as per specification, engineering drawings and local requirements.
- All access pipes shall be accessible via removable panel to ensure easy maintenance access.
- All traps and gullies shall be filled with water to prevent sewer gases ingress into the building prior handover.
- Pipes shall be adequately installed with allowance for thermal expansion and movements Drainage systems.

SCOPE OF WORK.

- The work shall include design check, supply, installation, testing, commissioning, adjustment and setting to work of the drainage installation.
- By virtue of carrying out a check on the complete design, the contractor is to assume full responsibility for the correct functioning of the system and to carry any liability or guarantee as may be necessary to protect all parties in this regard.
- The drainage services subcontractor shall be responsible for coordination with other trades and services and shall provide all materials, labour and supervision, equipment, tools, appliance, services, etc. for carrying out the following items of works:

  a) Soil and waste drainage systems.

  b) Storm water drainage systems.

    - Noise suppression measures for plant and equipment installed within this section of the contract.
    - Protective painting of all materials installed within this section of the contract.

IDENTIFICATION OF ALL SERVICES.

- Electrical power and control panels for power supply to equipment including cabling from panels to respective equipment. Power supply terminating in an isolating switch terminating within 3 meters of each equipment shall be provided by the electrical subcontractor.
- All builder's work including breaking and making good of walls, floors, etc., equipment foundation pads, sleeves, etc.

- Hoisting of heavy equipment.

SERVICES DESCRIPTION

System Design

- The plumbing system shall be designed in accordance with the recommendations of the Chartered Institution of Building Services Engineers and to the requirements of the Local Authorities. The system shall be installed as detailed in the drawings.

Sewer Drainage Network

- The system shall be DOUBLE stack together with vent installation. All work shall be installed and tested in accordance with BS 5572 code of practice for sanitary works.

Surface water drainage network

- Rain water from the roof shall be piped down via separate pipe work to connect as shown on drawing
- Rain water from the balconies shall be piped down and connected to the waste pipes through running traps.
- Rain water shall be free discharge at ground floor.
- Condensate drain from air-conditioning equipment shall be connected to main condensate water tank. Contractor to coordinate with A/C layouts for exact locations of Air-conditioning equipment

EXTERNAL DRAINAGE

GENERAL REQUIREMENTS

- Pipe connections to manholes, collection tanks and percolating pits shall be made in a completely watertight and approved manner.
- Pipes shall be kept clean until final acceptance of the work. Exposed ends of all incomplete lines shall be closed with wooden plugs and adequately secured at all times when pipe laying is not actually in progress.
- Pipes shall be installed on a good foundation and adequate means taken to prevent settlement. Pipes laid in trenches shall be provided with a solid uniform bearing throughout their entire length.
- Pipes shall not be buried at less than 60 cm below finished grade for protection against mechanical damage. Pipes shall not be run closer than 1m to building bearing walls and footings for protection against building settlement.
- All pipes shall be laid to a uniform slope. Slopes of sewer drain shall be limited to 3% maximum. The free vertical drop of a sewer pipe into a manhole shall be limited to 45cm between the invert level of the pipe opening and the bottom of the manhole. Where conditions necessitate that the drop would exceed 45cm at the maximum slope of 3% a drop manhole shall be used.
- Trenches shall be kept free of water by pumping; use of well points, under drains or other

approved means during pipe laying operations so that all pipe joints are made in the dry.

- Precautions shall be taken to protect incomplete work from floating due to storms or from any other cause. All pipe lines or structures not stable against uplift during construction shall be well braced or otherwise protected.

- All completed underground lines shall be subject to the inspection and approval of the Engineer. All pipes shall be true to line and grade. The full circle of the pipe shall be visible at the manholes.

INSTALLATION

- The drains shall be laid truly straight in line and to an even gradient.
- Excavations shall be made true and even to falls. The bottoms being trimmed to correct level and well rammed. Remove mud, rock projections, boulders and hard spots and replace with approved fill material well consolidated.
- Minimum width of trench shall be 300mm greater than external diameter of pipe.
- Any trenches excavated in error to a greater depth than required shall be in-filled back to the required level with concrete mix, at the Contractor's expense.
- Before laying, all pipes and components shall be checked for defects and joint spaces cleared of dirt.
- Socketed pipes to be laid with sockets uphill.
- Flexible joints shall be made in strict accordance with manufacturer's instructions.
- Where lubrication of the joint is required the pipe manufacturer's recommended lubricant shall only be used.
- Spigot and socket cement joints where required shall be made with ring of tarred gasket well caulked in socket and remaining socket space filled with 1:3 cement mortar, finished in 45 deg. fillet.
- Lay and compact bed of granular material to provide 100mm thickness over the full width of the trench. Scoop out locally at pipe sockets where socketed pipes are used. Adjust pipes to line and level and ensure that pipe barrels rest uniformly on the bedding.
- Add granular side fill uniformly up each side of pipes compacting by hand.
- Any trench sheeting should be lifted before the fill is compacted.
- The granular material shall be compacted in 100mm layers by hand up to 100mm minimum distance above top of pipes.
- Granular material shall be locally available crushed rock graded in relation to pipe size as follows:

10mm nominal size aggregate	-	100mm dia. pipes 10
or 14mm. ditto.	-	150mm dia. pipes
10, 14 or 20mm. ditto.	-	200mm dia. pipes & above

- During bad weather, or in wet fine-grained soils such as clays, silts or sands, it is important to prevent the trench bottom being churned up by mean working in the trench. In such cases a blanket of granular material 75mm thick laid over the trench bottom immediately after excavation, or alternatively a sealing layer of weak concrete 50mm thick, is required.

- Provide concrete bed Grade A (CP 301 Table 1 Specified Works Cube Strength at 21 N/mm2 at 28 days) to drain pipes and concrete infill (Concrete Mix 1:15) where drain track is below level of bottom of foundation base and within 900mm horizontal distance from foundation base. The concrete infill to be carried up to the level of the bottom of the foundation base

- Drain tracks beyond 900mm of adjacent foundations and where the bottom of the trench is lower than a depth beneath the foundation equal to the horizontal distance between the nearside of the trench and the foundation less 150mm, shall be in filled with concrete up to that depth.

- Concrete bed and surround shall incorporate movement joints at each drain pipe joint.

- Each joint shall form a plane surface in the concrete above and below the drain and vertical to the drain and centre line and shall separate the lengths of concrete with 13mm thick resilient fibre board or expanded polystyrene pre-cut to pipe diameter and concrete cross-section height and width and left in-situ.

- Drains below roadways, car parks, and any area subject to vehicular traffic where less than 900mm of cover shall be bedded and surrounded with concrete Grade 'A' (CP 301 Table Specified Works Cube Strength at 21 N/mm at 28days) 150mm minimum thickness all round with provision for movement joints.

INTERNAL DRAINAGE

INSTALLATION

- Vent pipes shall extend through roof and terminate 600mm above it with vent cowl. Vent pipes passage through the roof shall be made watertight by proper flashing.
- All changes of direction shall be gradual and not abrupt, 45-degree fittings shall be used wherever possible, and 90-degree fittings shall be of the long sweep type. All unnecessary turns and off sets shall be carefully avoided and run as directly as possible from the sanitary fixtures to the vertical stacks.
- Concealed pipes shall be installed in such a manner as to permit easy accessibility for maintenance this applies particularly to valve locations.
- All pipes shall be fixed in neatly arranged lines, and adequately pitched horizontal lines to allow the system to be properly vented and drained. Air pockets, traps and sags shall be carefully avoided.
- Supports, clamps and hangers shall be made of galvanized steel with rubber internal rubber seal, fixed with drilled plugs. Cutting and pinning of fixings will not be permitted.
- Run building drains at a minimum grade of 1% (1:100) pitch unless otherwise noted, downward in the direction of flow. Pitch branch connections to stacks from fixtures at 2% (1:50) where possible
- Provide all the required appurtenances to make the drainage system complete in compliance with code requirements including traps, pipe fittings, hangers, and the like.
- Wherever possible, vent stack offsets shall be made with 45-degree fittings.
- Take special care in setting roof drains to ensure that they are set at an elevation which will prelude formation of puddles.
- Install connections to roof drains in conjunction with the roofing specified under civil works, so that the building is adequately protected during construction from damage by storm water.
- Short radius bends shall not be used. Use short "Tee-Wyes" fittings in vertical piping only.

- Any piping passing through roofs shall be so arranged to be a minimum of 300mm (1 ft.) from walls or other obstructions so as to permit proper flashings which are provided by another trade.
- Where drainage pipe work crosses fire rated partitions, walls and floors, provide proprietary fire rated intumescent sleeves with a fire rating equal to or greater than the fire rating of the respective wall or floor.
- Long runs of horizontally installed drainage pipes shall be vented from the top end, whether shown or not on layouts.

CLEANOUTS ON SOIL WASTE VENT AND STORMWATER INSTALLATION

- Cleanouts shall be installed at each change of direction of drainage pipes, greater than 45 degrees, inside the building, and where indicated on the drawings. Cleanouts shall be not more than 10m apart in horizontal lines. A cleanout shall be provided at or near the foot of each vertical waste or soil stack.
- Cleanouts on concealed piping shall be extended through and terminate flush with finished wall or floor. Pits or chases may be left in the wall or floor, provided they are of sufficient size to permit removal of the cleanout plug and proper cleaning of the system.
- Where it is necessary to conceal a cleanout plug, a heavy-duty covering plate shall be provided, which will permit ready access to the plug.
- Cleanout plugs shall be of heavy duty with seal and lock. Final finish shall be to the approval of the Architect.
- Cleanouts shall be of the same nominal size as the pipes up to 100mm pipe diameter and not less than 100mm for larger piping.
- Cleanouts shall be so installed that there is a clearance of not less than 45cm for the purpose of rodding and cleaning.
- Provide cleanouts at foot of all stacks, changes of directions, at the ends of branch runs, in straight runs as required and where indicated. Terminate as specified under "Cleanouts".
- shall not be used until there is at least 300mm cover over the pipes.
- Construction vehicles shall not be allowed to cross drain trenches until the final surface is placed except where timber sleepers or steel plates are positioned to bridge the trench.

MATERIAL

- Unless otherwise indicated drainage, vent and rain water pipes and fittings inside the uilding (except kitchen and laundry pipes) shall be of U-PVC to BS 4514 for pipe sizes up to 80mm dia. and MU-PVC / ABS to BS 5255 for pipe sizes 50mm dia. and smaller. Fittings shall be push fit type.
- drainage pipes for discharged from the sump pumps shall be of HDPE.
- Acoustic lagging shall be provided for all horizontal drainage pipes above false ceiling, as detailed in drawings.
- Drainage pipes buried under ground up to manhole shall be U-PVC to BS 4660.
- Sewage pipe between manholes shall be U-PVC to BS 4660 protected with concrete encasement.
- All pipes under building and those subject to traffic especially under roads shall be protected with reinforced concrete.

FLOOR / ROOF / BALCONY DRAINS

- Floor drains unless otherwise indicated shall have traps of a minimum water seal of 7cm and shall be provided with adjustable and removable strainers. The open area of strainer shall be at least two thirds of the cross-sectional area of the drain line to which it connects. Floor drains shall have heavy-duty stainless-steel cover with a removable strainer and cover plate. Floor drain shall have built in Roding eye.

- All floor drains / cleanouts must be coordinated with floor tiling layout.
- Roof Rainwater outlet shall be doomed outlets or side type, cast iron similar to wade series or approved equal.
- Balcony outlet shall be of C.P Nickel Bronze or stainless steel, with trapped or non- trapped sump.

PIPE WORK:
INSTALLATION:

- Vent pipe shall be extend through roof and terminate 1500 mm above it. Vent pipe passage through the roof shall be made watertight by proper flashing.
- All changers of direction shall be gradual and not abrupt, 45-degree fittings shall be used wherever possible and 90-degree fittings shall be of the long sweep type. All unnecessary turns and off sets shall be carefully avoided and run as directly as possible from the sanitary fixtures to the vertical stacks.
- Concealed pipes shall be installed in such a manner as to permit easy accessibility for maintenance this applies particularly to valve locations.
- All pipes shall be fixed in neatly arranged lines, and adequately pitched horizontal lines to allow the system to be properly vented and drained. Air pockets, traps and sags shall be carefully avoided.
- Storm Drainage: Provide a complete system of storm drainage piping for all roofs, set back areas and canopies.
- Sanitary Drainage: Provide a complete system of sanitary drainage piping for plumbing fixtures, kitchen wastes, air conditioning wastes and the like from ground level which can be connected to the city drainage scheme.
- The storm, sanitary and vent piping shall be as shown on the drawings as specified herein and in accordance with the local authorities having jurisdiction.
- Take special care in setting roof drains to ensure that they are set at an elevation, which will prelude formation of puddles.
- Install connections to roof drains in conjunction with the roofing specified under civil works so that the building is adequately protected during construction from damage by storm water.
- No short radius bends to be used. Use short “Tee-Wye” fittings in vertical piping only.
- Provide cleanouts at foot of all stacks, changes of directions, at the ends of branch runs in straight runs as required and where indicated. Terminate as specified under “Cleanouts”.
- Run building drains at a minimum grade of 2 % (1:50) pitch unless otherwise noted, downward in the direction of flow. Pitch branch connections to stacks from fixtures at 2 % (1:50) where possible.
- Provide all the required appurtenances to make the drainage system complete in compliance with code requirements including traps, pipe fittings hangers and the like.
- Wherever possible, vent stack offsets shall be made with 45-degree fittings.
- Extend the tops of ventilating stacks independently through the roof or collect together and run through the roof in series of larger pipes as shown on the drawings.
- Any piping passing through roofs shall be so arranged to be a minimum of 300mm (1ft) from walls

or other obstructions so as to permit proper flashings which are provided by another trade.

- All the high level drainage pipes shall be covered with sound proof insulation.

CLEANOUTS ON SOIL, WASTE & VENT INSTALLATION:

- Cleanouts shall be installed at each change of direction of drainage pipes greater than 45 degrees, inside the building and where indicated on the drawings. Cleanouts shall be not more than 10m apart in horizontal lines. A cleanout shall be provided at or near the foot of each vertical waste or soil stack.
- Cleanouts on concealed piping shall be extended through and terminate flush with finished wall or floor. Pits or chases may be left in the wall or floor, provided they are of sufficient size to permit removal of the cleanout plug and proper cleaning of the system.
- Wherever it is necessary to conceal a cleanout plug, a chrome plated, brass covering plate shall be provided which will permit ready access to the plug.
- Cleanout plugs shall be of heavy cast brass of the screwed type.
- Cleanout plugs shall be of same nominal size as the pipes upto 100mm pipe diameter and not less than 100mm for larger piping.
- Cleanout shall be so installed that there is a clearance of not less than 45 cm for the purpose of rodding and cleaning.

PIPING MATERIAL:

- Soil, vent and rain water pipes above ground inside the building shall be of UPVC to BS 4514. Waste pipes shall be UPVC to BS 5255.
- Soil, waste and rain water pipes buried under the buildings up to manholes and the pipes between manholes shall be of UPVC to BS 4660, protected with concrete except the joints which shall be left flexible.
- Fittings shall be push-fit type with approved quality ring seal.
- All pipes subject to traffic specially under shall be protected with reinforced concrete.
- Drainage pipe supports shall be of electrolytically zinc plated clip with factory rubber lining complete with 8mm threaded GI rod, fixed with drilled plugs.

FLOOR DRAINS:

- Floor drains shall be UPVC traps of a minimum water seal of 7cm and shall be provided with adjustable and removable strainers. The open area of strainer shall be at least two thirds of the cross-sectional area of the drain line to which it connects.
- Floor drains shall have brass chrome plated or heavy duty stainless steel floor plates connected to the UPVC trap with a removable strainer and cover plate. Floor drains shall have a built in rodding eye.

ROOF DRAIN:

- Roof drain and open to sky drains shall be die cast aluminum of Dome/Parapet type connected to the rain water pipe without any trap. Size of drain shall be as per drawings.Free discharge outlets shall be back draft SS316 cover.

CHANNEL GRATING:

- Channel grating shall be of heavy duty galvanized cast iron construction to the sizes shown on drawings. The channel shall be set on angle frame of the same construction all to the relevant British Standard.

GULLY TRAPS:

- Supply and install wherever shown on the drawing of UPVC gully traps with 4" dia spigot outlet, round plastic grid.

CATCH BASINS:

- Catch basins of 450x450 mm size shall be provided as shown on the drawings complete with GRP perforated bucket heavy duty cast iron grating and frame.

FLOOR TRAPS AND COVERS:

- Deep seal type UPVC floor trap with 3 inlets (2”/11/2”) and 3” outlets complete with stainless steel SS316 cover and grating.

KITCHEN SINK:

- Shall be stainless steel of satin polished chrome nickel 18/10 material, double bowl double drain complete with strainer, chain & plug. Blanco-Germany make or equivalent.

TRAPS FOR KITCHEN SINKS & WASH BASINS:

- Traps shall be UPVC, telescopic bottle type, with cleaning eye and adjustable heights.

SINK MIXER:

- Chrome plated, wall type, stainless steel, with high efficient as per green building regulation. From Grohe-Germany, Ideal standard – UK, Tyford – UK.

ABLUTION MIXER:

- Chrome plated, push type, stainless steel, with high efficient as per green building regulation. From Grohe-Germany.

MANHOLES AND COVERS:

- Internal manholes will be constructed as part of the concrete ground floor slab channels and ¾ section branch bends shall comply with BS 65 and 450. Concrete benching between the branch channels, shall be trowelled smooth and slope towards the main channel at a slope of not less than 1:12.
- External manholes shall be constructed as shown on drawings.
- Internal Dry manhole covers shall be bolt down air tight, light duty, double seal, recessed top covers are to be institute concrete filled with surface finish to match surrounding floor. The visible edges shall be provided with brass trim covering.
- Covers for external manholes subject to vehicular traffic shall be double seal, cast iron, heavy duty. Covers for external manholes not subject to vehicles traffic shall be medium duty, double seal, cast iron.
- Manholes Covers and Frames:
- All covers and frames shall be Cast iron, double seal of 600 x 600mm. Covers shall be either heavy duty grade – A or medium duty grade – B as shown on the drawings. All covers are to be bitumen coated.
- Covers and frames for Gully Traps:
- Shall be Cast iron, double seal of 300x300 mm size, covers shall be of heavy duty grade-A with

bitumen coated.

- Manholes and Gully traps to be complete with:
  - Sealing grooves to be filled with grease.
  - 3 sets of cover keys to be provided.
  - GRP sealing of 6mm thickness to be provided when MH is connecting to main server line.

TESTING OF DRAINAGE AND VENT PIPE SYSTEMS:

- Before the sanitary fixtures are installed, drainage and vent pipes systems shall be subjected to a water pressure test to ensure and prove their tightness and to a flow test to ensure their freedom from obstruction.
- The water pressure test shall be applied to the system in its entirely or in sections. All openings in the piping shall be tightly closed with special cast iron pipe plugs or other suitable means and the system filled with water to the point of overflow from the highest opening. The plugs shall be temporarily opened to make sure that all air has been vented and that has reached all parts of the system.
- No section shall be tested to less than a 3-meter head of water. In testing successive sections at least the upper 3 meters of the next preceding section shall be tested so that no joint or pipe, except the uppermost 3 meters of the whole system shall have been subjected to a test of less than a 3-meter head of water.
- The water shall be kept in the system or in the portion under test for at least 4 hours before inspection starts. While the system is under pressure, a careful inspection shall be made of all pipes and joints and if any leaks in joints or evidence of defective pipe or fitting is disclosed the defective work shall be corrected immediately by replacing defective parts with new joints and materials. No make shift repairs or application of any repair compound will be permitted.
- After the correction is made the pressure test shall be repeated until the system is proved tight.
- Underground drainage pipes shall be tested by plugging the end of the pipes and fittings with water to a minimum head of 3 meters. The test pressure shall be maintained for 24 hours pipes and joints shall be inspected and approved before backfilling the trench.
- All drainage systems shall be tested for proper flow to ensure their freedom from any obstruction. The Contractor shall disassemble, clear repair and re-assemble obstructed piping at his own expenses. After re-assembly the piping shall again be subjected to the pressure test.

SWIMMING POOL

- Water connections will be provided from the site wide potable water network, and will extend to the swimming pool plant rooms as required by the swimming pool Consultant. Drainage connections will also be provided to meet the requirements of the swimming pool plant room systems.
- The design of any swimming pool tanks, pumps, plant or distribution systems will be the responsibility of the swimming pool Consultant and should be encompassed within their scope of works or their specialist sub-Consultant.
- The spatial requirements for these elements should be provided by the swimming pool consultant

to the project architect for incorporation to the base plans space planning.

- The swimming pool design Consultant must supply WSP with their detailed water supply and drainage requirements, at the next design stage, to enable progress of the Public Health design.

IRRIGATION SYSTEMS

- The design of any irrigation tanks, low-evaporative irrigation systems, irrigation pump systems, plant or feeder systems will be the responsibility of the Landscape Consultant and should be encompassed within their scope of works or their specialist sub-Consultant.
- Water connections will be provided from the site wide potable water network to the irrigation plant rooms as required by the Landscape Consultant. A drain connection will also be provided to meet the requirements of the irrigation system. The main water make-up for the irrigation water tank shall be via TSE water. A condensate recovery system shall be provided to supplement the irrigation water demand requirement.
- The spatial requirements for these elements should be provided by the Landscape Consultant to theproject architect.
- The landscape design Consultant must supply WSP with their detailed water supply and drainage requirements, at the next design stage, to enable progress of the Public Health design.

CONDENSATE DRAINAGE

- The condensate drainage for all FCU's/AHU's from each unit shall be conveyed to the basement condensate collection tanks or sump pumps and then transferred to the condensate collection tan.
- Collected condensate shall be used for irrigation.
- All condensate pipes and condensate sump tanks shall be insulated to offer reliable protection against condensation.
- The condensate drainage recycling will act only, as a supplement to the TSE supply used for irrigation.

BASEMENT DRAINAGE

- Separate basement drainage disposal system will be provided to convey the discharge from the car park, plant rooms and HVAC shafts.
- A combination of channel drains, gully traps, floor clean-outs, pressure break assemblies will be provided to collect and discharge wastewater from the building and connect to the external drainage network.
- The drainage from all car parking and vehicle areas will be provided with the appropriate sized sand traps and petrol interceptors. These interceptors will be located in the basement and in the areas with access for maintenance.
- The piping and the fitting material will be HDPE for the underground foul and storm water drainage.
- Granular fill shall be laid in 100mm layers and had compacted to a level 300mm minimum above top of pipe followed by main backfill material placed and compacted in 300mm layers any trench sheeting being withdrawn as the work proceeds. Heavy mechanical compactors

MANHOLES

- All manholes shall be constructed to the requirements of the Drainage Department.
- Internal manhole covers shall be bolt down airtight double seal, recessed top.
- Covers are to be in-situ concrete filled, with surface finish to match surrounding floor. All covers are to be medium duty, locking and sealed, with a clear opening of 600mm x 600mm.
- Covers for external manholes subject to vehicular traffic shall be kite marked, heavy ductile iron single seal, non-rock heavy duty with square push fit seal plate. Covers to be complete with lifting key-holes, and generally constructed in accordance with BS. 497:76 and shall have a clear opening of 600x600mm unless otherwise indicated. Double seal covers shall be provided where indicated on drawings.
- All covers shall be of ductile iron and shall be grey - epoxy coated.

GULLY GRATES

- Where indicated on drawings provide kite-marked kerb gully unit of ductile iron hinged grate, road retaining bar with clear opening size 532 x 405mm. Unit to be black bitumen coated and shall be designed to deter ingress and help prevent blockage. A GRP grid shall be provided as standard.
- Storm water gratings shall be kite-marked, heavy duty ductile iron, non-rock clear openings size 600 x 600mm, grey - epoxy coated and shall comply with BS 497.

TESTING OF DRAINAGE AND VENT PIPE SYSTEMS.

- Before the sanitary fixtures are installed, drainage and vent pipe systems shall be subjected to a water pressure test to ensure and prove their tightness and to a flow test to ensure their freedom from obstructions.
- The water pressure test shall be applied to the system in its entirety or in sections. All openings in the piping shall be tightly closed with special cast iron pipe plugs or other suitable means and the system filled with water to the point of overflow from the highest opening. The plugs shall be temporarily opened to make sure that all air has been vented and that water has reached all parts of the system.
- No section shall be tested to less than a 3-metre head of water. In testing successive sections at least, the upper 3 metres of the next preceding section shall be tested so that no joint or pipe, except the uppermost 3 metres of the whole system, shall have been subjected to a test of less than a 3-metre head of water.
- The water shall be kept in the system or in the portion under test for at least 4 hours before inspection starts. While the system is under pressure, a careful inspection shall be made of all pipes and joints and if any leaks in joints or evidence of defective pipe or fitting is disclosed the defective work shall be corrected immediately by replacing defective parts with new joints and materials. No makeshift repairs or application of any repair compound will be permitted.
- After the correction is made the pressure test shall be repeated until the system is proved tight.
- Underground drainage pipes shall be tested by plugging the end of the pipe and filling with water to a minimum head of 3 metres. The test pressure shall be maintained for 24 hours. Pipes and joints shall be inspected and approved before backfilling the trench.

- All drainage systems shall be tested for proper flow to ensure their freedom from any obstruction. The Contractor shall disassemble, clear, repair and re-assemble obstructed piping at his own expense. After re-assembly the piping shall again be subjected to the pressure test.

SECTION 6

GAS (LPG) SYSTEM

SECTION – 6

GAS SYSTEMS – INDEX

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- **FIELD ADJUSTMENTS**

GAS SYSTEMS:

- LPG will be stored in tanks and will be accompanied by vaporizers and ancillary equipment.
- LPG tanks shall be located as follows:
  3m from neighbouring building, building façade and door/window openings
  6.1m from air intakes
  7.6m from edge of roof
- All fill and supply pipework shall be run externally. Pipes runs into the building shall be at the level of supply only and shall be ventilated and protected as per NFPA 58 requirements.
- The LPG plant will be fed from the Fill Point located above the grade level. The Fill Point will be conveniently located for the operators and incorporated into the external façade. The tanker access will be from the service road at Level 00 level.
- In order to minimize the impact of filling pipes (for liquid and vapor) on the ascetics of the external façade, it has been proposed to run the pipes within ventilated shaft in the external façade with appropriate markings.
- Distribution of the LPG gas will be via a number of OTS (Open to Sky) risers, made from a blockwork. The blockwork will be made from materials with min. fire rating of 2 hr, and as approved by Dubai Civil Defense.
- All OTS risers will be vertical and will vent directly to the atmosphere. For additional safety each OTS risers shall be provided with a number of gas sensors linked to the fire alarm. The alarm shall also trigger the gas main supply to be shut down.
- Each apartment unit will have its own gas meter, located within wall cabinet in kitchen. These meters would need to be read by the Landlord (at one central location through the BMS) and bills calculated and invoiced to the individual tenants. All supplies will be equipped with gas detectors and automatic alarm and isolation system to comply with local codes.
- All gas pipework will be clearly identified.
- The main distribution pipework will provided with the leak detection system which will be interfaced with fire alarm system and BMS. The LPG system will be provided with the main gas control panel located in the main control room to monitor dedicated tenant panel, entire leak detection system and bulk storage tank for automatic operation in all respect.
- The gas installation shall comply with the Engineer’s specifications and local authorities requirements.
- The gas pipework shall be capable of meeting the design gas demands of the gas appliances, including future expansion demands.
- The system shall be automatically controlled preventing potential misuse and shall be provided with remote consumption monitoring.

PRODUCTS
GENERAL

- Supply, install, test, commission, regulate and set to work the Works.

WORK INCLUDED

- 1. Sections of Division 22 Plumbing are not intended to delegate functions or to delegate Work to any specific trade.
- 2. The Work shall include all labor, materials, equipment, permits and tools required for a complete and working installation as described but not necessarily limited to items, in the Work Sections of Division 22 Plumbing.

SCOPE OF WORK

- This section establishes the requirements for operation and maintenance of all plumbing and drainage systems in accordance with the Contract Documents.

- Provide all necessary materials, labour, equipment, tools, appliances, services, hoisting, scaffolding, supports and supervision to provide complete safe installations in conformity with the Contract Documents.
- A complete plumbing and sanitary drainage system including all final connections to plumbing fixtures, floor trenches, rainwater collection points, grease, sand and oil interceptors, sumps, sump pumps, manholes and catch basins.
- Provision of the utility services as necessary to complete the maintenance works as detailed in the Contract Documents. All proposed works shall be fully detailed by the Contractor in Method Statements and submitted for acceptance. No work associated with building services and utility services shall be undertaken until acceptance from the Engineer.

EQUIPMENT NAMEPLATES

- Provide new apparatus, if required, (including electric motors) with proper nameplates affixed thereto, showing the size, name of equipment, serial number and all information usually provided, which also includes voltage, cycle, phase and horsepower of motors and the name and address of the manufacturer.
- Nameplates shall be in both English and Arabic languages.

PRODUCT SELECTION
MANUFACTURERS:

- Provide systems and products from one of the manufacturers from the approved Vendor List.

EXPEDITING OF EQUIPMENT

- Continuously check and expedite delivery of equipment and materials. If necessary, inspect at the source of manufacture.
- Continuously check and expedite the flow of necessary information to and from all parties involved.
- Immediately inform the Engineer if information is required from him. Provide equipment delivery records updated monthly or more frequently as required by the Engineer.

MATERIALS

- Standard of Materials and Equipment
- Refer to applicable Divisions of the Specification.
- Provide new material and equipment of specified quality, delivered, erected, connected and finished in every detail and supplied with the acceptance of the Engineer.
- Where a manufacturer's name and model number is mentioned, it is for the purpose of setting a standard of quality, performance, capacity, appearance and serviceability.
- Assume full responsibility for ensuring that when providing other acceptable alternative manufacturers
- all space, weight, connections, power and wiring requirements, etc. are considered and costs therefore
- included in the tender. Equipment requiring greater than specified energy requirements or unduly
- limiting service space requirements shall not be accepted.

PIPING, FITTINGS AND VALVES.

General

This section includes basic design provisions and material specifications for pipe, tubing, pipe and tubing fittings, valves used to connect container appurtenances with the balance of the LP Gas system.

LPG fill line shall be seamless steel to ASTM A53 Sch 80
Distribution pipework shall be seamless steel to ASTM A53, Sch 40.
Final connections to equipment shall be soft copper tubes to BS 2871 Table

Pipe and Tubing Fittings

Pipe joints in steel, brass or copper pipe shall be welded or brazed.
Fittings used at pressure higher than container pressure, such as on the discharge of liquid transfer pumps, shall be suitable for a working pressure of at least 350 psig (2.4 MPa gauge).
Fittings used with liquid LP-Gas, or with vapor LP-Gas at operating pressures over 125 psig (0.9 MPa gauge) shall be suitable for a working pressure of at least 250 psig (1.7 MPa gauge).
Fittings for use with vapor LP-Gas at pressure not exceeding 125 psig (0.9 MPa gauge) shall be suitable for a working pressure of 125 psig (0.9 MPa gauge).
Soldering or brazing filler material shall have a melting point exceeding 1,000 deg. F (538 deg. C).

Valves

Pressure containing metal parts of valves (except appliance valves), including manual positive shut off valves, excess flow check valves (either manually or automatically operated), used in piping systems shall be of steel, ductile (nodular) iron, malleable iron or brass. Ductile iron shall meet the requirements of ANSI/ASTM A 395 or equivalent t and malleable iron the requirements of ANSI/ASTM A 47 or equivalent. All materials used to include valve seat discs, packing, seals and diaphragms shall be resistant to the action of LP Gas under service conditions.

Valves shall be suitable for the appropriate working pressure, as follows:

a) Valves used at pressures higher than container pressure, such as on the discharge of liquid transfer pumps shall be suitable for a working pressure of at least 350 psig (2.4 MPa gauge). [400 psig (2.8 MPa gauge) WOG valves comply with this provision].
b) Valves to be used with liquid LP Gas, or with vapor LP-Gas at pressures in excess of 125 psig (0.9 MP a gauge), but not to exceed 250 psig (1,.7 MPa gauge), shall be suitable for a working pressure of at least 250 psig (1.7 MPa gauge).
c) Valves (except appliance valves) to be used with vapor LP Gas at pressures not to exceed 125 psig (0.9 MPa gauge) shall be suitable for a working pressure of at least 125 psig (0.9 MPa gauge).

Manual shut off valves, excess flow check valves and backflow check valves used in piping systems shall comply with the provisions for container valves.

Emergency shut off valves shall be approved and incorporate all of the following means of closing.

d) Automatic shut off through thermal (fire) actuation. When fusible elements are used they shall have a melting point not exceeding 250 deg.F (121 deg. C).

e) Manual shut off from a remote location. Manual push button switches for solenoid valves.

f) Manual shut off at the installed location.

4.2.2 Solenoid valves shall be intrinsically safe and explosion proof construction suitable for 24V DC power supply.

Design or Service Pressure

For Gases with Vapor Pressure in psig (MPa gauge) at 100 deg.F Exceed	Minimum Design Pressure in psig (MPa gauge) ASME Code Section VIII, Division 1, (37.8 deg.C) not to 1980 Edition (Note 1)
80 (0.6)	100
100 (0.7)	125
125 (0.9)	156
150 (1.0)	187
175 (1.2)	219
215 (1.5)	250
215 (1.5)	312.5 (2.2)

LPG VAPORIZERS

Vaporizers shall be indirectly electrically heated, FM approved, meeting the requirements of NFPA 58 and shall be suitable for Class 1, Division 2, Group D, hazardous locations as verified per FM approval requirements.

Vaporizers shall be suitable for outdoor installation and shall incorporate suitable automatic means to prevent liquid from passing from the vaporizer to the vapour discharge piping.

Vaporizers shall be equipped, at or near the discharge, with a spring- loa de d pressure relief valve providing a relieving capacity as dictated by NFPA 58.

Direct fired vaporizers with an inside diameter of more than 152mm shall be constructed in accordance with the applicable provision of the ASME code for a design pressure of 1,700kPa. Those having an inside diameter of 152mm or less shall be exempted from the ASME code but still be constructed for a minimum design pressure of 1,700 kPa.

GAS LEAK DETECTION SYSTEM

A gas leak detection system shall be installed to monitor for gas leaks at different locations of the gas network.

Control Panel

a) The central control panel shall be located in the entrance of the building and shall easily be accessible in the event of an emergency.
b) The central panel shall have a modular design incorporating input alarm/display module for each gas detector. Each channel shall have indication for low alarm at 15% LEL, high alarm at 30% LEL and fault at 0- 100% LEL indicator. Main LPG supply shall be shut off at 30% LEL indication.
c) The concentration of the gas shall be displayed on a common indicator with selector switches to display the concentration of the desired channel or zone indicator.
d) Below the gas control panel is the system control panel which shall have the following indications:
   - AC healthy
   - DC healthy
   - Sprinkler on
   - Gas pressure low
   - Gas pressure high
   - Gas solenoid valve operated
   - Common fire signals
   - Gas leak high

The panel shall have the following switches:

- Sprinkler manual operate
- Gas manual shut off
- Test lamp
- System reset

Gas Detectors

The gas detectors shall use a pellistor sensor to detect flammable gases and vapors. The sensor shall be housed in a stainless steel, flame proof housing, fitted onto an explosion proof junction box (Tank Area).

Sounder Flasher

A sounder cum flasher unit shall be installed in the vicinity of the panel located in a visible area.

Cables

MICC cables or FP 200 cables of at least 1.5 sq. mm area shall be used to interconnect between each detector and the control panel.

Logic of operation

When low level gas leak is detected in any zone, local internal alarm and indication shall show o the control panel. When high level gas leak is detected in any zone, the following sequence shal occur:

e) The solenoid valve in the gas distribution channel shall be closed. b) External alarm shall be operated.

Connection to the fire alarm system:

In the event of a fire alarm signal, the following sequence shall occur:

a) The solenoid valve in the gas distribution channel shall be closed. b) The sprinkler system shall operate.

In the event of an emergency, it shall be possible to bypass the automatic system and operate the system manually, this shall include:

a) Manual shut off / on of the gas.
b) Manual operation of the sprinkler system on / off

SOURCE QUALITY CONTROL

INSPECTION AND TESTING DURING MANUFACTURE

- The Employer's and the Engineer's representatives shall be entitled at all reasonable times during manufacture to inspect, examine and test on the manufacturer's premises the materials and workmanship of all plant to be supplied.
- In accordance with the Contract Documents, notify the Employer's and the Engineer of the time and location of his manufacturing shop tests for the following liste Equipment's:
  a) Drainage interceptors.
  b) Domestic booster water pumps, filters, water meters and localized water heaters.
  c) Anti-vibration and seismic restraints.
- Such inspection, examination or testing, if made, shall not release the Contractor from any obligation under the Contract.
- Where the Contract calls for optional witness tests on the premises of the manufacturer, the manufacturer shall provide labor, materials, electricity, water, fuel, stores, apparatus and instruments as may be required and as may be reasonably demanded to carry out such tests.
- Results, recommendations and reports of all such inspections shall be documented as witnessed in writing with originals issued to the Employer.
- Pay for all costs incurred by the Engineer in traveling and expenses to witness the tests on equipment defined in this Section or as referenced in applicable Work Sections.
- Shop witness tests shall be optional and Contractor's costs shall be listed individually permitting deduct should witness test be waived by the Engineer.

a) The filling line operating pressure up to 20 bar.
b) The tank and high-pressure stage operating pressure 16.5 bar
c) The first stage regulator operating pressure 1.5 bar.
d) The second stage pressure, operating pressure 37 mbar.

Test pressure shall be at least 1.5 times the operating pressure. The leak test duration shall be 1 hour. The specified test pressure shall be monitored for a minimum of one hour during which the soap water method shall be used to leak test every joint. If no leak is detected and the pressured gauge indicates no drop in the test pressure then the system will be proven to be leak free.

If a leak is found, the leaking joint shall be repaired or the leaking equipment shall be replaced. The testing procedure shall be repeated to prove the complete system as leak free.

A test report shall be submitted to the consultant for approval.

EXECUTION EXAMINATION

- A. Verification of Conditions: Examine areas for compliance with requirements for installation and conditions affecting performance of the Work. Identify conditions detrimental to a proper and timely completion and notify the Engineer of the unsatisfactory conditions. Proceed with installation only after unsatisfactory conditions have been corrected.

SUPERINTENDENCE STAFF

Maintain at the Site, at all times, qualified personnel and supporting staff with proven experience in erecting, supervising, testing and adjusting plumbing projects of comparable nature and complexity.

- The Contractor's Site supervisory personnel and their qualifications are subject to the acceptance of the Engineer.

SYSTEM ACCEPTANCE

- Submit original copies of letters from the manufacturers of all systems indicating that their technical representatives have inspected and tested the respective systems and are satisfied with the method of installation, connection and operation.
- These letters shall state the names of persons present at testing, the methods used, and a list of functions performed with location and room numbers where applicable.

OPERATION AND MAINTENANCE:

GENERAL:

- Operation and maintenance shall comply with the Section 22 Plumbing specifications and manufacturer's recommendations.
- All systems shall be released to the Engineer certified ready for use and working in a satisfactory manner to meet the performance requirements of the Specification.
- All floor traps and traps of plumbing fixtures shall be filled with water to prevent the escape of sewer gases.
- All sumps with pumps shall be filled with water to the minimum level.
- All control systems shall be operational in accordance with the Specification.

INSPECTION INTERVALS

- A. Inspection intervals shall comply with the manufacturer's recommendations.
- B. Site conditions shall be taken into consideration, while establishing inspection intervals, especially if the drysolid content or sandy liquid is pumped.
- If not specifically indicated by the manufacturer, pumps and other mechanical equipment used in Plumbing systems shall be inspected at least once a year.

REPLACEMENT OF DEFECTIVE MATERIAL OR EQUIPMENT

- Repair or replace any material or equipment found defective or cannot pass the tests specified in this Section at no additional cost to the Employer.
- Complete correction of defective material or equipment and retesting within the Contract period.
- If the equipment or material cannot pass the second test, remove the defective equipment and replace it with equivalent equipment that meets the Specification. Such replacement at no additional cost to the Employer.
- Remove defective equipment or material from the Site no later than 15 days from the date of
- notification by the Engineer or his representative.

FIELD ADJUSTMENTS

- Properly remedy defects in Work or equipment that appear during tests and test again. Continue until acceptance by the Engineer.

MATERIAL APPROVALS

ALL MATERIAL ARE SHOULD BE DCD AND APPROVED FROM CONSULTANT TO BE UPTAINED .

DCD APPROVALS

GAS System and LPG contractor scope for approvals ..

END OF SECTION

SECTION 7

FIRE FIGHTING

SECTION 7

FIRE FIGHTING - INDEX

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FIRE SUPPRESSION
COMMON WORK RESULTS FOR FIRE SUPPRESSION

GENERAL
Fire suppression systems shall be provided as detailed in these specifications, A. drawings and other referenced documents.

Applicable codes and standards for all the Fire Suppression systems shall be as follows:

- 1. UAE Fire & Life Safety Code of Practice
- 2. National Fire Protection Association (NFPA) Standard Regulations

SCOPE OF WORKS
The Contractor shall supply, install, test and commission the complete Fire Suppression services installation as described within this specification, the standard specification and as indicated on the design drawings and equipment schedules. The Fire Suppression services associated with this package shall include

- Sprinkler Systems
- Standpipe Systems
- Hydrant System
- Clean Agent Gaseous Fire Suppression Systems
- Foam/Deluge Water Systems
- Portable Fire Extinguishers

DESCRIPTION OF WORKS
The fire suppression system descriptions associated with this package are described in a brief manner in the below sub-clauses. For more detailed general specifications for each system please refer to the Standard workmanship and materials specification for this project

FIRE SUPPRESSION WATER STORAGE

- The Contractor shall supply, install, test, commission and set to work a complete and fully operational fire suppression water storage installation in accordance with the true intent of this particular specification, the standard specification and as indicated on the design drawings.
- The design will be in accordance with NFPA 22 and will be constructed from RCC complete with two (2) equal compartments and supplied with appropriate isolation facilities and segregation to allow periodic cleaning and maintenance works.
- The tank shall be provided with a filling connection directly from incoming water services line with a float operated valve for automatic refilling. The tank shall be provided with drain arrangement, overflow connections, level indicators, level switches, and other necessary accessories as indicated within the standard specification and detailed within the drawings.

FIRE PUMPS
- The Contractor shall supply, install, test, commission and set to work a complete and fully operational fire pump installation in accordance with the true intent of this particular specification, the standard specification and as indicated on the design drawings and equipment schedules.
- Dedicated fire pump set shall serve the shall comprise of the following, as detailed within the drawings:

- One electric motor driven (duty) fire pump
- One diesel engine driven (standby) fire pump
- One electric pressure maintenance (jockey

SPRINKLER SYSTEM

- The Contractor shall supply, install, test, commission and set to work a complete and fully operational automatic wet pipe fire sprinkler system throughout in accordance with the true intent of this particular specification, the standard specification and as indicated on the design drawings and equipment schedules.
- buildings shall be sprinkler protected throughout in accordance with NFPA 13 and the UAE Fire Code as detailed within the drawings and equipment schedules:
- The main sprinkler system will be a wet pipe system wherein sprinkler risers will be provided to cover each individual fire zone in the buildings. All control valves and system risers will be supplied and connected with fire pumps at the basement.
- In addition, fire zones on all levels will be provided with a Zone Control Valve assembly (ZCV) that comprises of a supervised control valve, water flow alarm switch, pressure gauges and test/drain valve connection.
- A drain riser shall be provided within each fire systems riser shaft, feeding into the drainage system, or free discharge outside the building.
- The sprinkler pipework serving each zone shall be preferably looped (where possible) to provide enhanced system hydraulics, sized as shown on the design drawings.

TYPES OF SPRINKLERS TO BE USED ARE AS FOLLOWS:

1). Concealed Pendent Sprinkler – installed in all areas false ceiling is available, except for the back of the house (BOH) areas
2). Recessed Pendent Sprinkler – installed in back of the house (BOH) areas, with false ceiling
3). Upright Sprinkler – installed in areas where false ceiling is not available and ceiling voids
4). Sidewall Sprinkler – installed in areas where piping across the ceiling is not desirable due to aesthetics or building construction considerations (i.e. residential bedrooms, elevator shaft, etc.)

- Sprinklers shall be provided in ceiling voids where the ceiling void depth is 800mm and greater.
- Deluge (water spray) system will also be provided to serve the LPG tanks on the roof level in accordance with the UAE Fire Code 2011. From the fire pump system manifold, a separate connection with deluge valve will be provided to supply the spray system.

STANDPIPE SYSTEM

- The Contractor shall supply, install, test, commission and set to work a complete and fully operational standpipe system in accordance with the true intent of this particular specification, the standard specification and as indicated on the design drawings and equipment schedules.
- buildings shall be protected by a Class III standpipe system as defined in NFPA 14 and in compliance with NFPA 5000, NFPA 101 and the UAE Fire
  Code.
- For the stairwells throughout the building, the Class III system will be made up of the following:
- Within the exit stairs:
  1). 2 ½” Ø (65mm Ø) landing valve / hose connection On the perimeter of each exit stair core, a cabinet (schedule/drawing reference FHC-2) containing:
- 1”Ø (25mm Ø) with 30m length Hose Reel
- Portable Fire Extinguishers – 1x DCP and 1x CO2 types each with 4.5 kg capacity Further hose stations will be required to be installed such that all portions of each floor level will be within

36 meters (30 m hose length and 6 m water jet length) of a hose connection as indicated on the drawings.

- These Fire Hose Cabinets (schedule/drawing reference FHC-3) shall contain the following:
  1). 2 ½” Ø (65mm Ø) landing valve / hose connection with 30m length of hose.
  2). 1”Ø (25mm Ø) with 30m length Hose Reel
  3). Portable Fire Extinguishers – 1x DCP and 1x CO_2 types each with 4.5 kg capacity

FIRE DEPARTMENT CONNECTIONS

- The Contractor shall supply, install, test, commission and set to work complete and fully operational fire department connections in accordance with the true intent of this particular specification, the standard specification and as indicated on the design drawings and equipment schedules.
- Fire department connection shall be located on the external wall above ground level nearest to the vertical run of the standpipe and should be visible and recognisable from the nearest point of fire department access.
- The fire department connections shall not be sited more than 18m away from the fire engine access road.
- The fire department connections shall be located not more than 100 ft (30.5 m) from the nearest fire hydrant connected to an approved water supply.
- Fire department connection shall be provided for the main building only. Each set of fire department connections will consist of four-way (4) type breeching inlets complete with listed non return/check valves.

FIRE HYDRANTS

- The Contractor shall supply, install, test, commission and set to work complete and fully operational fire hydrant system in accordance with the true intent of this particular specification, the standard specification and as indicated on the design drawings and equipment schedules.
- b. Fire hydrants shall be provided within the development in accordance with NFPA 24 and the UAE Fire Code. Connection shall be made from the existing site wide infrastructure network and shall be extended to supply the additional fire hydrants for the buildings.

FIRE EXTINGUISHERS

- The Contractor shall supply, install, test, commission and set to work complete and fully operational
- Hand held / portable fire extinguishers shall be provided throughout the building in accordance with NFPA 10 and the UAE Fire Code. All Portable fire extinguishers complete with inspection tag shall be left in a fully charged condition and inspected and approved by Dubai Civil Defence at the completion of the project.
- Clean Agent Gaseous Fire Suppression System
- The Contractor shall design in detail, supply, install, test, commission and set to work a complete and fully operational clean agent gaseous fire suppression installation in accordance with the true intent of this particular specification, the standard specification and as indicated on the design drawings and equipment schedules.
- The following areas of the building shall be served by a clean agent gaseous fire suppression system:
  1). LV rooms
  2). Telecommunication/GSM rooms
  3). Control rooms

- The clean agent gaseous fire suppression system provided for the proposed development shall be covered by Novec 1230 system.

FOAM WATER FIRE SUPPRESSION SYSTEM

- The Contractor shall design in detail, supply, install, test, commission and set to work a complete and fully operational foam water fire suppression system in accordance with the true intent of this particular specification, the standard specification and as indicated on the design drawings and equipment schedules.
- A foam water fire suppression system will be provided to serve the generator rooms. From the sprinkler system, a separate connection with deluge valve will be provided to supply the foam water fire suppression system.

DESIGN CRITERIA AND STANDARDS

FIRE A. SUPPRESSION WATER STORAGE

- The dedicated fire water storage shall comply with NFPA 22 and the UAE Fire Code.
- The volume of stored water for the sprinkler and standpipe system shall be 228 m3 which has been calculated as sufficient for a minimum period of 60 minutes at the maximum design water flow rate of 1000 gallon per minute based on the calculated demand of the most onerous of the sprinkler and standpipe systems.

B. FIRE PUMP

- The fire pumps and their associated piping and electrical systems shall be in accordance with the relevant parts of the NFPA 20 and the UAE Fire Code. Fire pumps shall meet the following performance criteria:

- Shut off Head shall range from 101% to 140% of Rated Head NFPA 20: A.6.2 At 150% Rated Capacity, Head will range from a Minimum 65% to a maximum of just below the Rated Head Fire pump buildings or rooms enclosing fire pump drivers shall be protected with an automatic sprinkler system.

Design Criteria/Parameters	Reference
Shut off Head shall range from 101% to 140% of Rated Head	NFPA 20: A.6.2
At 150% Rated Capacity, Head will range from a Minimum 65% to a maximum of just below the Rated Head	
Fire pump buildings or rooms enclosing fire pump drivers shall be protected with an automatic sprinkler system.	

C. SPRINKLER SYSTEM

- The proposed buildings shall be sprinkler protected throughout in accordance with NFPA 13. The following design criteria shall be adhered to:

FOAM WATER G. FIRE SUPPRESSION SYSTEM

- The generator rooms shall be provided with a foam water fire suppression system in accordance with NFPA 16 and as detailed in the standard specification and drawings.

DELUGE WATER SPRAY SYSTEM

- The LPG tanks on the roof level shall be provided with a deluge water spray fire suppression system in accordance with NFPA 15 and as detailed in the standard specification and drawings.

Space	Occupancy Classification	Design Density (lpm/m²)	Area of Oper'n (m²)	Max. Sprinkler Coverage (m²)	Sprinkler Types	Standards Reference
Apartments	Light Hazard	4.1	139	21.0	Pendent, Sidewall, Quick Response	NFPA 13: Chapter 11 UAE Fire Code
Storage, Plant Rooms, Car Park	Ordinary Hazard Grp 1	6.1	139	12.1	Upright, Standard Response	
Retail shops	Ordinary Hazard Grp 2	8.1	139	12.1	Upright, Standard Response	
Fire Pump Room	Extra Hazard Grp 1	12.2	Entire Pump room	9.3	Upright, Standard Response	
Sprinkler system zone control valve (ZCV) provision: On each zone						NFPA 13: Chapter 8
Sprinkler system components will be rated for the maximum system working pressure to which they are exposed but will not be rated at less than 12.1 bar for aboveground components						NFPA 13: Chapter 6

STANDPIPE SYSTEM

- The main building shall be protected by Class III standpipe as defined in NFPA 14 and shall comply with the following design criteria:

Design Criteria/Parameters	Reference
Class III Type Standpipe Location: Throughout building such that all parts of the building are within 30 m of a Fire Hose Cabinet	NFPA 14
Minimum Residual Pressure: 6.9 Bar (100 psi) - Landing Valve Maximum Flow rate : 250 gpm (946 litre/ min.) - Individual Hose Connection Minimum Flow rate : 500 gpm (1892 litre/ min.) - Remotest Standpipe	NFPA 14: Chapter 7
Where the static pressure at a hose connection exceeds 12.1 bar, an approved pressure regulating device will be provided to limit static and residual pressures at the outlet of the hose connection to 12.1 bar (175 psi)	NFPA 14: 7.8.3.2.

FIRE EXTINGUISHERS

- The proposed buildings shall be provided with fire extinguishers throughout in accordance with NFPA 10 and the UAE Fire Code. The following design criteria shall be adhered to:

Design Criteria/Parameters	Reference
Occupancy Classification: Light, Ordinary, & Extra Hazard Occupancy. Selection for the class(es) of hazards to be protected, spacing and types	NFPA 10.6.2
Maximum travel distance to extinguisher: 75 Ft.(23 m)-criteria for Light, Ordinary & Extra Hazard Occupancy	

CLEAN AGENT GASEOUS FIRE SUPPRESSION SYSTEM

- Specific rooms within the proposed buildings, as listed above, shall be provided with a clean agent gaseous fire suppression system in accordance with NFPA 2001. The following design criteria shall be adhered to:

Occupancy Classification: Light & Ordinary Hazard Conditions/uses: Rooms as listed above Design, use, protection, installation, & limitations of Clean Agent gas suppression system.	NFPA 2001

CODES AND STANDARDS

- The complete fire suppression services have been designed and shall be installed, tested and maintained by the contractor in accordance with the below standards:
  - UAE Fire and Life Safety Code of Practice
  - NFPA 10: Standard for Portable Fire Extinguishers
  - NFPA 11: Standard for Low, Medium, and High-Expansion Foam
  - NFPA 13: Standard for the Installation of Sprinkler Systems
  - NFPA 14: Standard for Installation of Standpipe, and Hose Systems
  - NFPA 15: Standard for Water Spray Fixed Systems for Fire Protection
  - NFPA 16: Standard for the Installation of Foam-Water Sprinkler Systems
  - NFPA 20: Standard for the Installation of Stationary Pumps for Fire Protection
  - NFPA 22: Standard for Water Tanks for Private Fire Protection
  - NFPA 24: Standard for the Installation of Private Fire Service Mains and Their Appurtenances
  - NFPA 58: Liquefied Petroleum Gas Code
  - NFPA 70: National Electric Code
  - NFPA 72: National Fire Alarm Code
  - NFPA 2001: Standard on Clean Agent Fire Extinguishing Systems
- The final detail design and installation of the complete fire suppression systems shall be carried out by an approved specialist contractor who shall be on the approved list issued by the UAE Civil Defence Authorities.

OPERATION AND MAINTENANCE OF FIRE SUPPRESSION

GENERAL

SUMMARY

- General: Read this Section in conjunction with other related Sections, Division 01 General Requirements, the Particular Specification, Schedules, Design Drawings and the Contract Conditions.

SECTION INCLUDES

- General requirements for the operation and maintenance of Fire Suppression.

RELATED WORK SECTIONS AND DOCUMENTATION

- Division 21 Fire Suppression Work Sections.
- Division 22 Plumbing Work Sections.
- Division 23 Heating, Ventilating and Air Conditioning (HVAC) Work Sections.
- Division 25 Integrated Automation Work Sections.
- Division 26 Electrical Work Sections.
- Division 33 Utilities Work Sections.

SUBMITTALS

- General: Comply with the requirements of Section 013300 Submittal Procedures and submit the following.
- Provide the following items as referenced in applicable Work Sections for each related Division.

POST CONTRACT SUBMITTALS

- Shop Drawings.
- Product Data on materials and components for use.
- Supplementary Product Literature: Include a statement from the manufacturer for the design life of the system.
- List of tests included.
- Certified test data.
- Outline technical specifications reflecting proposed materials and systems.
- A list of proposed suppliers and Subcontractors intended to be used.
- Preliminary Method Statement.
- Preliminary Quality Plan.
- Summary of deviations from the Specification.
- Reports of factory witness tests as called for in each section and requested by the Engineer.
- Valve Tag Charts.
- Field Test Reports.
- Authorities Inspection Reports.
- System Verification Certificates.
- Special tools recommended by the product manufacturer and required for maintenance.
- Coordination Construction Drawings: Prepare and submit to the Engineer for acceptance a dimensioned layout with required sections of piping for each floor. These drawings shall be used as coordination drawings and be distributed to all Divisions whose Work is affected.
- Drawings shall be in a larger scale, usually 1:20 and shall include piping sections in accordance with planned fabrication, all pipe and conduit fittings, valves and cleanouts with all dimensions for erection.
- In addition to the above layouts and sections prepare details indicating a) sizes and locations of special details including but not limited to:
- Sleeves and openings.
- Inserts and anchors.
- Access panel openings.
- Pits.
- Housekeeping pads, vibration isolators and equipment bases.
- Post Contract Samples: As referenced in applicable Work Sections.

CLOSEOUT SUBMITTALS

GENERAL:

- Comply with the requirements of Section 017000 Execution and Closeout Requirement and submit the following:

WARRANTIES.

- Unless otherwise specified in other Work Sections, provide a) a manufacturer's full warranty covering all products of this Division, valid for 2 years from date of issue of Taking-Over Certificate, irrespective of delivery date and trial usage.
- Provide warranty certificates for all equipment supplied to this project.
- Include copies of warranties in the appropriate sections of the Operation and Maintenance (O+M) Manuals. Original certificates shall be submitted separately to the Employer.

GUARANTEE

- 2. Guarantee the system for 1 year free maintenance, 5 years for availability of spare parts and technical assistance. All other works for this section shall be in writing for 1 year, unless specifically stated in the Specification. All guarantees shall commence after completion of the testing and commissioning and final acceptance of the complete systems. Promptly repair or replace defective materials or equipment, workmanship and installation that develop within this period to the Engineer satisfaction at no cost to the Employer. Damage caused in making necessary repairs and replacements shall be remedied and corrected at no cost to the Employer.
- Comply with the guarantee requirements of the individual requirements of the individual Mechanical Work Sections of Division 21 Fire Suppression that exceed those of this paragraph.

OPERATION AND MAINTENANCE (O+M) MANUALS

- Include component list with manufacturer's reference numbers, descriptions of materials and procedures for repairing and cleaning of finishes and cleaning frequency.

Record Drawings: As referenced in applicable Work Sections.
Recommended Spare Parts list: As referenced in applicable Work Sections. Spare parts list shall identify by name, the part numbers and quantities recommended and the current prices

QUALITY ASSURANCE

- A. The manufacturer of all materials and equipment shall have at least 10 years of experience in the design and manufacture of the respective products unless otherwise stated in other sections of this Specification.

PERMITS, FEES AND INSPECTIONS

- Apply for, obtain, and pay for all permits, licenses, inspections, examinations and fees required for Work of Division 21 Fire Suppression.
- Arrange for inspection of all Work by all concerned Statutory Authorities having jurisdiction.
- On completion of the Work, present to the Engineer the final unconditional certificate of acceptance of all concerned Statutory Authorities having jurisdiction.
- The installation of gas, fire protection, fuel handling, domestic water and drainage systems shall be installed to the construction standards established by Statutory Authorities having jurisdiction. In no instance shall the Contractor reduce the standards or scope of Work established by the contract drawings and Specification.

Standards and other Codes of Practice:
As referenced in applicable Work Sections.

- Provide testing and inspections in accordance with Section 014000 Quality Requirements.

- Preconstruction Testing/ Reports: As referenced in applicable Work Sections.

DELIVERY, STORAGE AND HANDLING

DELIVERY

- Moving of Equipment: Ensure that materials and equipment are delivered to the Site at the proper time and in such assemblies and sizes so as to enter into the building and shall be moved into the spaces where they can be easily located.

STORAGE

- Store all equipment and supplies in a manner accepted by the Engineer.

PROJECT CONDITIONS

TEMPORARY SERVICE

- Refer to the applicable Division of the Contract Documents regarding temporary services, Contractor's shop, storage and other such facilities.
- Do not use any of the permanent Mechanical Systems during construction unless specific written acceptance is obtained from the Engineer. Accepted usage of the permanent Mechanical Systems shall not be construed as acceptance by the Employer.
- The use of permanent facilities for temporary construction service shall not affect in any way the commencement day of the warranty period.
- Be responsible for any cutting and patching involved in getting assemblies into place.

ACCESSIBILITY

- For operation, maintenance and repair minor deviations from Design Drawings shall be permissible.
- Group concealed valves, expansion joints, controls, dampers and equipment requiring access, shall be freely accessible through access doors.
- If required, provide catwalk for access to elements.

CONTRACT DESIGN DRAWINGS AND SITE EXAMINATION REQUIREMENTS

- The Design Drawings for Mechanical Work are performance drawings, diagrammatic, intended to convey the scope of Work and indicate general arrangement and approximate location of apparatus, fixtures and pipe runs. The Design Drawings do not intend to show architectural and structural details.
- If obvious ambiguities or omissions are noticed when tendering, they shall be referred to the Engineer for a ruling and obtain the ruling in writing in the form of an Addendum.
- Do not scale Design Drawings. Obtain information involving accurate dimensions from dimensions indicated on Architectural and Structural Design Drawings and by Site measurement.
- Visit and inspect the Site of the Work to verify location and elevation of existing services which may affect the Tender and Work of this Division including water, electrical, sanitary and storm water mains, irrigation water pipe work and telephone lines before submission of tender and again before proceeding with the Work.

PRODUCTS
GENERAL
WORK INCLUDED

- Sections of Division 21 Fire Suppression are not intended to delegate functions or to delegate Work to any specific trade.
- The Work shall include all labor, materials, equipment, permits and tools required for a complete and working installation as described but not necessarily limited to items, in the mechanical Work Sections of Division 21 Fire Suppression.

SCOPE OF WORK

- This section establishes the requirements for completing all fire protection in accordance with the Contract Documents.
- Provide all necessary materials, labor, equipment, tools, appliances, services, hoisting, scaffolding, supports and supervision to provide complete safe installations in conformity with the Contract Documents.
- Complete fire protection system comprising of breeching inlet connections, electric diesel and jockey fire pumps, standpipe wet risers, fire hydrants, fire hose cabinets, sprinkler system, pressure reducing valves, flow indicators, supervisory fire valves, sprinkler zone control valves, test drains, piping and accessories, portable and automatic fire extinguishers and controls interconnected to the Building Management System (BMS).
- Provision of the utility services as necessary to complete the new works as detailed in the Contract Documents. All proposed new works shall be fully detailed by the Contractor on Shop Drawings, in Method Statements and submitted for acceptance. No work associated with building services and utility services shall be undertaken until acceptance from the Engineer.

RELATED WORK SPECIFIED IN OTHER DIVISIONS

- 1. Finished painting and stenciling.
- 2. Access doors.
- 3. Fireproofing of access doors.
- 4. Manholes in concrete for access to all storage tanks. See Design Drawings.
- 5. Trench covers and frames. See Design Drawings.
- 6. Cutting and patching, except as noted in the applicable Division of the Specification.
- 7. Door louvers.
- 8. Undercutting doors.
- 9. Louvers and panels in masonry work.
- 10. Plenums other than sheet metal.
- 11. Flashing.
- 12. Gratings.
- 13. Electrical Service.
- 14. Fire and Life Safety Systems.
- 15. Clean-up.
- 16. Painting.
- 17. Site Drainage.
- 18. Site Utilities.
- 19. Equipment platforms unless specifically called for in the Specification.

CLEARANCE FROM ELECTRICAL EQUIPMENT

PIPING:

Piping shall not pass through the a) following locations:

- 1. HV Switchgear rooms.
- 2. Transformer rooms.
- 3. LV Switchgear room
- 4. Electric rooms and closets.
- 5. Telephone rooms and closets.
- 6. Elevator machine rooms.
- 7. BMS and Security control rooms.
- 8. Over electrical panels and bus way.

Where it is unavoidable and pipework must be routed in areas housing electrical work these services shall be installed no closer than 1.6m of:

- 1. Transformers.
- 2. Switchboards.
- 3. Motor control centers.
- 4. Motors, except for branch piping to equipment.
- 5. Electrical panels and bus way.

In exceptional circumstances when over or within 1.6m is unavoidable provide drip pans under piping, subject to the acceptance of the Engineer.
All Work within these areas shall be closely coordinated with Division 26 Electrical Work Sections.

DRIP PANS UNDER PIPING:

- 1. 18 gauge Type 304 stainless steel.
- 2. Reinforced and independently supported.
- 3. Watertight.
- 4. Provide 30mm drain outlet piped to hub floor drain.

EQUIPMENT NAMEPLATES

- Provide apparatus (including electric motors) with proper nameplates affixed thereto, showing the size, name of equipment, serial number and all information usually provided, which also includes voltage, cycle, phase and horsepower of motors and the name and address of the manufacturer.
- Nameplates shall be in both English and Arabic languages.

SYSTEM REQUIREMENTS

- Water-Based Fire Suppression Systems and Equipment, and Fire Extinguishing Systems.
- Test, commission, regulate and set to work the Works.
- Design all necessary facilities required for setting to work commissioning and testing of the completed installations.
- Appoint an independent specialist commissioning Contractor responsible for the testing and commissioning.

- The Contractor shall ensure that fixed extinguishing system components and agents are designed and approved for use on the specific fire hazards they are expected to control or extinguish.
- Provide all necessary attendance and attendance of Specialist Contractors/ Suppliers to allow familiarization of the local operating personnel in the Operation and Maintenance (O+M) of the Services Installation.
- Ensure that the commissioning requirements are compatible with any project restraints concerning sectional handover/ phasing.
- Provide complete Operation and Maintenance (O+M) instruction manuals.
- All aspects of system operation shall be detailed, including piping isometrics, wiring diagrams of all circuits, a written description of the system design, sequence of operation and drawing(s) illustrating control logic and equipment used in the system.
- Provide checklist and procedures for troubleshooting techniques, maintenance operations and procedures to be included in the manual.
- Provide drawing(s) showing actual installation details including all equipment locations.
- Review all designs to ensure that systems are commissionable.
- Provide procedures presented in the specification and contract drawings.
- Members of the designated Commissioning specialist shall witness various portions of the commissioning process.
- The Engineer shall witness various portions of the commissioning process.
- The Engineer shall sign-off on appropriate sections after verifying installation, operation and relevant documentation.
- Final sign-off shall be by the Civil Defence Authority having jurisdiction.
- Any test ports, gauges, test equipment, etc. needed to accomplish the functional performance tests shall be provided by the Contractor.
- The Contractor shall provide all documentation of calibration for all controls and equipment being used.
- Documentation shall include dates, setpoints, calibration coefficients, control loop verification, and other data required to verify system check-out.
- Documentation shall be dated and initialed by field engineer or technician performing the work.
- The Contractor shall provide system narrative descriptions supported by flow diagrams, one line diagrams, and appropriate specification sections for major systems to be commissioned.
- Coordinate "system description" meetings with members of facility management and maintenance department groups to review system description documentation.
- The Contractor shall provide an overview of major system features, components, and arrangements.
- The Contractor shall provide required training to operational staff after the system description meetings have occurred.
- Training sessions shall provide a more detailed analogy of systems Operation and Maintenance (O+M).
- Provide a statement on appointment and prior to commencement of the Works.
- Highlight for review by the Engineer any considerations in respect of commissioning.
- The Contractor shall incorporate into the systems design the essential components and features necessary to enable the proper preparation and commissioning of the systems.

- The Contractor shall produce comprehensive commissioning method statements including procedures, logic diagrams and risk assessments for the pre-commissioning checks, setting to work, commissioning, system proving and environmental testing of the Works.
- Produce flushing, chemical cleaning and water treatment method statements, logic diagrams and programmed for integration into the building contractors construction, commissioning and finishes programmers.
- Establish procedures with all parties to allow the demonstration of normal, emergency, shutdown and standby mode operation of plant and systems.
- Provision of all necessary facilities to enable tests to be witnessed and inspections carried out including all necessary instruments and recorders to monitor systems during the commissioning and environmental proving period.
- Produce record pro-forma documentation relating to the commissioning and testing of plant and systems.
- Ensure all requirements such as cleanliness, protection from harmful external and internal elements etc. are provided prior to commencement of commissioning.
- Coordinate the activities of all specialized personnel, including manufacturers together with providing necessary attendance.
- Ensure all certification is attained and witnessed as necessary for inclusion in the record documentation.
- Maintain a log of all significant activities during the testing, commissioning and system proving process.
- Ensure that all plant and system settings are recorded.
- Provide a report signed, by the Commissioning Specialist confirming, prior to installation that all system designs can be commissioned.
- Provide and submit a report for every test, demonstration, balance or commissioning activity witnessed, together with an engineering appraisal on the performance, either on or off-site.
- Provide a final commissioning report detailing the results of the commissioning and commenting on the performance of systems signed by the Commissioning Specialist.
- The report shall highlight and confirm that each installation is correctly tested and commissioned and achieving the specified system performance.
- Demonstrate that the equipment is capable of the performance and method of operation specified.
- Demonstrate that the overall and complete systems perform correctly in the required manner and as intended by the specification.

PORTABLE FIRE EXTINGUISHERS

- Fire extinguishers and fire blankets shall be maintained at regular intervals all in accordance and as specially detailed within the said manufacturers Service and maintenance literature and criteria as stipulated.
- Inspections and thorough examinations by a competent and accredited contractor inclusive of all necessary repairs, recharging or replacement of same, such that extinguisher(s) and blankets operate effectively and safely when duly called upon.
- All fire extinguishers and fire blankets shall be provided complete with a monthly inspection tag and shall be duly dated and signed, by the said contractor following inspection of same.

- All fire extinguishers shall be inspected and approved by the Civil defiance Authority having jurisdiction, at the time of completion and left in a fully charged condition.
- It is a requirement of the specification that the Contractor is appropriately trained, accredited and approved by the Civil Defense Authority having jurisdiction, at the time and during the period of construction and that all relevant certificates of training and approval be made available for inspection by the Clients appointed representatives when called upon.
- It shall be the responsibility of the Contractor to check for any damage or misuse prior to inspection inclusive of replacement of same.
- All fire blankets shall be inspected and approved by the Civil Defense Authority having jurisdiction, at the time of completion and shall be checked by the Contractor for damage or misuse prior to inspection inclusive of replacement or recharging of same.
- Any costs associated with non-conformance, compliance or non-approval of the Civil Defense Authority having jurisdiction shall be borne by the Contractor.
- Subsequent inspections shall be carried out by the said Contractor within 30 days following initial installation and approval.
- The Contractor shall allow for all costs associated with the inspection, examination and testing of same.
- In addition the Contractor shall allow for all costs associated with the provision of all manufacturers Warranties and for the repair or replacement of any and all fire extinguisher(s) and fire blankets that fail in materials or workmanship within the specified warranty period at no extra cost to the Client.

FAILURES SHALL INCLUDE, HOWEVER NOT LIMITED TO, THE FOLLOWING REQUIREMENT:

- Failure of hydrostatic a) test according to NFPA 10.
- Faulty operation of valves or release levers.
- Damage that could impair the safe operation or the life of the product.
- Use of the product prior to handover and receipt of authority approval.
- Illegible text and operating instructions.
- 13. It shall be at the Client's discretion that all fire extinguishers and fire blankets inclusive of associated ancillaries be replaced should a failure rate of 10% of all products be established and it shall be the responsibility of the Contractor to replace all such products, in full, at no extra cost to the Client.

PRODUCT SELECTION

Manufacturers: Provide systems and products from one of the listed manufacturers within the Vendor List.

Expediting of Equipment

- 1. Continuously check and expedite delivery of equipment and materials. If necessary, inspect at the source of manufacture.
- 2. Continuously check and expedite the flow of necessary information to and from all parties involved.
- 3. Immediately inform the Engineer if information is required from him.
- 4. Provide equipment delivery records updated monthly or more frequently as required by the Engineer.

MATERIALS

- A. Standard of Materials and Equipment
- Refer to applicable Divisions of the Specification.
- Provide new material and equipment of specified quality, delivered, erected, connected and finished in every detail and supplied with the acceptance of the Engineer.
- Where a manufacturer's name and model number is mentioned, it is for the purpose of setting a standard of quality, performance, capacity, appearance and serviceability.
- Assume full responsibility for ensuring that when providing other acceptable alternative manufacturers all space, weight, connections, power and wiring requirements, etc. are considered and costs therefore included in the tender. Equipment requiring greater than specified energy requirements or unduly limiting service space requirements shall not be accepted.
- All materials shall bear a clear cut mark indicating that they are eco-friendly and free from ozone layer depleting contents.

PATENTS

- Pay all royalties and license fees, and defend all suits or claims for infringement of any patent rights, and save the Employer and Engineer harmless of loss or annoyance on account of suit, or claims of any kind of violation or infringement of any letters patent or patent rights, by this Contractor or anyone directly or indirectly employed by him or by reason of the use by him or them of any part, machine, manufacture or composition of matter on the Work, in violation or infringement of such letters patent or rights.

SOURCE QUALITY CONTROL

Inspection and Testing during Manufacture

- The Employer's and the Engineer's representatives shall be entitled at all reasonable times during manufacture to inspect, examine and test on the manufacturer's premises the materials and workmanship of all plant to be supplied.
- In accordance with the Contract Documents, notify the Employer and the Engineer of the time and location of his manufacturing shop tests for the following listed equipment: Electric, diesel and jockey fire pumps, gas fire suppression system, a) kitchen hood fire suppression system, foam deluge system.

ANTI-VIBRATION AND SEISMIC RESTRAINTS.

- Such inspection, examination or testing, if made, shall not release the Contractor from any obligation under the Contract.
- Where the Contract calls for optional witness tests on the premises of the manufacturer, the manufacturer shall provide labor, materials, electricity, water, fuel, stores, apparatus and instruments as may be required and as may be reasonably demanded to carry out such tests.
- Results, recommendations and reports of all such inspections shall be documented as witnessed in writing with originals issued to the Employer's.
- Pay for all costs incurred by the Engineer in traveling and expenses to witness the tests on equipment defined in this Section or as referenced in applicable Work Sections.

- Shop witness tests shall be optional and Contractor's costs shall be listed individually permitting deduction should witness test be waived by the Engineer.

EXECUTION

EXAMINATION

Verification of Conditions: Examine areas for compliance with requirements for installation and conditions affecting performance of the Work. Identify conditions detrimental to a proper and timely completion and notify the Engineer of the unsatisfactory conditions. Proceed with installation only after unsatisfactory conditions have been corrected.

SITE QUALITY CONTROL

SUPERINTENDENCE STAFF

- Maintain at the Site, at all times, qualified personnel and supporting staff with proven experience in erecting, supervising, testing and adjusting mechanical projects of comparable nature and complexity.
- The Contractor's Site supervisory personnel and their qualifications are subject to the acceptance of the Engineer.

TRIAL USAGE

- The Employer has the privilege of trial usage of Mechanical Systems or parts thereof for the purpose of testing and learning operational procedures.
- Carry out trial usage over a maximum period of time as instructed by the Engineer.
- Carry out the operations only with the express knowledge and under supervision of the trade specialist who shall not waive any responsibility because of trial usage.
- Trial usage shall not be construed as Taking-Over of the Work.

SYSTEM ACCEPTANCE

- Submit original copies of letters from the manufacturers of all systems indicating that their technical representatives have inspected and tested the respective systems and are satisfied with the method of installation, connection and operation.
- These letters shall state the names of persons present at testing, the methods used, and a list of functions performed with location and room numbers where applicable.
- Submit such letters for the following:

FIRE

- protection systems.
- Automatic fire extinguishing systems.
- Gas supply system.
- Control and building management systems.

COMMISSIONING

GENERAL:

- The final start-up operation of all systems shall be coordinated with other trades participating in the construction.

- All systems shall be released to the Engineer certified ready for use and working in a satisfactory manner to meet the performance requirements of the Specification.
- All control systems shall be operational in accordance with the Specification.

FIRE PROTECTION SYSTEM

- The fire protection shall be complete with hoses and portable extinguishers all ready for use.
- The sprinkler and standpipe systems shall be filled with water
- Safety valves and pressure reducing valves shall all be set to correct operating pressures. Fire pumps shall be ready to operate via pressure switch control, and the jockey pump shall be operational maintaining design static pressure in pipe systems.
- All normally open gate valves in the system shall be opened with valves in bypass lines completely closed.

General Testing: Furnish all labor, material, instruments, supplies and services and bear all costs for the accomplishment of the test as listed, correct all defects appearing under tests and repeat the tests until no defects are disclosed; leave the equipment clean and ready for use:

- 1. Fire protection system test.
- 2. Fire pumps test.
- 3. Noise and vibration measurement test.
- 4. Room condition test.
- 5. Controls and building management test.

Electrical Equipment Tests: After completion of all electrical power and control wiring the Contractor shall in conjunction with Division 26 Electrical Work Sections make the following checks:
1. Correct rotation of mechanical equipment.
2. Correct interconnection of control devices.

- Furnish all necessary supplies, equipment and labor for cleaning, testing, adjusting and startingup of equipment, provisional and removal of all temporary piping connections required for tests and dispose of test water and waste after tests.
- Provide Technicians capable of performing air and water balance, vibration and noise measurements. The Engineer shall accept the Technicians.
- Each piece of equipment, including motors and controls shall be operated continuously for a period of not less than 2 hours in the presence of the Engineer before acceptance.
- Make tests of piping necessary during the progress of the Work. No Work and equipment shall be insulated or concealed until it is successfully tested.
- Each piece of apparatus shall be carefully checked for completeness of connections, accessories, wiring and controls and be placed in operation, tested and adjusted.
- Carry out final tests at a time suitable to and in the presence of the Employer and the Engineer.
- Provide all necessary labor, material and equipment necessary to assist in the performance of tests to be conducted by the local inspection authorities.

CLEANING

- Vacuum clean and remove debris from the inside of including but not limited to water reservoirs, tanks, air handling systems, fans, ducts, coils and terminal units before testing and operational start-up.

- Clean exposed surfaces mechanical equipment, ductwork, piping and polish plated work.
- Comb all bent fins to proper configuration on all coils of air handling units and fan coil units.

DEMONSTRATION

- Instructions to the Employer
- Instruct Employer selected personnel on all aspects of the operation of systems and equipment.
- Arrange and pay for services of manufacturers' representatives required for instruction on the following specialized portions of the installation:
- Fire Suppression Systems.
- Section 212000 Fire Extinguishing Systems.

PROVIDE THE FOLLOWING FOR EACH SESSION:

- Date of instructions given to the Employer's staff.
- Duration of instruction.
- Names of persons instructed.
- Other parties present (manufacturer's representative and Engineer).

OBTAIN THE FOLLOWING FOR EACH SESSION:

- Signature of the Employer's staff stating that they properly understand the system installation, operation and maintenance requirements.
- Employer's Written Acceptance: The Contractor shall submit for acceptance 3 copies of the following written acceptance by the Employer's acknowledging the Contractor's instruction of the Employer's Representative.
- The Undersigned Representative of the Employer has been instructed by the Mechanical Subcontractor in the proper operation of the Mechanical Systems and Equipment.

END OF SECTION

M.E.P. APPROVED MANUFACTURE LIST (MECHANICAL)

- **HVAC - MECHANICAL WORKS**
- **WATER SUPPLY WORKS**
- **FIRE FIGHTING**
- **DRIANGE**

M.E.P. APPROVED MANUFACTURE LIST (MECHANICAL)HVAC - MECHANICAL WORKS (APPROVED LIST)-URBAN HABITAT ARCH& CONSULTANT

HVAC WORKS (APPROVED LIST)-URBAN HABITAT ARCH& CONSULTANT			
S. NO.	**MATERIAL**	**MANUFACTURER**	**ORGIN**
1	VRF UNITS WITH R 410A REFRIGERANT	LG	KOREA
2	DUCTED SPLIT UNIT WITH R 410A REFRIGERANT	LG	KOREA
3	FAHU	VTS AIRCHEL TROSTEN	
4	COPPER PIPES AND FITTINGS	MUELLER MAKSAL	USA USA
5	STRAINERS	CRANE ECONOSTO HATTERSLEY	UK EUROPE UK
6	VIBRATION ISOLATORS	KINETICS WEICCO NTEICO	USA INDIA UAE
7	FIBERGLASS INSULATION	GULFOFLEX AFICO MANSON	UAE KSA USA
8	RUBBER INSULATION / ELASTOMERIC INSULATION	GULFOFFLEX ARMAFLEX – AP KFLEX ISOLINE	
9	DUCT LINER	AFICO KFLEX GULFOFLEX	
10	VOLUME CONTROL DAMPERS	ALDES GMAMCO TECNALCO BETA	UAE UAE UAE UAE
11	SOUND ATTENUATORS	BIN AIR ATAI ENERGY INTERNATIONAL GMAMCO	UAE UAE UAE UAE
12	FIRE DAMPERS,MSFD,MD	TECNALCO BETA FIREGAURD (SHOULD BE APPROVED BY DCD)	UAE UAE UAE UAE
13	G.I DUCT	HADEED STEEL JINDAL STEEL FLAME GAURD	KSA INDIA UAE

S. NO.	MATERIAL	MANUFACTURER	ORGIN
14	FLEXIBLE DUCT CONNECTOR	DURODYNE CLIMATECH CAIN	USA BELGIUM USA
15	PHENOLIC DUCT	PAL EASY P3	
16	ELECTROSTATIC PRECIPITATORS	EXCELAIR AIRQUALITY-AIRMAKER	
17	FLEXIBLE DUCTS	ATCO THERMAFLEX HAFLEX	
18	AIROUTLETS GRILLES / DIFFUSERS / LOUVERS	AIRMASTER BETA FLAME GAURD	UAE UAE UAE
19	DISC VALES	AIRMASTER TECNALCO BETA	UAE UAE UAE
20	ACCESS DOORS	AIRMASTER SAFID	UAE KSA
21	ADHESIVE AND COATING	FOSTER PEDILITE MIRACLE	USA INDIA USA
22	AIR CURTAIN	HITACHI MITSUBISHI VTS	JAPAN JAPAN POLAND
23	UPVC / PRESSURE PVC	GEORGE FISCHER UPONOR MUELLER	GERMANY GERMANY USA
24	EXTRACT FANS (CENTRIFUGAL/AXIAL) WALL MOUNTED EXTRACT FANS	PENNE EXPLAIRE S & P	USA UK SPAIN
25	SMOKE EXTRACT FAN	GREENHECK ELTA FANS OR EQUIVALENT APPROVED BY LOCAL CIVIL DEFENSE DEPT.	USA ITALY
26	DUCT AND PIPE SUPPORTS	GRINNEL ISOFIX WEICCO	
27	SMOKE EXTRACT DUCT (FIRE RATED G.I DUCT)	HADEED STEEL JINDAL STEEL FLAME GAURD	KSA INDIA UAE
28	CLADDING	PROCLAD VENTURE CLAD	
29	THERMOSTAT	HONEYWELL SIEMENS	USA GERMANY

- Any item which is not mentioned in the vender list Contractor shall submit RFI.

NOTE: All the HVAC System to be properly tested and Commissioning, air balancing reports to be provide for approvals.

All the AIR balancing test to be done by our approved Contractors only.

WATER SUPPLY WORKS

WATER SUPPLY WORKS (APPROVED LIST)-URBAN HABITAT ARCH& CONSULTANT			
S. NO.	**MATERIAL**	**MANUFACTURER**	**ORIGIN**
1.	WATER TANKS (INSULATE GRP PANEL HOT PRESSED)	PIPECO MITSUBISHI BALMORAL	MALAYASIA JAPAN UK
2.	DOMESTIC WATER TRANSFER AND BOOSTER PUMPS SETS	GRUNDFOS DEAVY FLYGT	UK AUSTRALIA SWEDEN
3.	WATER HEATERS (GLASS LINED, STORAGE TYPE)	ARISTON RHEEM THERMEX	GREECE MEXICO RUSSIA
4.	UPVC PRESSURE PIPES AND FITTINGS	HEPWORTH	UK
5.	COPPER PIPES AND FITTINGS	MULLER CAMBRIDGE LEE WEDNESBURY	USA USA UK, equivalent
6.	ABS PIPES	DURAPIPE GEORGE FISCHER OR EQUIVALENT	UK UK
7.	CROSS LINK POLYETHYLENE PIPES (PEX-PIPES) WITH PVC CONDUITS	KAN-THERM IVAPEX	EUROPE EUROPE
8.	PPR PIPES AND FITTINGS	KE KLIT KAN-THERM TORO 25	EUROPE EUROPE EUROPE
9.	GATE, GLOBE, CHECK, BALL AND FLOAT VALVES	HERZ CRANE NIBCO	EUROPE UK USA
10.	PLUMBING / BUTTERFLY /CHILLED WATER - VALVES	HERZ CRANE NIBCO	EUROPE UK USA

11.	PRESSURE REDUCING VALVE	HERZ DANFOSS CLA-VAL ZURN	EUROPE
12.	BLACK FLOW PREVENTORS	CLAVAL WATTS	SWITZERLAND UK EQUIVALENT
13.	WATER HAMMER ARRESTORS	SIOUX CHIEF WADE JR SMITH ZURN	USA UK
14.	Y-STRAINERS	SPIRAX SARCO CRANE ECONOSTO	USA UK EUR
15.	FLOAT SWITCHES	SIEMENS HONEYWELL	UK USA
16.	PICV/PRESSURE INDEPENDENT CONTROL VALVE	HERZ DANFOSS IMI/TA BELIMO	EUROPE EUROPE EUROPE EUROPE
17.	FLOW METERS, FLOW SWITCHES AND LEVEL SWITCHES	SMITH METER KROHNE SPARLING	GERM UK USA

- Any item which is not mentioned in the vender list Contractor shall submit RFI.

DRIANGE APPROVED MATERIAL (APPROVED LIST)-URBAN HABITAT ARCH& CONSULTANT		
S. NO.	MATERIAL	MANUFACTURER
1.	SUMP AND SEWAGE PUMPS	DEAVY -AUSTRALIA GRUNDFOS -(UK OR GERMANY)
2.	FLOOR TRAP COVERS / GRATINGS (STAINLESS STEEL316)	TERRAIN -UK HEPWORTH-UK EUMEL-PORTUGAL
3.	MANHOLE COVERS / GULLEY TRAP COVERS /CATCH BASIN / GRATINGS C.I.	RBA- INDIA DUCAST- INDIA
4.	ROOF DRAINS	TERRAIN-UK HEPWORTH -UK OR EQUIVALENT
5.	GRATING FOR CAR PARK	RBA- INDIA DUCAST- INDIA
6.	ACOUSTIC LINING FOR DRAINAGE PIPING/ INSULATION FOR UPVC PIPES	HODGSON & HODGSON-UK KIMMCO -KSA OR AS PER THE SAMPLE APPROVAL BY CONSULTANTAND CLIENT
7.	MIXERS / ANGLE VALVE	GROHE GERMANY WEBERT-ITALY RAK-UAE
8.	UPVC PIPES AND FITTINGS (GRAVITY)	HEPWORTH
9.	UPVC PIPES AND FITTINGS (PRESSURE)	HEPWORTH
10.	SOUNDPROOF PIPING.	WAVIN – GERMANY SKOLAN – GERMANY POLOPLAST – GERMANY
11.	GREASE TRAP	HEPWORTH TERRAIN
12.	PIPE SUPPORT	WEICCO NTEICO HEVIT

- Any item which is not mentioned in the vender list Contractor shall submit RFI.

FIRE FIGHTING APPROVED LIST)-URBAN HABITAT ARCH& CONSULTANT

1	Fire Pumps	Naffco	UAE
		Bristol	UAE
		Sibca	UAE
2	Diesel Engine (UL Listed and or FM Approved)	Clarke	
		Kriloskar	
3	Sprinkler System	Shield	UK
		Victaulic	USA
		Bristol	UAE
		Sibca	UAE
4	Hose Reel Cabinet/ Hosereels & Acessories	Naffco	UAE
		Bristol	UAE
		Sibca	UAE
5	Fire Fighting Extinguishers	Naffco	UAE
		Bristol	UAE
		Sibca	UAE
6	GI Pipes seamlesss UL/FM Grade Bschedule 40/80	Shield	China
		UTP	China
		Sibca	China
		Surya	India
7	Pipe Fittings	Shield	China
		Surya	India
		BIS	Thailand
		Flamex	Italy
8	Foam System	Naffco	UAE
		Bristol	UAE
		Shield	UK
9	Clean Agent Fire Extinguishing System	Stat x	USA
		Bristol	UAE
		Shield	UAE
		Sibca	UAE

10	Zone Control valve	Sibca	UAE
		Bristol	UAE
		Shield	UK
		Naffco	UAE

- Any item which is not mentioned in the vender list Contractor shall submit RFI.

LPG APPROVED LIST- (DCD SPECIFICATIONS) - URBAN HABITAT ARCH& CONSULTANT

Sno	Item/Equipments	Manufacturer	Country of Origin
1	LPG Tanks	Dogumak	Turkey
		Cysta	Mexico
		RCS	Italy
2	Gas Control Panel	Sietron	Italy
		Sensitron	Italy
		Gas Sense	Bulgaria
3	Gas Detectors	Sietron	Italy
		Sensitron	Italy
		Gas Sense	Bulgaria
4	Cables	Tekab	UAE
		Cabco	Turkey
		Ducab	UAE
6	Conduits	Decoduct	UAE
		Hotace	Oman
7	Ball Valves	Itap	Italy
		Apollo	USA
8	Solenoid valve	Madas	Italy
		Technogas	Italy
		Italpump	Italy